

# Draft

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December 2025

## Background

Delete everything in environment

```
rm(list = ls())
```

Load libraries

```
library(arrow)
library(broom)
library(CausalArima)
library(colorspace)
library(glue)
library(mgcv)
library(moderndiver)
library(scales)
library(scatterplot3d)
library(sf)
library(tidyverse)
library(tigris)
```

Load data for county shapefile (from library(tigris))

```
# Load county shapefile (includes FIPS codes and geometry)
counties_sf <- counties(cb = TRUE, resolution = "20m", year = 2020) %>%
  mutate(fips = paste0(STATEFP, COUNTYFP))
```

## |

|

```

# Filter for contiguous U.S.
contiguous_fips <- setdiff(sprintf("%02d", 1:56),
                             c("02", "15", "60", "66", "69", "72", "78"))
counties_sf <- counties_sf %>%
  filter(STATEFP %in% contiguous_fips)

```

Load data for crosswalk (between ZCTA and county)

```

xwalk22 <- read_parquet("us_xwalks__census__zcta2county_yearly__2022.parquet")
xwalk21 <- read_parquet("us_xwalks__census__zcta2county_yearly__2021.parquet")
xwalk20 <- read_parquet("us_xwalks__census__zcta2county_yearly__2020.parquet")
xwalk19 <- read_parquet("us_xwalks__census__zcta2county_yearly__2019.parquet")
xwalk18 <- read_parquet("us_xwalks__census__zcta2county_yearly__2018.parquet")
xwalk17 <- read_parquet("us_xwalks__census__zcta2county_yearly__2017.parquet")
xwalk16 <- read_parquet("us_xwalks__census__zcta2county_yearly__2016.parquet")
xwalk15 <- read_parquet("us_xwalks__census__zcta2county_yearly__2015.parquet")
xwalk14 <- read_parquet("us_xwalks__census__zcta2county_yearly__2014.parquet")
xwalk13 <- read_parquet("us_xwalks__census__zcta2county_yearly__2013.parquet")
xwalk12 <- read_parquet("us_xwalks__census__zcta2county_yearly__2012.parquet")
xwalk11 <- read_parquet("us_xwalks__census__zcta2county_yearly__2011.parquet")
xwalk10 <- read_parquet("us_xwalks__census__zcta2county_yearly__2010.parquet")
xwalk09 <- read_parquet("us_xwalks__census__zcta2county_yearly__2009.parquet")
xwalk08 <- read_parquet("us_xwalks__census__zcta2county_yearly__2008.parquet")
xwalk07 <- read_parquet("us_xwalks__census__zcta2county_yearly__2007.parquet")
xwalk06 <- read_parquet("us_xwalks__census__zcta2county_yearly__2006.parquet")
xwalk05 <- read_parquet("us_xwalks__census__zcta2county_yearly__2005.parquet")
xwalk04 <- read_parquet("us_xwalks__census__zcta2county_yearly__2004.parquet")
xwalk03 <- read_parquet("us_xwalks__census__zcta2county_yearly__2003.parquet")
xwalk02 <- read_parquet("us_xwalks__census__zcta2county_yearly__2002.parquet")
xwalk01 <- read_parquet("us_xwalks__census__zcta2county_yearly__2001.parquet")
xwalk00 <- read_parquet("us_xwalks__census__zcta2county_yearly__2000.parquet")

```

Load data for total CO2 emissions (at census block group level, in kilograms)

```

# Create path to the .gdb directory
co2_path <- "CMS_DARTE_V2_1735/data/DARTE_v2.gdb"

# List the available layers (feature classes) inside the .gdb
st_layers(dsn = co2_path)

```

```

## Driver: OpenFileGDB
## Available layers:
##               layer_name geometry_type features fields  crs_name
## 1 DARTE_v2_blockgroup_kgco2_1980_2017 Multi Polygon  220333    43 Vulcan_LCC

```

```
# Load the specific layer
co2_cbg <- st_read(dsn = co2_path, layer = "DARTE_v2_blockgroup_kgco2_1980_2017")

## Reading layer 'DARTE_v2_blockgroup_kgco2_1980_2017' from data source
##   '/n/home10/rtruong/CMS_DARTE_V2_1735/data/DARTE_v2.gdb' using driver 'OpenFileGDB'
## Simple feature collection with 220333 features and 43 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -7034829 ymin: -1957575 xmax: 3488418 ymax: 4595604
## Projected CRS: Vulcan_LCC
```

Load data for average PM2.5 concentration (at ZCTA level, in micrograms per cubic meter)

```
# Create path to the .parquet files
pm25_path <- "dataverse_files"

# List all .parquet files for years 2000-2020
file_list <- list.files(path = pm25_path,
                        pattern = "^pm25_ushap_zcta_yearly_20\\d{2}\\$.parquet$",
                        full.names = TRUE)

# Read and combine all files into data frame
pm25_zcta <- file_list %>%
  lapply(read_parquet) %>%
  bind_rows()
```

## Data wrangling for co2\_cbg

Convert co2\_cbg to tidy format

```
co2_cbg <- co2_cbg %>%
  pivot_longer(cols = starts_with("kgco2_"),
               names_to = "year",
               names_prefix = "kgco2_",
               values_to = "value") %>%
  mutate(year = as.integer(year))
```

Select for variables to avoid large data set

```
co2_cbg <- co2_cbg %>%
  ungroup() %>%
  st_drop_geometry() %>%
  as.data.frame() %>%
  rename(geoid = GEOID) %>%
  select(geoid, year, value)
```

## Data wrangling for xwalk

Combine everything into one crosswalk

```
years <- 2000:2022
file_paths <- paste0("us_xwalks__census__zcta2county_yearly__", years, ".parquet")
xwalk_list <- map(file_paths, read_parquet)
xwalk <- bind_rows(xwalk_list)
```

## Aggregation for pm25\_zcta

Join xwalk and pm25\_zcta

```
pm25_county <- inner_join(xwalk, pm25_zcta, by = c("zcta", "year"))
```

Aggregate from ZCTA level to county level

```
pm25_county <- pm25_county %>%
  group_by(year, county) %>%
  mutate(weight = (arealand_pct / 100)) %>%
  mutate(weight = weight / sum(weight, na.rm = TRUE)) %>%
  summarize(pm25 = sum(PM25 * weight, na.rm = TRUE),
    .groups = "drop")
```

## Aggregation for co2\_cbg

Create variable for county FIPS code

```
co2_cbg <- co2_cbg %>%
  mutate(county = substring(geoid, 1, 5))
```

Aggregate from census block group level to county level

```
co2_county <- co2_cbg %>%  
  group_by(year, county) %>%  
  summarize(co2 = sum(value, na.rm = TRUE),  
            .groups = "drop")
```

## Finalizing files

Filter for contiguous U.S.

```
pm25_county <- pm25_county %>%  
  filter(substr(county, 1, 2) %in% contiguous_fips)  
  
co2_county <- co2_county %>%  
  filter(substr(county, 1, 2) %in% contiguous_fips)
```

Create df by joining pm25\_county and co2\_county

```
df <- inner_join(pm25_county, co2_county, by = c("county", "year")) %>%  
  filter(substr(county, 1, 2) %in% contiguous_fips)
```

Filter for years less than or equal to 2010 (to avoid confounding)

```
df <- df %>%  
  filter(year <= 2010)
```

## Data visualization

Aggregate CO2

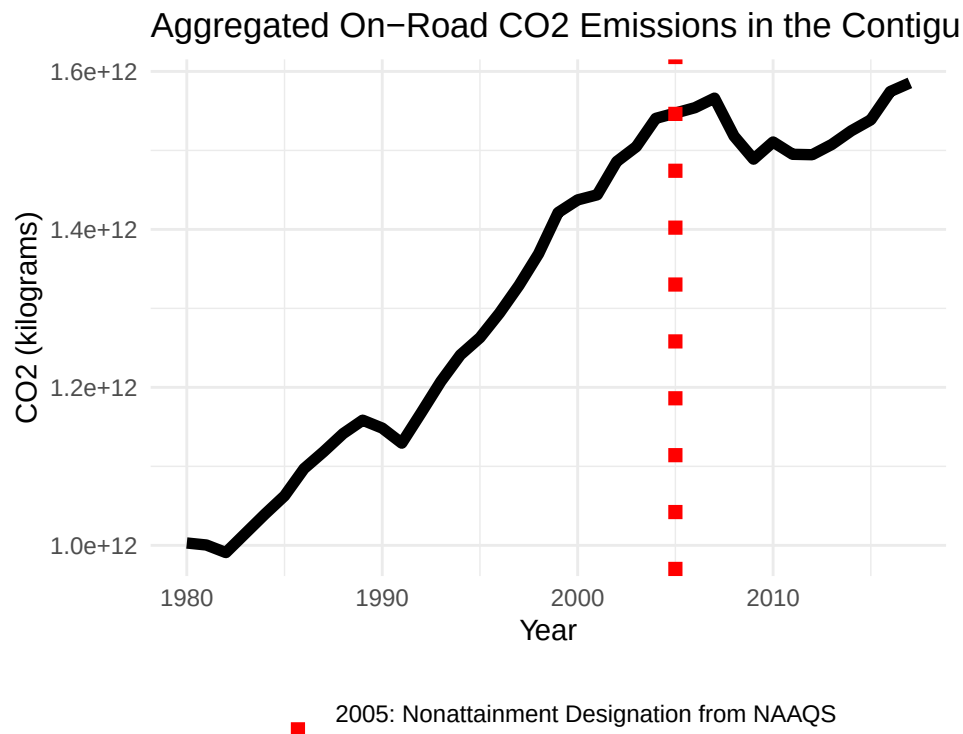
```
# Aggregate total CO2 for all counties by year  
co2_total_year <- co2_county %>%  
  group_by(year) %>%  
  summarize(total_co2 = sum(co2, na.rm = TRUE))  
  
# Check the result  
head(co2_total_year)
```

```
## # A tibble: 6 x 2
##   year total_co2
##   <int>     <dbl>
## 1  1980  1.00e12
## 2  1981  1.00e12
## 3  1982  9.91e11
## 4  1983  1.02e12
## 5  1984  1.04e12
## 6  1985  1.06e12
```

## Plot aggregate CO2

```
# Create graph of CO2 data
co2_plot <- co2_total_year %>%
  ggplot() +
  geom_line(aes(x = year, y = total_co2), linewidth = 2) +
  geom_vline(aes(xintercept = 2005, linetype = "2005: Nonattainment Designation from NAAQS"),
    color = "red",
    linewidth = 2.5) +
  labs(title = "Aggregated On-Road CO2 Emissions in the Contiguous U.S.",
    x = "Year",
    y = "CO2 (kilograms)",
    fill = "",
    linetype = "") +
  scale_linetype_manual(values = c("2005: Nonattainment Designation from NAAQS" = "dotted")) +
  theme_minimal() +
  # theme(legend.position = "bottom",
  #   plot.title = element_text(size = 44),
  #   axis.text = element_text(size = 32),
  #   axis.title = element_text(size = 40),
  #   legend.text = element_text(size = 36))
  theme(legend.position = "bottom")

# Plot and save graph
co2_plot
```



```
ggsave("co2_plot.png",
  plot = co2_plot,
  width = 20, height = 12, dpi = 300,
  bg = "white")
```

## Individual county-level analysis

Run CausalArima on specified county to estimate statistics

```
# FIPS "01053" is in Alabama (which had minimal legislation besides NAAQS)

# Specify FIPS code for individual county
my_fips <- "01053"

# Get county_df for specified FIPS
county_df <- co2_county %>%
  filter(county == my_fips)

# Filter for 2010
county_df <- county_df %>%
  filter(year <= 2010)

# Define intervention time point
intervention_date <- as.Date("2005-01-01")
```

```

# Create vector for dates
all_dates <- as.Date(paste0(county_df$year, "-01-01"))

# Create time series for outcome with yearly seasonality
y_ts <- ts(county_df$co2 * 100000, frequency = 1)

# Run CausalArima()
ce <- CausalArima(y = y_ts,
                  dates = all_dates,
                  int.date = intervention_date,
                  nboot = 1000)

# calculate (observed - forecast) / forecast
# (y1 - y0) / y0
# reduced co2 with respect what would have been

# calculate this year by year

# extract forecasted series after running model with 2010
# Create list to store relative effects

# Get impact as list
imp <- impact(ce)

# Extract statistics via impact_norm
norm_effect <- imp$impact_norm$sum
cum_effect <- norm_effect$estimate
avg_effect <- imp$impact_norm$average$estimate
sd_norm <- norm_effect$sd
ci_lower_norm <- cum_effect - 1.96 * sd_norm
ci_upper_norm <- cum_effect + 1.96 * sd_norm

# Extract statistics via impact_boot
boot_list <- imp$impact_boot
boot_effect <- boot_list$effect_cum[3, ]
rel_cum_effect <- imp$impact_boot$effect_cum$estimates[4]
p_value_boot_one_sided <- as.numeric(boot_list$p_values["p"])

?CausalArima

# Store data as tibble
object <- tibble(cum_effect = cum_effect,
                 rel_cum_effect = rel_cum_effect,
                 avg_effect = avg_effect,
                 sd_norm = sd_norm,
                 ci_lower_norm = ci_lower_norm,

```



```

        ci_upper_norm = ci_upper_norm,
        p_value_norm = norm_effect$p_value_bidirectional,
        sd_boot = boot_effect$sd,
        ci_lower_boot = boot_effect$inf,
        ci_upper_boot = boot_effect$sup,
        p_value_boot = min(1, 2 * p_value_boot_one_sided))
object

## # A tibble: 1 x 11
##   cum_effect rel_cum_effect avg_effect sd_norm ci_lower_norm ci_upper_norm
##   <dbl>         <dbl>         <dbl>   <dbl>         <dbl>         <dbl>
## 1   -2.16e12        -0.0103    -3.61e11 1.81e12        -5.71e12        1.39e12
## # i 5 more variables: p_value_norm <dbl>, sd_boot <dbl>, ci_lower_boot <dbl>,
## #   ci_upper_boot <dbl>, p_value_boot <dbl>

ce$dates

## [1] "1980-01-01" "1981-01-01" "1982-01-01" "1983-01-01" "1984-01-01"
## [6] "1985-01-01" "1986-01-01" "1987-01-01" "1988-01-01" "1989-01-01"
## [11] "1990-01-01" "1991-01-01" "1992-01-01" "1993-01-01" "1994-01-01"
## [16] "1995-01-01" "1996-01-01" "1997-01-01" "1998-01-01" "1999-01-01"
## [21] "2000-01-01" "2001-01-01" "2002-01-01" "2003-01-01" "2004-01-01"
## [26] "2005-01-01" "2006-01-01" "2007-01-01" "2008-01-01" "2009-01-01"
## [31] "2010-01-01"

```

## County-level analysis across counties

Run CausalArima on every available county to estimate statistics

```

# Initialize list to store results
results_list <- list()

# Define intervention time point
intervention_date <- as.Date("2005-01-01")

# Get unique counties (to avoid repeating counties over years)
fips_vec <- unique(df$county)

# Loop through each county
for (i in seq_along(fips_vec)) {
  my_fips <- fips_vec[i]

  # Get county_df for specified FIPS
  county_df <- co2_county %>%

```

```

filter(county == my_fips)

# Skip if no data
if (nrow(county_df) == 0) {
  warning(glue("No data for FIPS {my_fips}, skipping..."))
  next
}

# Create vector for dates
all_dates <- as.Date(paste0(county_df$year, "-01-01"))

# Create time series for outcome with yearly seasonality
y_ts <- ts(county_df$co2, frequency = 1)

# Use tryCatch to avoid full script crashing
tryCatch({
  # Run CausalArima()
  ce <- CausalArima(y = y_ts,
                    dates = all_dates,
                    int.date = intervention_date,
                    nboot = 1000)

  # Get impact as list
  imp <- impact(ce)

  # Extract statistics via impact_norm
  norm_effect <- imp$impact_norm$sum
  cum_effect <- norm_effect$estimate
  avg_effect <- imp$impact_norm$average$estimate
  sd_norm <- norm_effect$sd
  ci_lower_norm <- cum_effect - 1.96 * sd_norm
  ci_upper_norm <- cum_effect + 1.96 * sd_norm

  # Extract statistics via impact_boot
  boot_list <- imp$impact_boot
  boot_effect <- boot_list$effect_cum[3, ]
  rel_cum_effect <- imp$impact_boot$effect_cum$estimates[4]
  p_value_boot_one_sided <- as.numeric(boot_list$p_values["p"])

  # Store data as tibble
  result_row <- tibble(fips = my_fips,
                      avg_effect = avg_effect,
                      cum_effect = cum_effect,
                      rel_cum_effect = rel_cum_effect,
                      sd_norm = sd_norm,
                      ci_lower_norm = ci_lower_norm,
                      ci_upper_norm = ci_upper_norm,

```

```

        p_value_norm = norm_effect$p_value_bidirectional,
        sd_boot = boot_effect$sd,
        ci_lower_boot = boot_effect$inf,
        ci_upper_boot = boot_effect$sup,
        p_value_boot = min(1, 2 * p_value_boot_one_sided))

# Add to current results
results_list[[length(results_list) + 1]] <- result_row

# Print progress
message(glue("{i}/{length(fips_vec)}: FIPS {my_fips} processed."))

# Save checkpoint every 50 iterations
if (i %% 50 == 0 || i == length(fips_vec)) {
  checkpoint_df <- bind_rows(results_list)
  write_csv(checkpoint_df, "checkpoint_results.csv")
  message(glue("Checkpoint saved at iteration {i}"))
}

}, error = function(e) {
  warning(glue("Error for FIPS {my_fips}: {e$message}"))
})
}

```

```

## 1/3107: FIPS 01001 processed.

## 2/3107: FIPS 01003 processed.

## 3/3107: FIPS 01005 processed.

## 4/3107: FIPS 01007 processed.

## 5/3107: FIPS 01009 processed.

## 6/3107: FIPS 01011 processed.

## 7/3107: FIPS 01013 processed.

## 8/3107: FIPS 01015 processed.

## 9/3107: FIPS 01017 processed.

## 10/3107: FIPS 01019 processed.

## 11/3107: FIPS 01021 processed.

```

## 12/3107: FIPS 01023 processed.  
## 13/3107: FIPS 01025 processed.  
## 14/3107: FIPS 01027 processed.  
## 15/3107: FIPS 01029 processed.  
## 16/3107: FIPS 01031 processed.  
## 17/3107: FIPS 01033 processed.  
## 18/3107: FIPS 01035 processed.  
## 19/3107: FIPS 01037 processed.  
## 20/3107: FIPS 01039 processed.  
## 21/3107: FIPS 01041 processed.  
## 22/3107: FIPS 01043 processed.  
## 23/3107: FIPS 01045 processed.  
## 24/3107: FIPS 01047 processed.  
## 25/3107: FIPS 01049 processed.  
## 26/3107: FIPS 01051 processed.  
## 27/3107: FIPS 01053 processed.  
## 28/3107: FIPS 01055 processed.  
## 29/3107: FIPS 01057 processed.  
## 30/3107: FIPS 01059 processed.  
## 31/3107: FIPS 01061 processed.  
## 32/3107: FIPS 01063 processed.  
## 33/3107: FIPS 01065 processed.

## 34/3107: FIPS 01067 processed.

## 35/3107: FIPS 01069 processed.

## 36/3107: FIPS 01071 processed.

## 37/3107: FIPS 01073 processed.

## 38/3107: FIPS 01075 processed.

## 39/3107: FIPS 01077 processed.

## 40/3107: FIPS 01079 processed.

## 41/3107: FIPS 01081 processed.

## 42/3107: FIPS 01083 processed.

## 43/3107: FIPS 01085 processed.

## 44/3107: FIPS 01087 processed.

## 45/3107: FIPS 01089 processed.

## 46/3107: FIPS 01091 processed.

## 47/3107: FIPS 01093 processed.

## 48/3107: FIPS 01095 processed.

## 49/3107: FIPS 01097 processed.

## 50/3107: FIPS 01099 processed.

## Checkpoint saved at iteration 50

## 51/3107: FIPS 01101 processed.

## 52/3107: FIPS 01103 processed.

## 53/3107: FIPS 01105 processed.

## 54/3107: FIPS 01107 processed.

## 55/3107: FIPS 01109 processed.  
## 56/3107: FIPS 01111 processed.  
## 57/3107: FIPS 01113 processed.  
## 58/3107: FIPS 01115 processed.  
## 59/3107: FIPS 01117 processed.  
## 60/3107: FIPS 01119 processed.  
## 61/3107: FIPS 01121 processed.  
## 62/3107: FIPS 01123 processed.  
## 63/3107: FIPS 01125 processed.  
## 64/3107: FIPS 01127 processed.  
## 65/3107: FIPS 01129 processed.  
## 66/3107: FIPS 01131 processed.  
## 67/3107: FIPS 01133 processed.  
## 68/3107: FIPS 04001 processed.  
## 69/3107: FIPS 04003 processed.  
## 70/3107: FIPS 04005 processed.  
## 71/3107: FIPS 04007 processed.  
## 72/3107: FIPS 04009 processed.  
## 73/3107: FIPS 04011 processed.  
## 74/3107: FIPS 04012 processed.  
## 75/3107: FIPS 04013 processed.  
## 76/3107: FIPS 04015 processed.

## 77/3107: FIPS 04017 processed.  
## 78/3107: FIPS 04019 processed.  
## 79/3107: FIPS 04021 processed.  
## 80/3107: FIPS 04023 processed.  
## 81/3107: FIPS 04025 processed.  
## 82/3107: FIPS 04027 processed.  
## 83/3107: FIPS 05001 processed.  
## 84/3107: FIPS 05003 processed.  
## 85/3107: FIPS 05005 processed.  
## 86/3107: FIPS 05007 processed.  
## 87/3107: FIPS 05009 processed.  
## 88/3107: FIPS 05011 processed.  
## 89/3107: FIPS 05013 processed.  
## 90/3107: FIPS 05015 processed.  
## 91/3107: FIPS 05017 processed.  
## 92/3107: FIPS 05019 processed.  
## 93/3107: FIPS 05021 processed.  
## 94/3107: FIPS 05023 processed.  
## 95/3107: FIPS 05025 processed.  
## 96/3107: FIPS 05027 processed.  
## 97/3107: FIPS 05029 processed.  
## 98/3107: FIPS 05031 processed.

## 99/3107: FIPS 05033 processed.

## 100/3107: FIPS 05035 processed.

## Checkpoint saved at iteration 100

## 101/3107: FIPS 05037 processed.

## 102/3107: FIPS 05039 processed.

## 103/3107: FIPS 05041 processed.

## 104/3107: FIPS 05043 processed.

## 105/3107: FIPS 05045 processed.

## 106/3107: FIPS 05047 processed.

## 107/3107: FIPS 05049 processed.

## 108/3107: FIPS 05051 processed.

## 109/3107: FIPS 05053 processed.

## 110/3107: FIPS 05055 processed.

## 111/3107: FIPS 05057 processed.

## 112/3107: FIPS 05059 processed.

## 113/3107: FIPS 05061 processed.

## 114/3107: FIPS 05063 processed.

## 115/3107: FIPS 05065 processed.

## 116/3107: FIPS 05067 processed.

## 117/3107: FIPS 05069 processed.

## 118/3107: FIPS 05071 processed.

## 119/3107: FIPS 05073 processed.



## 120/3107: FIPS 05075 processed.  
## 121/3107: FIPS 05077 processed.  
## 122/3107: FIPS 05079 processed.  
## 123/3107: FIPS 05081 processed.  
## 124/3107: FIPS 05083 processed.  
## 125/3107: FIPS 05085 processed.  
## 126/3107: FIPS 05087 processed.  
## 127/3107: FIPS 05089 processed.  
## 128/3107: FIPS 05091 processed.  
## 129/3107: FIPS 05093 processed.  
## 130/3107: FIPS 05095 processed.  
## 131/3107: FIPS 05097 processed.  
## 132/3107: FIPS 05099 processed.  
## 133/3107: FIPS 05101 processed.  
## 134/3107: FIPS 05103 processed.  
## 135/3107: FIPS 05105 processed.  
## 136/3107: FIPS 05107 processed.  
## 137/3107: FIPS 05109 processed.  
## 138/3107: FIPS 05111 processed.  
## 139/3107: FIPS 05113 processed.  
## 140/3107: FIPS 05115 processed.  
## 141/3107: FIPS 05117 processed.

## 142/3107: FIPS 05119 processed.  
## 143/3107: FIPS 05121 processed.  
## 144/3107: FIPS 05123 processed.  
## 145/3107: FIPS 05125 processed.  
## 146/3107: FIPS 05127 processed.  
## 147/3107: FIPS 05129 processed.  
## 148/3107: FIPS 05131 processed.  
## 149/3107: FIPS 05133 processed.  
## 150/3107: FIPS 05135 processed.  
## Checkpoint saved at iteration 150  
## 151/3107: FIPS 05137 processed.  
## 152/3107: FIPS 05139 processed.  
## 153/3107: FIPS 05141 processed.  
## 154/3107: FIPS 05143 processed.  
## 155/3107: FIPS 05145 processed.  
## 156/3107: FIPS 05147 processed.  
## 157/3107: FIPS 05149 processed.  
## 158/3107: FIPS 06001 processed.  
## 159/3107: FIPS 06003 processed.  
## 160/3107: FIPS 06005 processed.  
## 161/3107: FIPS 06007 processed.  
## 162/3107: FIPS 06009 processed.

## 163/3107: FIPS 06011 processed.  
## 164/3107: FIPS 06013 processed.  
## 165/3107: FIPS 06015 processed.  
## 166/3107: FIPS 06017 processed.  
## 167/3107: FIPS 06019 processed.  
## 168/3107: FIPS 06021 processed.  
## 169/3107: FIPS 06023 processed.  
## 170/3107: FIPS 06025 processed.  
## 171/3107: FIPS 06027 processed.  
## 172/3107: FIPS 06029 processed.  
## 173/3107: FIPS 06031 processed.  
## 174/3107: FIPS 06033 processed.  
## 175/3107: FIPS 06035 processed.  
## 176/3107: FIPS 06037 processed.  
## 177/3107: FIPS 06039 processed.  
## 178/3107: FIPS 06041 processed.  
## 179/3107: FIPS 06043 processed.  
## 180/3107: FIPS 06045 processed.  
## 181/3107: FIPS 06047 processed.  
## 182/3107: FIPS 06049 processed.  
## 183/3107: FIPS 06051 processed.  
## 184/3107: FIPS 06053 processed.

## 185/3107: FIPS 06055 processed.

## 186/3107: FIPS 06057 processed.

## 187/3107: FIPS 06059 processed.

## 188/3107: FIPS 06061 processed.

## 189/3107: FIPS 06063 processed.

## 190/3107: FIPS 06065 processed.

## 191/3107: FIPS 06067 processed.

## 192/3107: FIPS 06069 processed.

## 193/3107: FIPS 06071 processed.

## 194/3107: FIPS 06073 processed.

## 195/3107: FIPS 06075 processed.

## 196/3107: FIPS 06077 processed.

## 197/3107: FIPS 06079 processed.

## 198/3107: FIPS 06081 processed.

## 199/3107: FIPS 06083 processed.

## 200/3107: FIPS 06085 processed.

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## 201/3107: FIPS 06087 processed.

## 202/3107: FIPS 06089 processed.

## 203/3107: FIPS 06091 processed.

## 204/3107: FIPS 06093 processed.

## 205/3107: FIPS 06095 processed.

## 206/3107: FIPS 06097 processed.  
## 207/3107: FIPS 06099 processed.  
## 208/3107: FIPS 06101 processed.  
## 209/3107: FIPS 06103 processed.  
## 210/3107: FIPS 06105 processed.  
## 211/3107: FIPS 06107 processed.  
## 212/3107: FIPS 06109 processed.  
## 213/3107: FIPS 06111 processed.  
## 214/3107: FIPS 06113 processed.  
## 215/3107: FIPS 06115 processed.  
## 216/3107: FIPS 08001 processed.  
## 217/3107: FIPS 08003 processed.  
## 218/3107: FIPS 08005 processed.  
## 219/3107: FIPS 08007 processed.  
## 220/3107: FIPS 08009 processed.  
## 221/3107: FIPS 08011 processed.  
## 222/3107: FIPS 08013 processed.  
## 223/3107: FIPS 08014 processed.  
## 224/3107: FIPS 08015 processed.  
## 225/3107: FIPS 08017 processed.  
## 226/3107: FIPS 08019 processed.  
## 227/3107: FIPS 08021 processed.

## 228/3107: FIPS 08023 processed.  
## 229/3107: FIPS 08025 processed.  
## 230/3107: FIPS 08027 processed.  
## 231/3107: FIPS 08029 processed.  
## 232/3107: FIPS 08031 processed.  
## 233/3107: FIPS 08033 processed.  
## 234/3107: FIPS 08035 processed.  
## 235/3107: FIPS 08037 processed.  
## 236/3107: FIPS 08039 processed.  
## 237/3107: FIPS 08041 processed.  
## 238/3107: FIPS 08043 processed.  
## 239/3107: FIPS 08045 processed.  
## 240/3107: FIPS 08047 processed.  
## 241/3107: FIPS 08049 processed.  
## 242/3107: FIPS 08051 processed.  
## 243/3107: FIPS 08053 processed.  
## 244/3107: FIPS 08055 processed.  
## 245/3107: FIPS 08057 processed.  
## 246/3107: FIPS 08059 processed.  
## 247/3107: FIPS 08061 processed.  
## 248/3107: FIPS 08063 processed.  
## 249/3107: FIPS 08065 processed.

## 250/3107: FIPS 08067 processed.

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## 251/3107: FIPS 08069 processed.

## 252/3107: FIPS 08071 processed.

## 253/3107: FIPS 08073 processed.

## 254/3107: FIPS 08075 processed.

## 255/3107: FIPS 08077 processed.

## 256/3107: FIPS 08079 processed.

## 257/3107: FIPS 08081 processed.

## 258/3107: FIPS 08083 processed.

## 259/3107: FIPS 08085 processed.

## 260/3107: FIPS 08087 processed.

## 261/3107: FIPS 08089 processed.

## 262/3107: FIPS 08091 processed.

## 263/3107: FIPS 08093 processed.

## 264/3107: FIPS 08095 processed.

## 265/3107: FIPS 08097 processed.

## 266/3107: FIPS 08099 processed.

## 267/3107: FIPS 08101 processed.

## 268/3107: FIPS 08103 processed.

## 269/3107: FIPS 08105 processed.

## 270/3107: FIPS 08107 processed.

## 271/3107: FIPS 08109 processed.  
## 272/3107: FIPS 08111 processed.  
## 273/3107: FIPS 08113 processed.  
## 274/3107: FIPS 08115 processed.  
## 275/3107: FIPS 08117 processed.  
## 276/3107: FIPS 08119 processed.  
## 277/3107: FIPS 08121 processed.  
## 278/3107: FIPS 08123 processed.  
## 279/3107: FIPS 08125 processed.  
## 280/3107: FIPS 09001 processed.  
## 281/3107: FIPS 09003 processed.  
## 282/3107: FIPS 09005 processed.  
## 283/3107: FIPS 09007 processed.  
## 284/3107: FIPS 09009 processed.  
## 285/3107: FIPS 09011 processed.  
## 286/3107: FIPS 09013 processed.  
## 287/3107: FIPS 09015 processed.  
## 288/3107: FIPS 10001 processed.  
## 289/3107: FIPS 10003 processed.  
## 290/3107: FIPS 10005 processed.  
## 291/3107: FIPS 11001 processed.  
## 292/3107: FIPS 12001 processed.



## 293/3107: FIPS 12003 processed.

## 294/3107: FIPS 12005 processed.

## 295/3107: FIPS 12007 processed.

## 296/3107: FIPS 12009 processed.

## 297/3107: FIPS 12011 processed.

## 298/3107: FIPS 12013 processed.

## 299/3107: FIPS 12015 processed.

## 300/3107: FIPS 12017 processed.

## Checkpoint saved at iteration 300

## 301/3107: FIPS 12019 processed.

## 302/3107: FIPS 12021 processed.

## 303/3107: FIPS 12023 processed.

## 304/3107: FIPS 12027 processed.

## 305/3107: FIPS 12029 processed.

## 306/3107: FIPS 12031 processed.

## 307/3107: FIPS 12033 processed.

## 308/3107: FIPS 12035 processed.

## 309/3107: FIPS 12037 processed.

## 310/3107: FIPS 12039 processed.

## 311/3107: FIPS 12041 processed.

## 312/3107: FIPS 12043 processed.

## 313/3107: FIPS 12045 processed.

## 314/3107: FIPS 12047 processed.  
## 315/3107: FIPS 12049 processed.  
## 316/3107: FIPS 12051 processed.  
## 317/3107: FIPS 12053 processed.  
## 318/3107: FIPS 12055 processed.  
## 319/3107: FIPS 12057 processed.  
## 320/3107: FIPS 12059 processed.  
## 321/3107: FIPS 12061 processed.  
## 322/3107: FIPS 12063 processed.  
## 323/3107: FIPS 12065 processed.  
## 324/3107: FIPS 12067 processed.  
## 325/3107: FIPS 12069 processed.  
## 326/3107: FIPS 12071 processed.  
## 327/3107: FIPS 12073 processed.  
## 328/3107: FIPS 12075 processed.  
## 329/3107: FIPS 12077 processed.  
## 330/3107: FIPS 12079 processed.  
## 331/3107: FIPS 12081 processed.  
## 332/3107: FIPS 12083 processed.  
## 333/3107: FIPS 12085 processed.  
## 334/3107: FIPS 12086 processed.  
## 335/3107: FIPS 12087 processed.

## 336/3107: FIPS 12089 processed.

## 337/3107: FIPS 12091 processed.

## 338/3107: FIPS 12093 processed.

## 339/3107: FIPS 12095 processed.

## 340/3107: FIPS 12097 processed.

## 341/3107: FIPS 12099 processed.

## 342/3107: FIPS 12101 processed.

## 343/3107: FIPS 12103 processed.

## 344/3107: FIPS 12105 processed.

## 345/3107: FIPS 12107 processed.

## 346/3107: FIPS 12109 processed.

## 347/3107: FIPS 12111 processed.

## 348/3107: FIPS 12113 processed.

## 349/3107: FIPS 12115 processed.

## 350/3107: FIPS 12117 processed.

## Checkpoint saved at iteration 350

## 351/3107: FIPS 12119 processed.

## 352/3107: FIPS 12121 processed.

## 353/3107: FIPS 12123 processed.

## 354/3107: FIPS 12125 processed.

## 355/3107: FIPS 12127 processed.

## 356/3107: FIPS 12129 processed.

## 357/3107: FIPS 12131 processed.  
## 358/3107: FIPS 12133 processed.  
## 359/3107: FIPS 13001 processed.  
## 360/3107: FIPS 13003 processed.  
## 361/3107: FIPS 13005 processed.  
## 362/3107: FIPS 13007 processed.  
## 363/3107: FIPS 13009 processed.  
## 364/3107: FIPS 13011 processed.  
## 365/3107: FIPS 13013 processed.  
## 366/3107: FIPS 13015 processed.  
## 367/3107: FIPS 13017 processed.  
## 368/3107: FIPS 13019 processed.  
## 369/3107: FIPS 13021 processed.  
## 370/3107: FIPS 13023 processed.  
## 371/3107: FIPS 13025 processed.  
## 372/3107: FIPS 13027 processed.  
## 373/3107: FIPS 13029 processed.  
## 374/3107: FIPS 13031 processed.  
## 375/3107: FIPS 13033 processed.  
## 376/3107: FIPS 13035 processed.  
## 377/3107: FIPS 13037 processed.  
## 378/3107: FIPS 13039 processed.

## 379/3107: FIPS 13043 processed.  
## 380/3107: FIPS 13045 processed.  
## 381/3107: FIPS 13047 processed.  
## 382/3107: FIPS 13049 processed.  
## 383/3107: FIPS 13051 processed.  
## 384/3107: FIPS 13053 processed.  
## 385/3107: FIPS 13055 processed.  
## 386/3107: FIPS 13057 processed.  
## 387/3107: FIPS 13059 processed.  
## 388/3107: FIPS 13061 processed.  
## 389/3107: FIPS 13063 processed.  
## 390/3107: FIPS 13065 processed.  
## 391/3107: FIPS 13067 processed.  
## 392/3107: FIPS 13069 processed.  
## 393/3107: FIPS 13071 processed.  
## 394/3107: FIPS 13073 processed.  
## 395/3107: FIPS 13075 processed.  
## 396/3107: FIPS 13077 processed.  
## 397/3107: FIPS 13079 processed.  
## 398/3107: FIPS 13081 processed.  
## 399/3107: FIPS 13083 processed.  
## 400/3107: FIPS 13085 processed.

## Checkpoint saved at iteration 400

## 401/3107: FIPS 13087 processed.

## 402/3107: FIPS 13089 processed.

## 403/3107: FIPS 13091 processed.

## 404/3107: FIPS 13093 processed.

## 405/3107: FIPS 13095 processed.

## 406/3107: FIPS 13097 processed.

## 407/3107: FIPS 13099 processed.

## 408/3107: FIPS 13101 processed.

## 409/3107: FIPS 13103 processed.

## 410/3107: FIPS 13105 processed.

## 411/3107: FIPS 13107 processed.

## 412/3107: FIPS 13109 processed.

## 413/3107: FIPS 13111 processed.

## 414/3107: FIPS 13113 processed.

## 415/3107: FIPS 13115 processed.

## 416/3107: FIPS 13117 processed.

## 417/3107: FIPS 13119 processed.

## 418/3107: FIPS 13121 processed.

## 419/3107: FIPS 13123 processed.

## 420/3107: FIPS 13125 processed.

## 421/3107: FIPS 13127 processed.

## 422/3107: FIPS 13129 processed.  
## 423/3107: FIPS 13131 processed.  
## 424/3107: FIPS 13133 processed.  
## 425/3107: FIPS 13135 processed.  
## 426/3107: FIPS 13137 processed.  
## 427/3107: FIPS 13139 processed.  
## 428/3107: FIPS 13141 processed.  
## 429/3107: FIPS 13143 processed.  
## 430/3107: FIPS 13145 processed.  
## 431/3107: FIPS 13147 processed.  
## 432/3107: FIPS 13149 processed.  
## 433/3107: FIPS 13151 processed.  
## 434/3107: FIPS 13153 processed.  
## 435/3107: FIPS 13155 processed.  
## 436/3107: FIPS 13157 processed.  
## 437/3107: FIPS 13159 processed.  
## 438/3107: FIPS 13161 processed.  
## 439/3107: FIPS 13163 processed.  
## 440/3107: FIPS 13165 processed.  
## 441/3107: FIPS 13167 processed.  
## 442/3107: FIPS 13169 processed.  
## 443/3107: FIPS 13171 processed.

## 444/3107: FIPS 13173 processed.  
## 445/3107: FIPS 13175 processed.  
## 446/3107: FIPS 13177 processed.  
## 447/3107: FIPS 13179 processed.  
## 448/3107: FIPS 13181 processed.  
## 449/3107: FIPS 13183 processed.  
## 450/3107: FIPS 13185 processed.  
## Checkpoint saved at iteration 450  
## 451/3107: FIPS 13187 processed.  
## 452/3107: FIPS 13189 processed.  
## 453/3107: FIPS 13191 processed.  
## 454/3107: FIPS 13193 processed.  
## 455/3107: FIPS 13195 processed.  
## 456/3107: FIPS 13197 processed.  
## 457/3107: FIPS 13199 processed.  
## 458/3107: FIPS 13201 processed.  
## 459/3107: FIPS 13205 processed.  
## 460/3107: FIPS 13207 processed.  
## 461/3107: FIPS 13209 processed.  
## 462/3107: FIPS 13211 processed.  
## 463/3107: FIPS 13213 processed.  
## 464/3107: FIPS 13215 processed.



## 465/3107: FIPS 13217 processed.  
## 466/3107: FIPS 13219 processed.  
## 467/3107: FIPS 13221 processed.  
## 468/3107: FIPS 13223 processed.  
## 469/3107: FIPS 13225 processed.  
## 470/3107: FIPS 13227 processed.  
## 471/3107: FIPS 13229 processed.  
## 472/3107: FIPS 13231 processed.  
## 473/3107: FIPS 13233 processed.  
## 474/3107: FIPS 13235 processed.  
## 475/3107: FIPS 13237 processed.  
## 476/3107: FIPS 13239 processed.  
## 477/3107: FIPS 13241 processed.  
## 478/3107: FIPS 13243 processed.  
## 479/3107: FIPS 13245 processed.  
## 480/3107: FIPS 13247 processed.  
## 481/3107: FIPS 13249 processed.  
## 482/3107: FIPS 13251 processed.  
## 483/3107: FIPS 13253 processed.  
## 484/3107: FIPS 13255 processed.  
## 485/3107: FIPS 13257 processed.  
## 486/3107: FIPS 13259 processed.

## 487/3107: FIPS 13261 processed.  
## 488/3107: FIPS 13263 processed.  
## 489/3107: FIPS 13265 processed.  
## 490/3107: FIPS 13267 processed.  
## 491/3107: FIPS 13269 processed.  
## 492/3107: FIPS 13271 processed.  
## 493/3107: FIPS 13273 processed.  
## 494/3107: FIPS 13275 processed.  
## 495/3107: FIPS 13277 processed.  
## 496/3107: FIPS 13279 processed.  
## 497/3107: FIPS 13281 processed.  
## 498/3107: FIPS 13283 processed.  
## 499/3107: FIPS 13285 processed.  
## 500/3107: FIPS 13287 processed.  
## Checkpoint saved at iteration 500  
## 501/3107: FIPS 13289 processed.  
## 502/3107: FIPS 13291 processed.  
## 503/3107: FIPS 13293 processed.  
## 504/3107: FIPS 13295 processed.  
## 505/3107: FIPS 13297 processed.  
## 506/3107: FIPS 13299 processed.  
## 507/3107: FIPS 13301 processed.

## 508/3107: FIPS 13303 processed.  
## 509/3107: FIPS 13305 processed.  
## 510/3107: FIPS 13307 processed.  
## 511/3107: FIPS 13309 processed.  
## 512/3107: FIPS 13311 processed.  
## 513/3107: FIPS 13313 processed.  
## 514/3107: FIPS 13315 processed.  
## 515/3107: FIPS 13317 processed.  
## 516/3107: FIPS 13319 processed.  
## 517/3107: FIPS 13321 processed.  
## 518/3107: FIPS 16001 processed.  
## 519/3107: FIPS 16003 processed.  
## 520/3107: FIPS 16005 processed.  
## 521/3107: FIPS 16007 processed.  
## 522/3107: FIPS 16009 processed.  
## 523/3107: FIPS 16011 processed.  
## 524/3107: FIPS 16013 processed.  
## 525/3107: FIPS 16015 processed.  
## 526/3107: FIPS 16017 processed.  
## 527/3107: FIPS 16019 processed.  
## 528/3107: FIPS 16021 processed.  
## 529/3107: FIPS 16023 processed.

## 530/3107: FIPS 16025 processed.

## 531/3107: FIPS 16027 processed.

## 532/3107: FIPS 16029 processed.

## 533/3107: FIPS 16031 processed.

## 534/3107: FIPS 16033 processed.

## 535/3107: FIPS 16035 processed.

## 536/3107: FIPS 16037 processed.

## 537/3107: FIPS 16039 processed.

## 538/3107: FIPS 16041 processed.

## 539/3107: FIPS 16043 processed.

## 540/3107: FIPS 16045 processed.

## 541/3107: FIPS 16047 processed.

## 542/3107: FIPS 16049 processed.

## 543/3107: FIPS 16051 processed.

## 544/3107: FIPS 16053 processed.

## 545/3107: FIPS 16055 processed.

## 546/3107: FIPS 16057 processed.

## 547/3107: FIPS 16059 processed.

## 548/3107: FIPS 16061 processed.

## 549/3107: FIPS 16063 processed.

## 550/3107: FIPS 16065 processed.

## Checkpoint saved at iteration 550

## 551/3107: FIPS 16067 processed.  
## 552/3107: FIPS 16069 processed.  
## 553/3107: FIPS 16071 processed.  
## 554/3107: FIPS 16073 processed.  
## 555/3107: FIPS 16075 processed.  
## 556/3107: FIPS 16077 processed.  
## 557/3107: FIPS 16079 processed.  
## 558/3107: FIPS 16081 processed.  
## 559/3107: FIPS 16083 processed.  
## 560/3107: FIPS 16085 processed.  
## 561/3107: FIPS 16087 processed.  
## 562/3107: FIPS 17001 processed.  
## 563/3107: FIPS 17003 processed.  
## 564/3107: FIPS 17005 processed.  
## 565/3107: FIPS 17007 processed.  
## 566/3107: FIPS 17009 processed.  
## 567/3107: FIPS 17011 processed.  
## 568/3107: FIPS 17013 processed.  
## 569/3107: FIPS 17015 processed.  
## 570/3107: FIPS 17017 processed.  
## 571/3107: FIPS 17019 processed.  
## 572/3107: FIPS 17021 processed.

## 573/3107: FIPS 17023 processed.  
## 574/3107: FIPS 17025 processed.  
## 575/3107: FIPS 17027 processed.  
## 576/3107: FIPS 17029 processed.  
## 577/3107: FIPS 17031 processed.  
## 578/3107: FIPS 17033 processed.  
## 579/3107: FIPS 17035 processed.  
## 580/3107: FIPS 17037 processed.  
## 581/3107: FIPS 17039 processed.  
## 582/3107: FIPS 17041 processed.  
## 583/3107: FIPS 17043 processed.  
## 584/3107: FIPS 17045 processed.  
## 585/3107: FIPS 17047 processed.  
## 586/3107: FIPS 17049 processed.  
## 587/3107: FIPS 17051 processed.  
## 588/3107: FIPS 17053 processed.  
## 589/3107: FIPS 17055 processed.  
## 590/3107: FIPS 17057 processed.  
## 591/3107: FIPS 17059 processed.  
## 592/3107: FIPS 17061 processed.  
## 593/3107: FIPS 17063 processed.  
## 594/3107: FIPS 17065 processed.

## 595/3107: FIPS 17067 processed.  
## 596/3107: FIPS 17069 processed.  
## 597/3107: FIPS 17071 processed.  
## 598/3107: FIPS 17073 processed.  
## 599/3107: FIPS 17075 processed.  
## 600/3107: FIPS 17077 processed.  
## Checkpoint saved at iteration 600  
## 601/3107: FIPS 17079 processed.  
## 602/3107: FIPS 17081 processed.  
## 603/3107: FIPS 17083 processed.  
## 604/3107: FIPS 17085 processed.  
## 605/3107: FIPS 17087 processed.  
## 606/3107: FIPS 17089 processed.  
## 607/3107: FIPS 17091 processed.  
## 608/3107: FIPS 17093 processed.  
## 609/3107: FIPS 17095 processed.  
## 610/3107: FIPS 17097 processed.  
## 611/3107: FIPS 17099 processed.  
## 612/3107: FIPS 17101 processed.  
## 613/3107: FIPS 17103 processed.  
## 614/3107: FIPS 17105 processed.  
## 615/3107: FIPS 17107 processed.

## 616/3107: FIPS 17109 processed.  
## 617/3107: FIPS 17111 processed.  
## 618/3107: FIPS 17113 processed.  
## 619/3107: FIPS 17115 processed.  
## 620/3107: FIPS 17117 processed.  
## 621/3107: FIPS 17119 processed.  
## 622/3107: FIPS 17121 processed.  
## 623/3107: FIPS 17123 processed.  
## 624/3107: FIPS 17125 processed.  
## 625/3107: FIPS 17127 processed.  
## 626/3107: FIPS 17129 processed.  
## 627/3107: FIPS 17131 processed.  
## 628/3107: FIPS 17133 processed.  
## 629/3107: FIPS 17135 processed.  
## 630/3107: FIPS 17137 processed.  
## 631/3107: FIPS 17139 processed.  
## 632/3107: FIPS 17141 processed.  
## 633/3107: FIPS 17143 processed.  
## 634/3107: FIPS 17145 processed.  
## 635/3107: FIPS 17147 processed.  
## 636/3107: FIPS 17149 processed.  
## 637/3107: FIPS 17151 processed.



## 638/3107: FIPS 17153 processed.

## 639/3107: FIPS 17155 processed.

## 640/3107: FIPS 17157 processed.

## 641/3107: FIPS 17159 processed.

## 642/3107: FIPS 17161 processed.

## 643/3107: FIPS 17163 processed.

## 644/3107: FIPS 17165 processed.

## 645/3107: FIPS 17167 processed.

## 646/3107: FIPS 17169 processed.

## 647/3107: FIPS 17171 processed.

## 648/3107: FIPS 17173 processed.

## 649/3107: FIPS 17175 processed.

## 650/3107: FIPS 17177 processed.

## Checkpoint saved at iteration 650

## 651/3107: FIPS 17179 processed.

## 652/3107: FIPS 17181 processed.

## 653/3107: FIPS 17183 processed.

## 654/3107: FIPS 17185 processed.

## 655/3107: FIPS 17187 processed.

## 656/3107: FIPS 17189 processed.

## 657/3107: FIPS 17191 processed.

## 658/3107: FIPS 17193 processed.

## 659/3107: FIPS 17195 processed.  
## 660/3107: FIPS 17197 processed.  
## 661/3107: FIPS 17199 processed.  
## 662/3107: FIPS 17201 processed.  
## 663/3107: FIPS 17203 processed.  
## 664/3107: FIPS 18001 processed.  
## 665/3107: FIPS 18003 processed.  
## 666/3107: FIPS 18005 processed.  
## 667/3107: FIPS 18007 processed.  
## 668/3107: FIPS 18009 processed.  
## 669/3107: FIPS 18011 processed.  
## 670/3107: FIPS 18013 processed.  
## 671/3107: FIPS 18015 processed.  
## 672/3107: FIPS 18017 processed.  
## 673/3107: FIPS 18019 processed.  
## 674/3107: FIPS 18021 processed.  
## 675/3107: FIPS 18023 processed.  
## 676/3107: FIPS 18025 processed.  
## 677/3107: FIPS 18027 processed.  
## 678/3107: FIPS 18029 processed.  
## 679/3107: FIPS 18031 processed.  
## 680/3107: FIPS 18033 processed.

## 681/3107: FIPS 18035 processed.  
## 682/3107: FIPS 18037 processed.  
## 683/3107: FIPS 18039 processed.  
## 684/3107: FIPS 18041 processed.  
## 685/3107: FIPS 18043 processed.  
## 686/3107: FIPS 18045 processed.  
## 687/3107: FIPS 18047 processed.  
## 688/3107: FIPS 18049 processed.  
## 689/3107: FIPS 18051 processed.  
## 690/3107: FIPS 18053 processed.  
## 691/3107: FIPS 18055 processed.  
## 692/3107: FIPS 18057 processed.  
## 693/3107: FIPS 18059 processed.  
## 694/3107: FIPS 18061 processed.  
## 695/3107: FIPS 18063 processed.  
## 696/3107: FIPS 18065 processed.  
## 697/3107: FIPS 18067 processed.  
## 698/3107: FIPS 18069 processed.  
## 699/3107: FIPS 18071 processed.  
## 700/3107: FIPS 18073 processed.  
## Checkpoint saved at iteration 700  
## 701/3107: FIPS 18075 processed.

## 702/3107: FIPS 18077 processed.  
## 703/3107: FIPS 18079 processed.  
## 704/3107: FIPS 18081 processed.  
## 705/3107: FIPS 18083 processed.  
## 706/3107: FIPS 18085 processed.  
## 707/3107: FIPS 18087 processed.  
## 708/3107: FIPS 18089 processed.  
## 709/3107: FIPS 18091 processed.  
## 710/3107: FIPS 18093 processed.  
## 711/3107: FIPS 18095 processed.  
## 712/3107: FIPS 18097 processed.  
## 713/3107: FIPS 18099 processed.  
## 714/3107: FIPS 18101 processed.  
## 715/3107: FIPS 18103 processed.  
## 716/3107: FIPS 18105 processed.  
## 717/3107: FIPS 18107 processed.  
## 718/3107: FIPS 18109 processed.  
## 719/3107: FIPS 18111 processed.  
## 720/3107: FIPS 18113 processed.  
## 721/3107: FIPS 18115 processed.  
## 722/3107: FIPS 18117 processed.  
## 723/3107: FIPS 18119 processed.

## 724/3107: FIPS 18121 processed.  
## 725/3107: FIPS 18123 processed.  
## 726/3107: FIPS 18125 processed.  
## 727/3107: FIPS 18127 processed.  
## 728/3107: FIPS 18129 processed.  
## 729/3107: FIPS 18131 processed.  
## 730/3107: FIPS 18133 processed.  
## 731/3107: FIPS 18135 processed.  
## 732/3107: FIPS 18137 processed.  
## 733/3107: FIPS 18139 processed.  
## 734/3107: FIPS 18141 processed.  
## 735/3107: FIPS 18143 processed.  
## 736/3107: FIPS 18145 processed.  
## 737/3107: FIPS 18147 processed.  
## 738/3107: FIPS 18149 processed.  
## 739/3107: FIPS 18151 processed.  
## 740/3107: FIPS 18153 processed.  
## 741/3107: FIPS 18155 processed.  
## 742/3107: FIPS 18157 processed.  
## 743/3107: FIPS 18159 processed.  
## 744/3107: FIPS 18161 processed.  
## 745/3107: FIPS 18163 processed.

## 746/3107: FIPS 18165 processed.  
## 747/3107: FIPS 18167 processed.  
## 748/3107: FIPS 18169 processed.  
## 749/3107: FIPS 18171 processed.  
## 750/3107: FIPS 18173 processed.  
## Checkpoint saved at iteration 750  
## 751/3107: FIPS 18175 processed.  
## 752/3107: FIPS 18177 processed.  
## 753/3107: FIPS 18179 processed.  
## 754/3107: FIPS 18181 processed.  
## 755/3107: FIPS 18183 processed.  
## 756/3107: FIPS 19001 processed.  
## 757/3107: FIPS 19003 processed.  
## 758/3107: FIPS 19005 processed.  
## 759/3107: FIPS 19007 processed.  
## 760/3107: FIPS 19009 processed.  
## 761/3107: FIPS 19011 processed.  
## 762/3107: FIPS 19013 processed.  
## 763/3107: FIPS 19015 processed.  
## 764/3107: FIPS 19017 processed.  
## 765/3107: FIPS 19019 processed.  
## 766/3107: FIPS 19021 processed.

## 767/3107: FIPS 19023 processed.  
## 768/3107: FIPS 19025 processed.  
## 769/3107: FIPS 19027 processed.  
## 770/3107: FIPS 19029 processed.  
## 771/3107: FIPS 19031 processed.  
## 772/3107: FIPS 19033 processed.  
## 773/3107: FIPS 19035 processed.  
## 774/3107: FIPS 19037 processed.  
## 775/3107: FIPS 19039 processed.  
## 776/3107: FIPS 19041 processed.  
## 777/3107: FIPS 19043 processed.  
## 778/3107: FIPS 19045 processed.  
## 779/3107: FIPS 19047 processed.  
## 780/3107: FIPS 19049 processed.  
## 781/3107: FIPS 19051 processed.  
## 782/3107: FIPS 19053 processed.  
## 783/3107: FIPS 19055 processed.  
## 784/3107: FIPS 19057 processed.  
## 785/3107: FIPS 19059 processed.  
## 786/3107: FIPS 19061 processed.  
## 787/3107: FIPS 19063 processed.  
## 788/3107: FIPS 19065 processed.

## 789/3107: FIPS 19067 processed.  
## 790/3107: FIPS 19069 processed.  
## 791/3107: FIPS 19071 processed.  
## 792/3107: FIPS 19073 processed.  
## 793/3107: FIPS 19075 processed.  
## 794/3107: FIPS 19077 processed.  
## 795/3107: FIPS 19079 processed.  
## 796/3107: FIPS 19081 processed.  
## 797/3107: FIPS 19083 processed.  
## 798/3107: FIPS 19085 processed.  
## 799/3107: FIPS 19087 processed.  
## 800/3107: FIPS 19089 processed.  
## Checkpoint saved at iteration 800  
## 801/3107: FIPS 19091 processed.  
## 802/3107: FIPS 19093 processed.  
## 803/3107: FIPS 19095 processed.  
## 804/3107: FIPS 19097 processed.  
## 805/3107: FIPS 19099 processed.  
## 806/3107: FIPS 19101 processed.  
## 807/3107: FIPS 19103 processed.  
## 808/3107: FIPS 19105 processed.  
## 809/3107: FIPS 19107 processed.



## 810/3107: FIPS 19109 processed.  
## 811/3107: FIPS 19111 processed.  
## 812/3107: FIPS 19113 processed.  
## 813/3107: FIPS 19115 processed.  
## 814/3107: FIPS 19117 processed.  
## 815/3107: FIPS 19119 processed.  
## 816/3107: FIPS 19121 processed.  
## 817/3107: FIPS 19123 processed.  
## 818/3107: FIPS 19125 processed.  
## 819/3107: FIPS 19127 processed.  
## 820/3107: FIPS 19129 processed.  
## 821/3107: FIPS 19131 processed.  
## 822/3107: FIPS 19133 processed.  
## 823/3107: FIPS 19135 processed.  
## 824/3107: FIPS 19137 processed.  
## 825/3107: FIPS 19139 processed.  
## 826/3107: FIPS 19141 processed.  
## 827/3107: FIPS 19143 processed.  
## 828/3107: FIPS 19145 processed.  
## 829/3107: FIPS 19147 processed.  
## 830/3107: FIPS 19149 processed.  
## 831/3107: FIPS 19151 processed.

## 832/3107: FIPS 19153 processed.  
## 833/3107: FIPS 19155 processed.  
## 834/3107: FIPS 19157 processed.  
## 835/3107: FIPS 19159 processed.  
## 836/3107: FIPS 19161 processed.  
## 837/3107: FIPS 19163 processed.  
## 838/3107: FIPS 19165 processed.  
## 839/3107: FIPS 19167 processed.  
## 840/3107: FIPS 19169 processed.  
## 841/3107: FIPS 19171 processed.  
## 842/3107: FIPS 19173 processed.  
## 843/3107: FIPS 19175 processed.  
## 844/3107: FIPS 19177 processed.  
## 845/3107: FIPS 19179 processed.  
## 846/3107: FIPS 19181 processed.  
## 847/3107: FIPS 19183 processed.  
## 848/3107: FIPS 19185 processed.  
## 849/3107: FIPS 19187 processed.  
## 850/3107: FIPS 19189 processed.  
## Checkpoint saved at iteration 850  
## 851/3107: FIPS 19191 processed.  
## 852/3107: FIPS 19193 processed.

## 853/3107: FIPS 19195 processed.

## 854/3107: FIPS 19197 processed.

## 855/3107: FIPS 20001 processed.

## 856/3107: FIPS 20003 processed.

## 857/3107: FIPS 20005 processed.

## 858/3107: FIPS 20007 processed.

## 859/3107: FIPS 20009 processed.

## 860/3107: FIPS 20011 processed.

## 861/3107: FIPS 20013 processed.

## 862/3107: FIPS 20015 processed.

## 863/3107: FIPS 20017 processed.

## 864/3107: FIPS 20019 processed.

## 865/3107: FIPS 20021 processed.

## 866/3107: FIPS 20023 processed.

## 867/3107: FIPS 20025 processed.

## 868/3107: FIPS 20027 processed.

## 869/3107: FIPS 20029 processed.

## 870/3107: FIPS 20031 processed.

## 871/3107: FIPS 20033 processed.

## 872/3107: FIPS 20035 processed.

## 873/3107: FIPS 20037 processed.

## 874/3107: FIPS 20039 processed.

## 875/3107: FIPS 20041 processed.  
## 876/3107: FIPS 20043 processed.  
## 877/3107: FIPS 20045 processed.  
## 878/3107: FIPS 20047 processed.  
## 879/3107: FIPS 20049 processed.  
## 880/3107: FIPS 20051 processed.  
## 881/3107: FIPS 20053 processed.  
## 882/3107: FIPS 20055 processed.  
## 883/3107: FIPS 20057 processed.  
## 884/3107: FIPS 20059 processed.  
## 885/3107: FIPS 20061 processed.  
## 886/3107: FIPS 20063 processed.  
## 887/3107: FIPS 20065 processed.  
## 888/3107: FIPS 20067 processed.  
## 889/3107: FIPS 20069 processed.  
## 890/3107: FIPS 20071 processed.  
## 891/3107: FIPS 20073 processed.  
## 892/3107: FIPS 20075 processed.  
## 893/3107: FIPS 20077 processed.  
## 894/3107: FIPS 20079 processed.  
## 895/3107: FIPS 20081 processed.  
## 896/3107: FIPS 20083 processed.

## 897/3107: FIPS 20085 processed.

## 898/3107: FIPS 20087 processed.

## 899/3107: FIPS 20089 processed.

## 900/3107: FIPS 20091 processed.

## Checkpoint saved at iteration 900

## 901/3107: FIPS 20093 processed.

## 902/3107: FIPS 20095 processed.

## 903/3107: FIPS 20097 processed.

## 904/3107: FIPS 20099 processed.

## 905/3107: FIPS 20101 processed.

## 906/3107: FIPS 20103 processed.

## 907/3107: FIPS 20105 processed.

## 908/3107: FIPS 20107 processed.

## 909/3107: FIPS 20109 processed.

## 910/3107: FIPS 20111 processed.

## 911/3107: FIPS 20113 processed.

## 912/3107: FIPS 20115 processed.

## 913/3107: FIPS 20117 processed.

## 914/3107: FIPS 20119 processed.

## 915/3107: FIPS 20121 processed.

## 916/3107: FIPS 20123 processed.

## 917/3107: FIPS 20125 processed.

## 918/3107: FIPS 20127 processed.  
## 919/3107: FIPS 20129 processed.  
## 920/3107: FIPS 20131 processed.  
## 921/3107: FIPS 20133 processed.  
## 922/3107: FIPS 20135 processed.  
## 923/3107: FIPS 20137 processed.  
## 924/3107: FIPS 20139 processed.  
## 925/3107: FIPS 20141 processed.  
## 926/3107: FIPS 20143 processed.  
## 927/3107: FIPS 20145 processed.  
## 928/3107: FIPS 20147 processed.  
## 929/3107: FIPS 20149 processed.  
## 930/3107: FIPS 20151 processed.  
## 931/3107: FIPS 20153 processed.  
## 932/3107: FIPS 20155 processed.  
## 933/3107: FIPS 20157 processed.  
## 934/3107: FIPS 20159 processed.  
## 935/3107: FIPS 20161 processed.  
## 936/3107: FIPS 20163 processed.  
## 937/3107: FIPS 20165 processed.  
## 938/3107: FIPS 20167 processed.  
## 939/3107: FIPS 20169 processed.

## 940/3107: FIPS 20171 processed.  
## 941/3107: FIPS 20173 processed.  
## 942/3107: FIPS 20175 processed.  
## 943/3107: FIPS 20177 processed.  
## 944/3107: FIPS 20179 processed.  
## 945/3107: FIPS 20181 processed.  
## 946/3107: FIPS 20183 processed.  
## 947/3107: FIPS 20185 processed.  
## 948/3107: FIPS 20187 processed.  
## 949/3107: FIPS 20189 processed.  
## 950/3107: FIPS 20191 processed.  
## Checkpoint saved at iteration 950  
## 951/3107: FIPS 20193 processed.  
## 952/3107: FIPS 20195 processed.  
## 953/3107: FIPS 20197 processed.  
## 954/3107: FIPS 20199 processed.  
## 955/3107: FIPS 20201 processed.  
## 956/3107: FIPS 20203 processed.  
## 957/3107: FIPS 20205 processed.  
## 958/3107: FIPS 20207 processed.  
## 959/3107: FIPS 20209 processed.  
## 960/3107: FIPS 21001 processed.

## 961/3107: FIPS 21003 processed.  
## 962/3107: FIPS 21005 processed.  
## 963/3107: FIPS 21007 processed.  
## 964/3107: FIPS 21009 processed.  
## 965/3107: FIPS 21011 processed.  
## 966/3107: FIPS 21013 processed.  
## 967/3107: FIPS 21015 processed.  
## 968/3107: FIPS 21017 processed.  
## 969/3107: FIPS 21019 processed.  
## 970/3107: FIPS 21021 processed.  
## 971/3107: FIPS 21023 processed.  
## 972/3107: FIPS 21025 processed.  
## 973/3107: FIPS 21027 processed.  
## 974/3107: FIPS 21029 processed.  
## 975/3107: FIPS 21031 processed.  
## 976/3107: FIPS 21033 processed.  
## 977/3107: FIPS 21035 processed.  
## 978/3107: FIPS 21037 processed.  
## 979/3107: FIPS 21039 processed.  
## 980/3107: FIPS 21041 processed.  
## 981/3107: FIPS 21043 processed.  
## 982/3107: FIPS 21045 processed.



## 983/3107: FIPS 21047 processed.

## 984/3107: FIPS 21049 processed.

## 985/3107: FIPS 21051 processed.

## 986/3107: FIPS 21053 processed.

## 987/3107: FIPS 21055 processed.

## 988/3107: FIPS 21057 processed.

## 989/3107: FIPS 21059 processed.

## 990/3107: FIPS 21061 processed.

## 991/3107: FIPS 21063 processed.

## 992/3107: FIPS 21065 processed.

## 993/3107: FIPS 21067 processed.

## 994/3107: FIPS 21069 processed.

## 995/3107: FIPS 21071 processed.

## 996/3107: FIPS 21073 processed.

## 997/3107: FIPS 21075 processed.

## 998/3107: FIPS 21077 processed.

## 999/3107: FIPS 21079 processed.

## 1000/3107: FIPS 21081 processed.

## Checkpoint saved at iteration 1000

## 1001/3107: FIPS 21083 processed.

## 1002/3107: FIPS 21085 processed.

## 1003/3107: FIPS 21087 processed.

## 1004/3107: FIPS 21089 processed.

## 1005/3107: FIPS 21091 processed.

## 1006/3107: FIPS 21093 processed.

## 1007/3107: FIPS 21095 processed.

## 1008/3107: FIPS 21097 processed.

## 1009/3107: FIPS 21099 processed.

## 1010/3107: FIPS 21101 processed.

## 1011/3107: FIPS 21103 processed.

## 1012/3107: FIPS 21105 processed.

## 1013/3107: FIPS 21107 processed.

## 1014/3107: FIPS 21109 processed.

## 1015/3107: FIPS 21111 processed.

## 1016/3107: FIPS 21113 processed.

## 1017/3107: FIPS 21115 processed.

## 1018/3107: FIPS 21117 processed.

## 1019/3107: FIPS 21119 processed.

## 1020/3107: FIPS 21121 processed.

## 1021/3107: FIPS 21123 processed.

## 1022/3107: FIPS 21125 processed.

## 1023/3107: FIPS 21127 processed.

## 1024/3107: FIPS 21129 processed.

## 1025/3107: FIPS 21131 processed.

## 1026/3107: FIPS 21133 processed.

## 1027/3107: FIPS 21135 processed.

## 1028/3107: FIPS 21137 processed.

## 1029/3107: FIPS 21139 processed.

## 1030/3107: FIPS 21141 processed.

## 1031/3107: FIPS 21143 processed.

## 1032/3107: FIPS 21145 processed.

## 1033/3107: FIPS 21147 processed.

## 1034/3107: FIPS 21149 processed.

## 1035/3107: FIPS 21151 processed.

## 1036/3107: FIPS 21153 processed.

## 1037/3107: FIPS 21155 processed.

## 1038/3107: FIPS 21157 processed.

## 1039/3107: FIPS 21159 processed.

## 1040/3107: FIPS 21161 processed.

## 1041/3107: FIPS 21163 processed.

## 1042/3107: FIPS 21165 processed.

## 1043/3107: FIPS 21167 processed.

## 1044/3107: FIPS 21169 processed.

## 1045/3107: FIPS 21171 processed.

## 1046/3107: FIPS 21173 processed.

## 1047/3107: FIPS 21175 processed.

## 1048/3107: FIPS 21177 processed.

## 1049/3107: FIPS 21179 processed.

## 1050/3107: FIPS 21181 processed.

## Checkpoint saved at iteration 1050

## 1051/3107: FIPS 21183 processed.

## 1052/3107: FIPS 21185 processed.

## 1053/3107: FIPS 21187 processed.

## 1054/3107: FIPS 21189 processed.

## 1055/3107: FIPS 21191 processed.

## 1056/3107: FIPS 21193 processed.

## 1057/3107: FIPS 21195 processed.

## 1058/3107: FIPS 21197 processed.

## 1059/3107: FIPS 21199 processed.

## 1060/3107: FIPS 21201 processed.

## 1061/3107: FIPS 21203 processed.

## 1062/3107: FIPS 21205 processed.

## 1063/3107: FIPS 21207 processed.

## 1064/3107: FIPS 21209 processed.

## 1065/3107: FIPS 21211 processed.

## 1066/3107: FIPS 21213 processed.

## 1067/3107: FIPS 21215 processed.

## 1068/3107: FIPS 21217 processed.

## 1069/3107: FIPS 21219 processed.

## 1070/3107: FIPS 21221 processed.

## 1071/3107: FIPS 21223 processed.

## 1072/3107: FIPS 21225 processed.

## 1073/3107: FIPS 21227 processed.

## 1074/3107: FIPS 21229 processed.

## 1075/3107: FIPS 21231 processed.

## 1076/3107: FIPS 21233 processed.

## 1077/3107: FIPS 21235 processed.

## 1078/3107: FIPS 21237 processed.

## 1079/3107: FIPS 21239 processed.

## 1080/3107: FIPS 22001 processed.

## 1081/3107: FIPS 22003 processed.

## 1082/3107: FIPS 22005 processed.

## 1083/3107: FIPS 22007 processed.

## 1084/3107: FIPS 22009 processed.

## 1085/3107: FIPS 22011 processed.

## 1086/3107: FIPS 22013 processed.

## 1087/3107: FIPS 22015 processed.

## 1088/3107: FIPS 22017 processed.

## 1089/3107: FIPS 22019 processed.

## 1090/3107: FIPS 22021 processed.

## 1091/3107: FIPS 22023 processed.

## 1092/3107: FIPS 22025 processed.

## 1093/3107: FIPS 22027 processed.

## 1094/3107: FIPS 22029 processed.

## 1095/3107: FIPS 22031 processed.

## 1096/3107: FIPS 22033 processed.

## 1097/3107: FIPS 22035 processed.

## 1098/3107: FIPS 22037 processed.

## 1099/3107: FIPS 22039 processed.

## 1100/3107: FIPS 22041 processed.

## Checkpoint saved at iteration 1100

## 1101/3107: FIPS 22043 processed.

## 1102/3107: FIPS 22045 processed.

## 1103/3107: FIPS 22047 processed.

## 1104/3107: FIPS 22049 processed.

## 1105/3107: FIPS 22051 processed.

## 1106/3107: FIPS 22053 processed.

## 1107/3107: FIPS 22055 processed.

## 1108/3107: FIPS 22057 processed.

## 1109/3107: FIPS 22059 processed.

## 1110/3107: FIPS 22061 processed.

## 1111/3107: FIPS 22063 processed.

## 1112/3107: FIPS 22065 processed.

## 1113/3107: FIPS 22067 processed.

## 1114/3107: FIPS 22069 processed.

## 1115/3107: FIPS 22071 processed.

## 1116/3107: FIPS 22073 processed.

## 1117/3107: FIPS 22075 processed.

## 1118/3107: FIPS 22077 processed.

## 1119/3107: FIPS 22079 processed.

## 1120/3107: FIPS 22081 processed.

## 1121/3107: FIPS 22083 processed.

## 1122/3107: FIPS 22085 processed.

## 1123/3107: FIPS 22087 processed.

## 1124/3107: FIPS 22089 processed.

## 1125/3107: FIPS 22091 processed.

## 1126/3107: FIPS 22093 processed.

## 1127/3107: FIPS 22095 processed.

## 1128/3107: FIPS 22097 processed.

## 1129/3107: FIPS 22099 processed.

## 1130/3107: FIPS 22101 processed.

## 1131/3107: FIPS 22103 processed.

## 1132/3107: FIPS 22105 processed.

## 1133/3107: FIPS 22107 processed.

## 1134/3107: FIPS 22109 processed.

## 1135/3107: FIPS 22111 processed.

## 1136/3107: FIPS 22113 processed.

## 1137/3107: FIPS 22115 processed.

## 1138/3107: FIPS 22117 processed.

## 1139/3107: FIPS 22119 processed.

## 1140/3107: FIPS 22121 processed.

## 1141/3107: FIPS 22123 processed.

## 1142/3107: FIPS 22125 processed.

## 1143/3107: FIPS 22127 processed.

## 1144/3107: FIPS 23001 processed.

## 1145/3107: FIPS 23003 processed.

## 1146/3107: FIPS 23005 processed.

## 1147/3107: FIPS 23007 processed.

## 1148/3107: FIPS 23009 processed.

## 1149/3107: FIPS 23011 processed.

## 1150/3107: FIPS 23013 processed.

## Checkpoint saved at iteration 1150

## 1151/3107: FIPS 23015 processed.

## 1152/3107: FIPS 23017 processed.

## 1153/3107: FIPS 23019 processed.

## 1154/3107: FIPS 23021 processed.



## 1155/3107: FIPS 23023 processed.

## 1156/3107: FIPS 23025 processed.

## 1157/3107: FIPS 23027 processed.

## 1158/3107: FIPS 23029 processed.

## 1159/3107: FIPS 23031 processed.

## 1160/3107: FIPS 24001 processed.

## 1161/3107: FIPS 24003 processed.

## 1162/3107: FIPS 24005 processed.

## 1163/3107: FIPS 24009 processed.

## 1164/3107: FIPS 24011 processed.

## 1165/3107: FIPS 24013 processed.

## 1166/3107: FIPS 24015 processed.

## 1167/3107: FIPS 24017 processed.

## 1168/3107: FIPS 24019 processed.

## 1169/3107: FIPS 24021 processed.

## 1170/3107: FIPS 24023 processed.

## 1171/3107: FIPS 24025 processed.

## 1172/3107: FIPS 24027 processed.

## 1173/3107: FIPS 24029 processed.

## 1174/3107: FIPS 24031 processed.

## 1175/3107: FIPS 24033 processed.

## 1176/3107: FIPS 24035 processed.

## 1177/3107: FIPS 24037 processed.

## 1178/3107: FIPS 24039 processed.

## 1179/3107: FIPS 24041 processed.

## 1180/3107: FIPS 24043 processed.

## 1181/3107: FIPS 24045 processed.

## 1182/3107: FIPS 24047 processed.

## 1183/3107: FIPS 24510 processed.

## 1184/3107: FIPS 25001 processed.

## 1185/3107: FIPS 25003 processed.

## 1186/3107: FIPS 25005 processed.

## 1187/3107: FIPS 25007 processed.

## 1188/3107: FIPS 25009 processed.

## 1189/3107: FIPS 25011 processed.

## 1190/3107: FIPS 25013 processed.

## 1191/3107: FIPS 25015 processed.

## 1192/3107: FIPS 25017 processed.

## 1193/3107: FIPS 25019 processed.

## 1194/3107: FIPS 25021 processed.

## 1195/3107: FIPS 25023 processed.

## 1196/3107: FIPS 25025 processed.

## 1197/3107: FIPS 25027 processed.

## 1198/3107: FIPS 26001 processed.

## 1199/3107: FIPS 26003 processed.

## 1200/3107: FIPS 26005 processed.

## Checkpoint saved at iteration 1200

## 1201/3107: FIPS 26007 processed.

## 1202/3107: FIPS 26009 processed.

## 1203/3107: FIPS 26011 processed.

## 1204/3107: FIPS 26013 processed.

## 1205/3107: FIPS 26015 processed.

## 1206/3107: FIPS 26017 processed.

## 1207/3107: FIPS 26019 processed.

## 1208/3107: FIPS 26021 processed.

## 1209/3107: FIPS 26023 processed.

## 1210/3107: FIPS 26025 processed.

## 1211/3107: FIPS 26027 processed.

## 1212/3107: FIPS 26029 processed.

## 1213/3107: FIPS 26031 processed.

## 1214/3107: FIPS 26033 processed.

## 1215/3107: FIPS 26035 processed.

## 1216/3107: FIPS 26037 processed.

## 1217/3107: FIPS 26039 processed.

## 1218/3107: FIPS 26041 processed.

## 1219/3107: FIPS 26043 processed.

## 1220/3107: FIPS 26045 processed.

## 1221/3107: FIPS 26047 processed.

## 1222/3107: FIPS 26049 processed.

## 1223/3107: FIPS 26051 processed.

## 1224/3107: FIPS 26053 processed.

## 1225/3107: FIPS 26055 processed.

## 1226/3107: FIPS 26057 processed.

## 1227/3107: FIPS 26059 processed.

## 1228/3107: FIPS 26061 processed.

## 1229/3107: FIPS 26063 processed.

## 1230/3107: FIPS 26065 processed.

## 1231/3107: FIPS 26067 processed.

## 1232/3107: FIPS 26069 processed.

## 1233/3107: FIPS 26071 processed.

## 1234/3107: FIPS 26073 processed.

## 1235/3107: FIPS 26075 processed.

## 1236/3107: FIPS 26077 processed.

## 1237/3107: FIPS 26079 processed.

## 1238/3107: FIPS 26081 processed.

## 1239/3107: FIPS 26083 processed.

## 1240/3107: FIPS 26085 processed.

## 1241/3107: FIPS 26087 processed.

## 1242/3107: FIPS 26089 processed.

## 1243/3107: FIPS 26091 processed.

## 1244/3107: FIPS 26093 processed.

## 1245/3107: FIPS 26095 processed.

## 1246/3107: FIPS 26097 processed.

## 1247/3107: FIPS 26099 processed.

## 1248/3107: FIPS 26101 processed.

## 1249/3107: FIPS 26103 processed.

## 1250/3107: FIPS 26105 processed.

## Checkpoint saved at iteration 1250

## 1251/3107: FIPS 26107 processed.

## 1252/3107: FIPS 26109 processed.

## 1253/3107: FIPS 26111 processed.

## 1254/3107: FIPS 26113 processed.

## 1255/3107: FIPS 26115 processed.

## 1256/3107: FIPS 26117 processed.

## 1257/3107: FIPS 26119 processed.

## 1258/3107: FIPS 26121 processed.

## 1259/3107: FIPS 26123 processed.

## 1260/3107: FIPS 26125 processed.

## 1261/3107: FIPS 26127 processed.

## 1262/3107: FIPS 26129 processed.

## 1263/3107: FIPS 26131 processed.

## 1264/3107: FIPS 26133 processed.

## 1265/3107: FIPS 26135 processed.

## 1266/3107: FIPS 26137 processed.

## 1267/3107: FIPS 26139 processed.

## 1268/3107: FIPS 26141 processed.

## 1269/3107: FIPS 26143 processed.

## 1270/3107: FIPS 26145 processed.

## 1271/3107: FIPS 26147 processed.

## 1272/3107: FIPS 26149 processed.

## 1273/3107: FIPS 26151 processed.

## 1274/3107: FIPS 26153 processed.

## 1275/3107: FIPS 26155 processed.

## 1276/3107: FIPS 26157 processed.

## 1277/3107: FIPS 26159 processed.

## 1278/3107: FIPS 26161 processed.

## 1279/3107: FIPS 26163 processed.

## 1280/3107: FIPS 26165 processed.

## 1281/3107: FIPS 27001 processed.

## 1282/3107: FIPS 27003 processed.

## 1283/3107: FIPS 27005 processed.

## 1284/3107: FIPS 27007 processed.

## 1285/3107: FIPS 27009 processed.

## 1286/3107: FIPS 27011 processed.

## 1287/3107: FIPS 27013 processed.

## 1288/3107: FIPS 27015 processed.

## 1289/3107: FIPS 27017 processed.

## 1290/3107: FIPS 27019 processed.

## 1291/3107: FIPS 27021 processed.

## 1292/3107: FIPS 27023 processed.

## 1293/3107: FIPS 27025 processed.

## 1294/3107: FIPS 27027 processed.

## 1295/3107: FIPS 27029 processed.

## 1296/3107: FIPS 27031 processed.

## 1297/3107: FIPS 27033 processed.

## 1298/3107: FIPS 27035 processed.

## 1299/3107: FIPS 27037 processed.

## 1300/3107: FIPS 27039 processed.

## Checkpoint saved at iteration 1300

## 1301/3107: FIPS 27041 processed.

## 1302/3107: FIPS 27043 processed.

## 1303/3107: FIPS 27045 processed.

## 1304/3107: FIPS 27047 processed.

## 1305/3107: FIPS 27049 processed.

## 1306/3107: FIPS 27051 processed.

## 1307/3107: FIPS 27053 processed.

## 1308/3107: FIPS 27055 processed.

## 1309/3107: FIPS 27057 processed.

## 1310/3107: FIPS 27059 processed.

## 1311/3107: FIPS 27061 processed.

## 1312/3107: FIPS 27063 processed.

## 1313/3107: FIPS 27065 processed.

## 1314/3107: FIPS 27067 processed.

## 1315/3107: FIPS 27069 processed.

## 1316/3107: FIPS 27071 processed.

## 1317/3107: FIPS 27073 processed.

## 1318/3107: FIPS 27075 processed.

## 1319/3107: FIPS 27077 processed.

## 1320/3107: FIPS 27079 processed.

## 1321/3107: FIPS 27081 processed.

## 1322/3107: FIPS 27083 processed.

## 1323/3107: FIPS 27085 processed.

## 1324/3107: FIPS 27087 processed.

## 1325/3107: FIPS 27089 processed.

## 1326/3107: FIPS 27091 processed.

## 1327/3107: FIPS 27093 processed.



## 1328/3107: FIPS 27095 processed.  
## 1329/3107: FIPS 27097 processed.  
## 1330/3107: FIPS 27099 processed.  
## 1331/3107: FIPS 27101 processed.  
## 1332/3107: FIPS 27103 processed.  
## 1333/3107: FIPS 27105 processed.  
## 1334/3107: FIPS 27107 processed.  
## 1335/3107: FIPS 27109 processed.  
## 1336/3107: FIPS 27111 processed.  
## 1337/3107: FIPS 27113 processed.  
## 1338/3107: FIPS 27115 processed.  
## 1339/3107: FIPS 27117 processed.  
## 1340/3107: FIPS 27119 processed.  
## 1341/3107: FIPS 27121 processed.  
## 1342/3107: FIPS 27123 processed.  
## 1343/3107: FIPS 27125 processed.  
## 1344/3107: FIPS 27127 processed.  
## 1345/3107: FIPS 27129 processed.  
## 1346/3107: FIPS 27131 processed.  
## 1347/3107: FIPS 27133 processed.  
## 1348/3107: FIPS 27135 processed.  
## 1349/3107: FIPS 27137 processed.

## 1350/3107: FIPS 27139 processed.

## Checkpoint saved at iteration 1350

## 1351/3107: FIPS 27141 processed.

## 1352/3107: FIPS 27143 processed.

## 1353/3107: FIPS 27145 processed.

## 1354/3107: FIPS 27147 processed.

## 1355/3107: FIPS 27149 processed.

## 1356/3107: FIPS 27151 processed.

## 1357/3107: FIPS 27153 processed.

## 1358/3107: FIPS 27155 processed.

## 1359/3107: FIPS 27157 processed.

## 1360/3107: FIPS 27159 processed.

## 1361/3107: FIPS 27161 processed.

## 1362/3107: FIPS 27163 processed.

## 1363/3107: FIPS 27165 processed.

## 1364/3107: FIPS 27167 processed.

## 1365/3107: FIPS 27169 processed.

## 1366/3107: FIPS 27171 processed.

## 1367/3107: FIPS 27173 processed.

## 1368/3107: FIPS 28001 processed.

## 1369/3107: FIPS 28003 processed.

## 1370/3107: FIPS 28005 processed.

## 1371/3107: FIPS 28007 processed.

## 1372/3107: FIPS 28009 processed.

## 1373/3107: FIPS 28011 processed.

## 1374/3107: FIPS 28013 processed.

## 1375/3107: FIPS 28015 processed.

## 1376/3107: FIPS 28017 processed.

## 1377/3107: FIPS 28019 processed.

## 1378/3107: FIPS 28021 processed.

## 1379/3107: FIPS 28023 processed.

## 1380/3107: FIPS 28025 processed.

## 1381/3107: FIPS 28027 processed.

## 1382/3107: FIPS 28029 processed.

## 1383/3107: FIPS 28031 processed.

## 1384/3107: FIPS 28033 processed.

## 1385/3107: FIPS 28035 processed.

## 1386/3107: FIPS 28037 processed.

## 1387/3107: FIPS 28039 processed.

## 1388/3107: FIPS 28041 processed.

## 1389/3107: FIPS 28043 processed.

## 1390/3107: FIPS 28045 processed.

## 1391/3107: FIPS 28047 processed.

## 1392/3107: FIPS 28049 processed.

## 1393/3107: FIPS 28051 processed.

## 1394/3107: FIPS 28053 processed.

## 1395/3107: FIPS 28055 processed.

## 1396/3107: FIPS 28057 processed.

## 1397/3107: FIPS 28059 processed.

## 1398/3107: FIPS 28061 processed.

## 1399/3107: FIPS 28063 processed.

## 1400/3107: FIPS 28065 processed.

## Checkpoint saved at iteration 1400

## 1401/3107: FIPS 28067 processed.

## 1402/3107: FIPS 28069 processed.

## 1403/3107: FIPS 28071 processed.

## 1404/3107: FIPS 28073 processed.

## 1405/3107: FIPS 28075 processed.

## 1406/3107: FIPS 28077 processed.

## 1407/3107: FIPS 28079 processed.

## 1408/3107: FIPS 28081 processed.

## 1409/3107: FIPS 28083 processed.

## 1410/3107: FIPS 28085 processed.

## 1411/3107: FIPS 28087 processed.

## 1412/3107: FIPS 28089 processed.

## 1413/3107: FIPS 28091 processed.

## 1414/3107: FIPS 28093 processed.

## 1415/3107: FIPS 28095 processed.

## 1416/3107: FIPS 28097 processed.

## 1417/3107: FIPS 28099 processed.

## 1418/3107: FIPS 28101 processed.

## 1419/3107: FIPS 28103 processed.

## 1420/3107: FIPS 28105 processed.

## 1421/3107: FIPS 28107 processed.

## 1422/3107: FIPS 28109 processed.

## 1423/3107: FIPS 28111 processed.

## 1424/3107: FIPS 28113 processed.

## 1425/3107: FIPS 28115 processed.

## 1426/3107: FIPS 28117 processed.

## 1427/3107: FIPS 28119 processed.

## 1428/3107: FIPS 28121 processed.

## 1429/3107: FIPS 28123 processed.

## 1430/3107: FIPS 28125 processed.

## 1431/3107: FIPS 28127 processed.

## 1432/3107: FIPS 28129 processed.

## 1433/3107: FIPS 28131 processed.

## 1434/3107: FIPS 28133 processed.

## 1435/3107: FIPS 28135 processed.

## 1436/3107: FIPS 28137 processed.

## 1437/3107: FIPS 28139 processed.

## 1438/3107: FIPS 28141 processed.

## 1439/3107: FIPS 28143 processed.

## 1440/3107: FIPS 28145 processed.

## 1441/3107: FIPS 28147 processed.

## 1442/3107: FIPS 28149 processed.

## 1443/3107: FIPS 28151 processed.

## 1444/3107: FIPS 28153 processed.

## 1445/3107: FIPS 28155 processed.

## 1446/3107: FIPS 28157 processed.

## 1447/3107: FIPS 28159 processed.

## 1448/3107: FIPS 28161 processed.

## 1449/3107: FIPS 28163 processed.

## 1450/3107: FIPS 29001 processed.

## Checkpoint saved at iteration 1450

## 1451/3107: FIPS 29003 processed.

## 1452/3107: FIPS 29005 processed.

## 1453/3107: FIPS 29007 processed.

## 1454/3107: FIPS 29009 processed.

## 1455/3107: FIPS 29011 processed.

## 1456/3107: FIPS 29013 processed.

## 1457/3107: FIPS 29015 processed.

## 1458/3107: FIPS 29017 processed.

## 1459/3107: FIPS 29019 processed.

## 1460/3107: FIPS 29021 processed.

## 1461/3107: FIPS 29023 processed.

## 1462/3107: FIPS 29025 processed.

## 1463/3107: FIPS 29027 processed.

## 1464/3107: FIPS 29029 processed.

## 1465/3107: FIPS 29031 processed.

## 1466/3107: FIPS 29033 processed.

## 1467/3107: FIPS 29035 processed.

## 1468/3107: FIPS 29037 processed.

## 1469/3107: FIPS 29039 processed.

## 1470/3107: FIPS 29041 processed.

## 1471/3107: FIPS 29043 processed.

## 1472/3107: FIPS 29045 processed.

## 1473/3107: FIPS 29047 processed.

## 1474/3107: FIPS 29049 processed.

## 1475/3107: FIPS 29051 processed.

## 1476/3107: FIPS 29053 processed.

## 1477/3107: FIPS 29055 processed.

## 1478/3107: FIPS 29057 processed.

## 1479/3107: FIPS 29059 processed.

## 1480/3107: FIPS 29061 processed.

## 1481/3107: FIPS 29063 processed.

## 1482/3107: FIPS 29065 processed.

## 1483/3107: FIPS 29067 processed.

## 1484/3107: FIPS 29069 processed.

## 1485/3107: FIPS 29071 processed.

## 1486/3107: FIPS 29073 processed.

## 1487/3107: FIPS 29075 processed.

## 1488/3107: FIPS 29077 processed.

## 1489/3107: FIPS 29079 processed.

## 1490/3107: FIPS 29081 processed.

## 1491/3107: FIPS 29083 processed.

## 1492/3107: FIPS 29085 processed.

## 1493/3107: FIPS 29087 processed.

## 1494/3107: FIPS 29089 processed.

## 1495/3107: FIPS 29091 processed.

## 1496/3107: FIPS 29093 processed.

## 1497/3107: FIPS 29095 processed.

## 1498/3107: FIPS 29097 processed.

## 1499/3107: FIPS 29099 processed.

## 1500/3107: FIPS 29101 processed.



## Checkpoint saved at iteration 1500

## 1501/3107: FIPS 29103 processed.

## 1502/3107: FIPS 29105 processed.

## 1503/3107: FIPS 29107 processed.

## 1504/3107: FIPS 29109 processed.

## 1505/3107: FIPS 29111 processed.

## 1506/3107: FIPS 29113 processed.

## 1507/3107: FIPS 29115 processed.

## 1508/3107: FIPS 29117 processed.

## 1509/3107: FIPS 29119 processed.

## 1510/3107: FIPS 29121 processed.

## 1511/3107: FIPS 29123 processed.

## 1512/3107: FIPS 29125 processed.

## 1513/3107: FIPS 29127 processed.

## 1514/3107: FIPS 29129 processed.

## 1515/3107: FIPS 29131 processed.

## 1516/3107: FIPS 29133 processed.

## 1517/3107: FIPS 29135 processed.

## 1518/3107: FIPS 29137 processed.

## 1519/3107: FIPS 29139 processed.

## 1520/3107: FIPS 29141 processed.

## 1521/3107: FIPS 29143 processed.

## 1522/3107: FIPS 29145 processed.  
## 1523/3107: FIPS 29147 processed.  
## 1524/3107: FIPS 29149 processed.  
## 1525/3107: FIPS 29151 processed.  
## 1526/3107: FIPS 29153 processed.  
## 1527/3107: FIPS 29155 processed.  
## 1528/3107: FIPS 29157 processed.  
## 1529/3107: FIPS 29159 processed.  
## 1530/3107: FIPS 29161 processed.  
## 1531/3107: FIPS 29163 processed.  
## 1532/3107: FIPS 29165 processed.  
## 1533/3107: FIPS 29167 processed.  
## 1534/3107: FIPS 29169 processed.  
## 1535/3107: FIPS 29171 processed.  
## 1536/3107: FIPS 29173 processed.  
## 1537/3107: FIPS 29175 processed.  
## 1538/3107: FIPS 29177 processed.  
## 1539/3107: FIPS 29179 processed.  
## 1540/3107: FIPS 29181 processed.  
## 1541/3107: FIPS 29183 processed.  
## 1542/3107: FIPS 29185 processed.  
## 1543/3107: FIPS 29186 processed.

## 1544/3107: FIPS 29187 processed.  
## 1545/3107: FIPS 29189 processed.  
## 1546/3107: FIPS 29195 processed.  
## 1547/3107: FIPS 29197 processed.  
## 1548/3107: FIPS 29199 processed.  
## 1549/3107: FIPS 29201 processed.  
## 1550/3107: FIPS 29203 processed.  
## Checkpoint saved at iteration 1550  
## 1551/3107: FIPS 29205 processed.  
## 1552/3107: FIPS 29207 processed.  
## 1553/3107: FIPS 29209 processed.  
## 1554/3107: FIPS 29211 processed.  
## 1555/3107: FIPS 29213 processed.  
## 1556/3107: FIPS 29215 processed.  
## 1557/3107: FIPS 29217 processed.  
## 1558/3107: FIPS 29219 processed.  
## 1559/3107: FIPS 29221 processed.  
## 1560/3107: FIPS 29223 processed.  
## 1561/3107: FIPS 29225 processed.  
## 1562/3107: FIPS 29227 processed.  
## 1563/3107: FIPS 29229 processed.  
## 1564/3107: FIPS 29510 processed.

## 1565/3107: FIPS 30001 processed.

## 1566/3107: FIPS 30003 processed.

## 1567/3107: FIPS 30005 processed.

## 1568/3107: FIPS 30007 processed.

## 1569/3107: FIPS 30009 processed.

## 1570/3107: FIPS 30011 processed.

## 1571/3107: FIPS 30013 processed.

## 1572/3107: FIPS 30015 processed.

## 1573/3107: FIPS 30017 processed.

## 1574/3107: FIPS 30019 processed.

## 1575/3107: FIPS 30021 processed.

## 1576/3107: FIPS 30023 processed.

## 1577/3107: FIPS 30025 processed.

## 1578/3107: FIPS 30027 processed.

## 1579/3107: FIPS 30029 processed.

## 1580/3107: FIPS 30031 processed.

## 1581/3107: FIPS 30033 processed.

## 1582/3107: FIPS 30035 processed.

## 1583/3107: FIPS 30037 processed.

## 1584/3107: FIPS 30039 processed.

## 1585/3107: FIPS 30041 processed.

## 1586/3107: FIPS 30043 processed.

## 1587/3107: FIPS 30045 processed.

## 1588/3107: FIPS 30047 processed.

## 1589/3107: FIPS 30049 processed.

## 1590/3107: FIPS 30051 processed.

## 1591/3107: FIPS 30053 processed.

## 1592/3107: FIPS 30055 processed.

## 1593/3107: FIPS 30057 processed.

## 1594/3107: FIPS 30059 processed.

## 1595/3107: FIPS 30061 processed.

## 1596/3107: FIPS 30063 processed.

## 1597/3107: FIPS 30065 processed.

## 1598/3107: FIPS 30067 processed.

## 1599/3107: FIPS 30069 processed.

## 1600/3107: FIPS 30071 processed.

## Checkpoint saved at iteration 1600

## 1601/3107: FIPS 30073 processed.

## 1602/3107: FIPS 30075 processed.

## 1603/3107: FIPS 30077 processed.

## 1604/3107: FIPS 30079 processed.

## 1605/3107: FIPS 30081 processed.

## 1606/3107: FIPS 30083 processed.

## 1607/3107: FIPS 30085 processed.

## 1608/3107: FIPS 30087 processed.

## 1609/3107: FIPS 30089 processed.

## 1610/3107: FIPS 30091 processed.

## 1611/3107: FIPS 30093 processed.

## 1612/3107: FIPS 30095 processed.

## 1613/3107: FIPS 30097 processed.

## 1614/3107: FIPS 30099 processed.

## 1615/3107: FIPS 30101 processed.

## 1616/3107: FIPS 30103 processed.

## 1617/3107: FIPS 30105 processed.

## 1618/3107: FIPS 30107 processed.

## 1619/3107: FIPS 30109 processed.

## 1620/3107: FIPS 30111 processed.

## 1621/3107: FIPS 31001 processed.

## 1622/3107: FIPS 31003 processed.

## 1623/3107: FIPS 31005 processed.

## 1624/3107: FIPS 31007 processed.

## 1625/3107: FIPS 31009 processed.

## 1626/3107: FIPS 31011 processed.

## 1627/3107: FIPS 31013 processed.

## 1628/3107: FIPS 31015 processed.

## 1629/3107: FIPS 31017 processed.

## 1630/3107: FIPS 31019 processed.  
## 1631/3107: FIPS 31021 processed.  
## 1632/3107: FIPS 31023 processed.  
## 1633/3107: FIPS 31025 processed.  
## 1634/3107: FIPS 31027 processed.  
## 1635/3107: FIPS 31029 processed.  
## 1636/3107: FIPS 31031 processed.  
## 1637/3107: FIPS 31033 processed.  
## 1638/3107: FIPS 31035 processed.  
## 1639/3107: FIPS 31037 processed.  
## 1640/3107: FIPS 31039 processed.  
## 1641/3107: FIPS 31041 processed.  
## 1642/3107: FIPS 31043 processed.  
## 1643/3107: FIPS 31045 processed.  
## 1644/3107: FIPS 31047 processed.  
## 1645/3107: FIPS 31049 processed.  
## 1646/3107: FIPS 31051 processed.  
## 1647/3107: FIPS 31053 processed.  
## 1648/3107: FIPS 31055 processed.  
## 1649/3107: FIPS 31057 processed.  
## 1650/3107: FIPS 31059 processed.  
## Checkpoint saved at iteration 1650

## 1651/3107: FIPS 31061 processed.

## 1652/3107: FIPS 31063 processed.

## 1653/3107: FIPS 31065 processed.

## 1654/3107: FIPS 31067 processed.

## 1655/3107: FIPS 31069 processed.

## 1656/3107: FIPS 31071 processed.

## 1657/3107: FIPS 31073 processed.

## 1658/3107: FIPS 31075 processed.

## 1659/3107: FIPS 31077 processed.

## 1660/3107: FIPS 31079 processed.

## 1661/3107: FIPS 31081 processed.

## 1662/3107: FIPS 31083 processed.

## 1663/3107: FIPS 31085 processed.

## 1664/3107: FIPS 31087 processed.

## 1665/3107: FIPS 31089 processed.

## 1666/3107: FIPS 31091 processed.

## 1667/3107: FIPS 31093 processed.

## 1668/3107: FIPS 31095 processed.

## 1669/3107: FIPS 31097 processed.

## 1670/3107: FIPS 31099 processed.

## 1671/3107: FIPS 31101 processed.

## 1672/3107: FIPS 31103 processed.



## 1673/3107: FIPS 31105 processed.

## 1674/3107: FIPS 31107 processed.

## 1675/3107: FIPS 31109 processed.

## 1676/3107: FIPS 31111 processed.

## 1677/3107: FIPS 31113 processed.

## 1678/3107: FIPS 31115 processed.

## 1679/3107: FIPS 31117 processed.

## 1680/3107: FIPS 31119 processed.

## 1681/3107: FIPS 31121 processed.

## 1682/3107: FIPS 31123 processed.

## 1683/3107: FIPS 31125 processed.

## 1684/3107: FIPS 31127 processed.

## 1685/3107: FIPS 31129 processed.

## 1686/3107: FIPS 31131 processed.

## 1687/3107: FIPS 31133 processed.

## 1688/3107: FIPS 31135 processed.

## 1689/3107: FIPS 31137 processed.

## 1690/3107: FIPS 31139 processed.

## 1691/3107: FIPS 31141 processed.

## 1692/3107: FIPS 31143 processed.

## 1693/3107: FIPS 31145 processed.

## 1694/3107: FIPS 31147 processed.

## 1695/3107: FIPS 31149 processed.

## 1696/3107: FIPS 31151 processed.

## 1697/3107: FIPS 31153 processed.

## 1698/3107: FIPS 31155 processed.

## 1699/3107: FIPS 31157 processed.

## 1700/3107: FIPS 31159 processed.

## Checkpoint saved at iteration 1700

## 1701/3107: FIPS 31161 processed.

## 1702/3107: FIPS 31163 processed.

## 1703/3107: FIPS 31165 processed.

## 1704/3107: FIPS 31167 processed.

## 1705/3107: FIPS 31169 processed.

## 1706/3107: FIPS 31171 processed.

## 1707/3107: FIPS 31173 processed.

## 1708/3107: FIPS 31175 processed.

## 1709/3107: FIPS 31177 processed.

## 1710/3107: FIPS 31179 processed.

## 1711/3107: FIPS 31181 processed.

## 1712/3107: FIPS 31183 processed.

## 1713/3107: FIPS 31185 processed.

## 1714/3107: FIPS 32001 processed.

## 1715/3107: FIPS 32003 processed.

## 1716/3107: FIPS 32005 processed.

## 1717/3107: FIPS 32007 processed.

## 1718/3107: FIPS 32009 processed.

## 1719/3107: FIPS 32011 processed.

## 1720/3107: FIPS 32013 processed.

## 1721/3107: FIPS 32015 processed.

## 1722/3107: FIPS 32017 processed.

## 1723/3107: FIPS 32019 processed.

## 1724/3107: FIPS 32021 processed.

## 1725/3107: FIPS 32023 processed.

## 1726/3107: FIPS 32027 processed.

## 1727/3107: FIPS 32029 processed.

## 1728/3107: FIPS 32031 processed.

## 1729/3107: FIPS 32033 processed.

## 1730/3107: FIPS 32510 processed.

## 1731/3107: FIPS 33001 processed.

## 1732/3107: FIPS 33003 processed.

## 1733/3107: FIPS 33005 processed.

## 1734/3107: FIPS 33007 processed.

## 1735/3107: FIPS 33009 processed.

## 1736/3107: FIPS 33011 processed.

## 1737/3107: FIPS 33013 processed.

## 1738/3107: FIPS 33015 processed.

## 1739/3107: FIPS 33017 processed.

## 1740/3107: FIPS 33019 processed.

## 1741/3107: FIPS 34001 processed.

## 1742/3107: FIPS 34003 processed.

## 1743/3107: FIPS 34005 processed.

## 1744/3107: FIPS 34007 processed.

## 1745/3107: FIPS 34009 processed.

## 1746/3107: FIPS 34011 processed.

## 1747/3107: FIPS 34013 processed.

## 1748/3107: FIPS 34015 processed.

## 1749/3107: FIPS 34017 processed.

## 1750/3107: FIPS 34019 processed.

## Checkpoint saved at iteration 1750

## 1751/3107: FIPS 34021 processed.

## 1752/3107: FIPS 34023 processed.

## 1753/3107: FIPS 34025 processed.

## 1754/3107: FIPS 34027 processed.

## 1755/3107: FIPS 34029 processed.

## 1756/3107: FIPS 34031 processed.

## 1757/3107: FIPS 34033 processed.

## 1758/3107: FIPS 34035 processed.

## 1759/3107: FIPS 34037 processed.

## 1760/3107: FIPS 34039 processed.

## 1761/3107: FIPS 34041 processed.

## 1762/3107: FIPS 35001 processed.

## 1763/3107: FIPS 35003 processed.

## 1764/3107: FIPS 35005 processed.

## 1765/3107: FIPS 35006 processed.

## 1766/3107: FIPS 35007 processed.

## 1767/3107: FIPS 35009 processed.

## 1768/3107: FIPS 35011 processed.

## 1769/3107: FIPS 35013 processed.

## 1770/3107: FIPS 35015 processed.

## 1771/3107: FIPS 35017 processed.

## 1772/3107: FIPS 35019 processed.

## 1773/3107: FIPS 35021 processed.

## 1774/3107: FIPS 35023 processed.

## 1775/3107: FIPS 35025 processed.

## 1776/3107: FIPS 35027 processed.

## 1777/3107: FIPS 35028 processed.

## 1778/3107: FIPS 35029 processed.

## 1779/3107: FIPS 35031 processed.

## 1780/3107: FIPS 35033 processed.

## 1781/3107: FIPS 35035 processed.  
## 1782/3107: FIPS 35037 processed.  
## 1783/3107: FIPS 35039 processed.  
## 1784/3107: FIPS 35041 processed.  
## 1785/3107: FIPS 35043 processed.  
## 1786/3107: FIPS 35045 processed.  
## 1787/3107: FIPS 35047 processed.  
## 1788/3107: FIPS 35049 processed.  
## 1789/3107: FIPS 35051 processed.  
## 1790/3107: FIPS 35053 processed.  
## 1791/3107: FIPS 35055 processed.  
## 1792/3107: FIPS 35057 processed.  
## 1793/3107: FIPS 35059 processed.  
## 1794/3107: FIPS 35061 processed.  
## 1795/3107: FIPS 36001 processed.  
## 1796/3107: FIPS 36003 processed.  
## 1797/3107: FIPS 36005 processed.  
## 1798/3107: FIPS 36007 processed.  
## 1799/3107: FIPS 36009 processed.  
## 1800/3107: FIPS 36011 processed.  
## Checkpoint saved at iteration 1800  
## 1801/3107: FIPS 36013 processed.

## 1802/3107: FIPS 36015 processed.

## 1803/3107: FIPS 36017 processed.

## 1804/3107: FIPS 36019 processed.

## 1805/3107: FIPS 36021 processed.

## 1806/3107: FIPS 36023 processed.

## 1807/3107: FIPS 36025 processed.

## 1808/3107: FIPS 36027 processed.

## 1809/3107: FIPS 36029 processed.

## 1810/3107: FIPS 36031 processed.

## 1811/3107: FIPS 36033 processed.

## 1812/3107: FIPS 36035 processed.

## 1813/3107: FIPS 36037 processed.

## 1814/3107: FIPS 36039 processed.

## 1815/3107: FIPS 36041 processed.

## 1816/3107: FIPS 36043 processed.

## 1817/3107: FIPS 36045 processed.

## 1818/3107: FIPS 36047 processed.

## 1819/3107: FIPS 36049 processed.

## 1820/3107: FIPS 36051 processed.

## 1821/3107: FIPS 36053 processed.

## 1822/3107: FIPS 36055 processed.

## 1823/3107: FIPS 36057 processed.

## 1824/3107: FIPS 36059 processed.

## 1825/3107: FIPS 36061 processed.

## 1826/3107: FIPS 36063 processed.

## 1827/3107: FIPS 36065 processed.

## 1828/3107: FIPS 36067 processed.

## 1829/3107: FIPS 36069 processed.

## 1830/3107: FIPS 36071 processed.

## 1831/3107: FIPS 36073 processed.

## 1832/3107: FIPS 36075 processed.

## 1833/3107: FIPS 36077 processed.

## 1834/3107: FIPS 36079 processed.

## 1835/3107: FIPS 36081 processed.

## 1836/3107: FIPS 36083 processed.

## 1837/3107: FIPS 36085 processed.

## 1838/3107: FIPS 36087 processed.

## 1839/3107: FIPS 36089 processed.

## 1840/3107: FIPS 36091 processed.

## 1841/3107: FIPS 36093 processed.

## 1842/3107: FIPS 36095 processed.

## 1843/3107: FIPS 36097 processed.

## 1844/3107: FIPS 36099 processed.

## 1845/3107: FIPS 36101 processed.



## 1846/3107: FIPS 36103 processed.

## 1847/3107: FIPS 36105 processed.

## 1848/3107: FIPS 36107 processed.

## 1849/3107: FIPS 36109 processed.

## 1850/3107: FIPS 36111 processed.

## Checkpoint saved at iteration 1850

## 1851/3107: FIPS 36113 processed.

## 1852/3107: FIPS 36115 processed.

## 1853/3107: FIPS 36117 processed.

## 1854/3107: FIPS 36119 processed.

## 1855/3107: FIPS 36121 processed.

## 1856/3107: FIPS 36123 processed.

## 1857/3107: FIPS 37001 processed.

## 1858/3107: FIPS 37003 processed.

## 1859/3107: FIPS 37005 processed.

## 1860/3107: FIPS 37007 processed.

## 1861/3107: FIPS 37009 processed.

## 1862/3107: FIPS 37011 processed.

## 1863/3107: FIPS 37013 processed.

## 1864/3107: FIPS 37015 processed.

## 1865/3107: FIPS 37017 processed.

## 1866/3107: FIPS 37019 processed.

## 1867/3107: FIPS 37021 processed.

## 1868/3107: FIPS 37023 processed.

## 1869/3107: FIPS 37025 processed.

## 1870/3107: FIPS 37027 processed.

## 1871/3107: FIPS 37029 processed.

## 1872/3107: FIPS 37031 processed.

## 1873/3107: FIPS 37033 processed.

## 1874/3107: FIPS 37035 processed.

## 1875/3107: FIPS 37037 processed.

## 1876/3107: FIPS 37039 processed.

## 1877/3107: FIPS 37041 processed.

## 1878/3107: FIPS 37043 processed.

## 1879/3107: FIPS 37045 processed.

## 1880/3107: FIPS 37047 processed.

## 1881/3107: FIPS 37049 processed.

## 1882/3107: FIPS 37051 processed.

## 1883/3107: FIPS 37053 processed.

## 1884/3107: FIPS 37055 processed.

## 1885/3107: FIPS 37057 processed.

## 1886/3107: FIPS 37059 processed.

## 1887/3107: FIPS 37061 processed.

## 1888/3107: FIPS 37063 processed.

## 1889/3107: FIPS 37065 processed.

## 1890/3107: FIPS 37067 processed.

## 1891/3107: FIPS 37069 processed.

## 1892/3107: FIPS 37071 processed.

## 1893/3107: FIPS 37073 processed.

## 1894/3107: FIPS 37075 processed.

## 1895/3107: FIPS 37077 processed.

## 1896/3107: FIPS 37079 processed.

## 1897/3107: FIPS 37081 processed.

## 1898/3107: FIPS 37083 processed.

## 1899/3107: FIPS 37085 processed.

## 1900/3107: FIPS 37087 processed.

## Checkpoint saved at iteration 1900

## 1901/3107: FIPS 37089 processed.

## 1902/3107: FIPS 37091 processed.

## 1903/3107: FIPS 37093 processed.

## 1904/3107: FIPS 37095 processed.

## 1905/3107: FIPS 37097 processed.

## 1906/3107: FIPS 37099 processed.

## 1907/3107: FIPS 37101 processed.

## 1908/3107: FIPS 37103 processed.

## 1909/3107: FIPS 37105 processed.

## 1910/3107: FIPS 37107 processed.

## 1911/3107: FIPS 37109 processed.

## 1912/3107: FIPS 37111 processed.

## 1913/3107: FIPS 37113 processed.

## 1914/3107: FIPS 37115 processed.

## 1915/3107: FIPS 37117 processed.

## 1916/3107: FIPS 37119 processed.

## 1917/3107: FIPS 37121 processed.

## 1918/3107: FIPS 37123 processed.

## 1919/3107: FIPS 37125 processed.

## 1920/3107: FIPS 37127 processed.

## 1921/3107: FIPS 37129 processed.

## 1922/3107: FIPS 37131 processed.

## 1923/3107: FIPS 37133 processed.

## 1924/3107: FIPS 37135 processed.

## 1925/3107: FIPS 37137 processed.

## 1926/3107: FIPS 37139 processed.

## 1927/3107: FIPS 37141 processed.

## 1928/3107: FIPS 37143 processed.

## 1929/3107: FIPS 37145 processed.

## 1930/3107: FIPS 37147 processed.

## 1931/3107: FIPS 37149 processed.

## 1932/3107: FIPS 37151 processed.

## 1933/3107: FIPS 37153 processed.

## 1934/3107: FIPS 37155 processed.

## 1935/3107: FIPS 37157 processed.

## 1936/3107: FIPS 37159 processed.

## 1937/3107: FIPS 37161 processed.

## 1938/3107: FIPS 37163 processed.

## 1939/3107: FIPS 37165 processed.

## 1940/3107: FIPS 37167 processed.

## 1941/3107: FIPS 37169 processed.

## 1942/3107: FIPS 37171 processed.

## 1943/3107: FIPS 37173 processed.

## 1944/3107: FIPS 37175 processed.

## 1945/3107: FIPS 37177 processed.

## 1946/3107: FIPS 37179 processed.

## 1947/3107: FIPS 37181 processed.

## 1948/3107: FIPS 37183 processed.

## 1949/3107: FIPS 37185 processed.

## 1950/3107: FIPS 37187 processed.

## Checkpoint saved at iteration 1950

## 1951/3107: FIPS 37189 processed.

## 1952/3107: FIPS 37191 processed.

## 1953/3107: FIPS 37193 processed.

## 1954/3107: FIPS 37195 processed.

## 1955/3107: FIPS 37197 processed.

## 1956/3107: FIPS 37199 processed.

## 1957/3107: FIPS 38001 processed.

## 1958/3107: FIPS 38003 processed.

## 1959/3107: FIPS 38005 processed.

## 1960/3107: FIPS 38007 processed.

## 1961/3107: FIPS 38009 processed.

## 1962/3107: FIPS 38011 processed.

## 1963/3107: FIPS 38013 processed.

## 1964/3107: FIPS 38015 processed.

## 1965/3107: FIPS 38017 processed.

## 1966/3107: FIPS 38019 processed.

## 1967/3107: FIPS 38021 processed.

## 1968/3107: FIPS 38023 processed.

## 1969/3107: FIPS 38025 processed.

## 1970/3107: FIPS 38027 processed.

## 1971/3107: FIPS 38029 processed.

## 1972/3107: FIPS 38031 processed.

## 1973/3107: FIPS 38033 processed.

## 1974/3107: FIPS 38035 processed.

## 1975/3107: FIPS 38037 processed.

## 1976/3107: FIPS 38039 processed.

## 1977/3107: FIPS 38041 processed.

## 1978/3107: FIPS 38043 processed.

## 1979/3107: FIPS 38045 processed.

## 1980/3107: FIPS 38047 processed.

## 1981/3107: FIPS 38049 processed.

## 1982/3107: FIPS 38051 processed.

## 1983/3107: FIPS 38053 processed.

## 1984/3107: FIPS 38055 processed.

## 1985/3107: FIPS 38057 processed.

## 1986/3107: FIPS 38059 processed.

## 1987/3107: FIPS 38061 processed.

## 1988/3107: FIPS 38063 processed.

## 1989/3107: FIPS 38065 processed.

## 1990/3107: FIPS 38067 processed.

## 1991/3107: FIPS 38069 processed.

## 1992/3107: FIPS 38071 processed.

## 1993/3107: FIPS 38073 processed.

## 1994/3107: FIPS 38075 processed.

## 1995/3107: FIPS 38077 processed.

## 1996/3107: FIPS 38079 processed.

## 1997/3107: FIPS 38081 processed.

## 1998/3107: FIPS 38083 processed.

## 1999/3107: FIPS 38085 processed.

## 2000/3107: FIPS 38087 processed.

## Checkpoint saved at iteration 2000

## 2001/3107: FIPS 38089 processed.

## 2002/3107: FIPS 38091 processed.

## 2003/3107: FIPS 38093 processed.

## 2004/3107: FIPS 38095 processed.

## 2005/3107: FIPS 38097 processed.

## 2006/3107: FIPS 38099 processed.

## 2007/3107: FIPS 38101 processed.

## 2008/3107: FIPS 38103 processed.

## 2009/3107: FIPS 38105 processed.

## 2010/3107: FIPS 39001 processed.

## 2011/3107: FIPS 39003 processed.

## 2012/3107: FIPS 39005 processed.

## 2013/3107: FIPS 39007 processed.

## 2014/3107: FIPS 39009 processed.

## 2015/3107: FIPS 39011 processed.

## 2016/3107: FIPS 39013 processed.

## 2017/3107: FIPS 39015 processed.



## 2018/3107: FIPS 39017 processed.

## 2019/3107: FIPS 39019 processed.

## 2020/3107: FIPS 39021 processed.

## 2021/3107: FIPS 39023 processed.

## 2022/3107: FIPS 39025 processed.

## 2023/3107: FIPS 39027 processed.

## 2024/3107: FIPS 39029 processed.

## 2025/3107: FIPS 39031 processed.

## 2026/3107: FIPS 39033 processed.

## 2027/3107: FIPS 39035 processed.

## 2028/3107: FIPS 39037 processed.

## 2029/3107: FIPS 39039 processed.

## 2030/3107: FIPS 39041 processed.

## 2031/3107: FIPS 39043 processed.

## 2032/3107: FIPS 39045 processed.

## 2033/3107: FIPS 39047 processed.

## 2034/3107: FIPS 39049 processed.

## 2035/3107: FIPS 39051 processed.

## 2036/3107: FIPS 39053 processed.

## 2037/3107: FIPS 39055 processed.

## 2038/3107: FIPS 39057 processed.

## 2039/3107: FIPS 39059 processed.

## 2040/3107: FIPS 39061 processed.

## 2041/3107: FIPS 39063 processed.

## 2042/3107: FIPS 39065 processed.

## 2043/3107: FIPS 39067 processed.

## 2044/3107: FIPS 39069 processed.

## 2045/3107: FIPS 39071 processed.

## 2046/3107: FIPS 39073 processed.

## 2047/3107: FIPS 39075 processed.

## 2048/3107: FIPS 39077 processed.

## 2049/3107: FIPS 39079 processed.

## 2050/3107: FIPS 39081 processed.

## Checkpoint saved at iteration 2050

## 2051/3107: FIPS 39083 processed.

## 2052/3107: FIPS 39085 processed.

## 2053/3107: FIPS 39087 processed.

## 2054/3107: FIPS 39089 processed.

## 2055/3107: FIPS 39091 processed.

## 2056/3107: FIPS 39093 processed.

## 2057/3107: FIPS 39095 processed.

## 2058/3107: FIPS 39097 processed.

## 2059/3107: FIPS 39099 processed.

## 2060/3107: FIPS 39101 processed.

## 2061/3107: FIPS 39103 processed.

## 2062/3107: FIPS 39105 processed.

## 2063/3107: FIPS 39107 processed.

## 2064/3107: FIPS 39109 processed.

## 2065/3107: FIPS 39111 processed.

## 2066/3107: FIPS 39113 processed.

## 2067/3107: FIPS 39115 processed.

## 2068/3107: FIPS 39117 processed.

## 2069/3107: FIPS 39119 processed.

## 2070/3107: FIPS 39121 processed.

## 2071/3107: FIPS 39123 processed.

## 2072/3107: FIPS 39125 processed.

## 2073/3107: FIPS 39127 processed.

## 2074/3107: FIPS 39129 processed.

## 2075/3107: FIPS 39131 processed.

## 2076/3107: FIPS 39133 processed.

## 2077/3107: FIPS 39135 processed.

## 2078/3107: FIPS 39137 processed.

## 2079/3107: FIPS 39139 processed.

## 2080/3107: FIPS 39141 processed.

## 2081/3107: FIPS 39143 processed.

## 2082/3107: FIPS 39145 processed.

## 2083/3107: FIPS 39147 processed.

## 2084/3107: FIPS 39149 processed.

## 2085/3107: FIPS 39151 processed.

## 2086/3107: FIPS 39153 processed.

## 2087/3107: FIPS 39155 processed.

## 2088/3107: FIPS 39157 processed.

## 2089/3107: FIPS 39159 processed.

## 2090/3107: FIPS 39161 processed.

## 2091/3107: FIPS 39163 processed.

## 2092/3107: FIPS 39165 processed.

## 2093/3107: FIPS 39167 processed.

## 2094/3107: FIPS 39169 processed.

## 2095/3107: FIPS 39171 processed.

## 2096/3107: FIPS 39173 processed.

## 2097/3107: FIPS 39175 processed.

## 2098/3107: FIPS 40001 processed.

## 2099/3107: FIPS 40003 processed.

## 2100/3107: FIPS 40005 processed.

## Checkpoint saved at iteration 2100

## 2101/3107: FIPS 40007 processed.

## 2102/3107: FIPS 40009 processed.

## 2103/3107: FIPS 40011 processed.

## 2104/3107: FIPS 40013 processed.

## 2105/3107: FIPS 40015 processed.

## 2106/3107: FIPS 40017 processed.

## 2107/3107: FIPS 40019 processed.

## 2108/3107: FIPS 40021 processed.

## 2109/3107: FIPS 40023 processed.

## 2110/3107: FIPS 40025 processed.

## 2111/3107: FIPS 40027 processed.

## 2112/3107: FIPS 40029 processed.

## 2113/3107: FIPS 40031 processed.

## 2114/3107: FIPS 40033 processed.

## 2115/3107: FIPS 40035 processed.

## 2116/3107: FIPS 40037 processed.

## 2117/3107: FIPS 40039 processed.

## 2118/3107: FIPS 40041 processed.

## 2119/3107: FIPS 40043 processed.

## 2120/3107: FIPS 40045 processed.

## 2121/3107: FIPS 40047 processed.

## 2122/3107: FIPS 40049 processed.

## 2123/3107: FIPS 40051 processed.

## 2124/3107: FIPS 40053 processed.

## 2125/3107: FIPS 40055 processed.

## 2126/3107: FIPS 40057 processed.

## 2127/3107: FIPS 40059 processed.

## 2128/3107: FIPS 40061 processed.

## 2129/3107: FIPS 40063 processed.

## 2130/3107: FIPS 40065 processed.

## 2131/3107: FIPS 40067 processed.

## 2132/3107: FIPS 40069 processed.

## 2133/3107: FIPS 40071 processed.

## 2134/3107: FIPS 40073 processed.

## 2135/3107: FIPS 40075 processed.

## 2136/3107: FIPS 40077 processed.

## 2137/3107: FIPS 40079 processed.

## 2138/3107: FIPS 40081 processed.

## 2139/3107: FIPS 40083 processed.

## 2140/3107: FIPS 40085 processed.

## 2141/3107: FIPS 40087 processed.

## 2142/3107: FIPS 40089 processed.

## 2143/3107: FIPS 40091 processed.

## 2144/3107: FIPS 40093 processed.

## 2145/3107: FIPS 40095 processed.

## 2146/3107: FIPS 40097 processed.

## 2147/3107: FIPS 40099 processed.

## 2148/3107: FIPS 40101 processed.

## 2149/3107: FIPS 40103 processed.

## 2150/3107: FIPS 40105 processed.

## Checkpoint saved at iteration 2150

## 2151/3107: FIPS 40107 processed.

## 2152/3107: FIPS 40109 processed.

## 2153/3107: FIPS 40111 processed.

## 2154/3107: FIPS 40113 processed.

## 2155/3107: FIPS 40115 processed.

## 2156/3107: FIPS 40117 processed.

## 2157/3107: FIPS 40119 processed.

## 2158/3107: FIPS 40121 processed.

## 2159/3107: FIPS 40123 processed.

## 2160/3107: FIPS 40125 processed.

## 2161/3107: FIPS 40127 processed.

## 2162/3107: FIPS 40129 processed.

## 2163/3107: FIPS 40131 processed.

## 2164/3107: FIPS 40133 processed.

## 2165/3107: FIPS 40135 processed.

## 2166/3107: FIPS 40137 processed.

## 2167/3107: FIPS 40139 processed.

## 2168/3107: FIPS 40141 processed.

## 2169/3107: FIPS 40143 processed.

## 2170/3107: FIPS 40145 processed.

## 2171/3107: FIPS 40147 processed.

## 2172/3107: FIPS 40149 processed.

## 2173/3107: FIPS 40151 processed.

## 2174/3107: FIPS 40153 processed.

## 2175/3107: FIPS 41001 processed.

## 2176/3107: FIPS 41003 processed.

## 2177/3107: FIPS 41005 processed.

## 2178/3107: FIPS 41007 processed.

## 2179/3107: FIPS 41009 processed.

## 2180/3107: FIPS 41011 processed.

## 2181/3107: FIPS 41013 processed.

## 2182/3107: FIPS 41015 processed.

## 2183/3107: FIPS 41017 processed.

## 2184/3107: FIPS 41019 processed.

## 2185/3107: FIPS 41021 processed.

## 2186/3107: FIPS 41023 processed.

## 2187/3107: FIPS 41025 processed.

## 2188/3107: FIPS 41027 processed.

## 2189/3107: FIPS 41029 processed.

## 2190/3107: FIPS 41031 processed.



## 2191/3107: FIPS 41033 processed.

## 2192/3107: FIPS 41035 processed.

## 2193/3107: FIPS 41037 processed.

## 2194/3107: FIPS 41039 processed.

## 2195/3107: FIPS 41041 processed.

## 2196/3107: FIPS 41043 processed.

## 2197/3107: FIPS 41045 processed.

## 2198/3107: FIPS 41047 processed.

## 2199/3107: FIPS 41049 processed.

## 2200/3107: FIPS 41051 processed.

## Checkpoint saved at iteration 2200

## 2201/3107: FIPS 41053 processed.

## 2202/3107: FIPS 41055 processed.

## 2203/3107: FIPS 41057 processed.

## 2204/3107: FIPS 41059 processed.

## 2205/3107: FIPS 41061 processed.

## 2206/3107: FIPS 41063 processed.

## 2207/3107: FIPS 41065 processed.

## 2208/3107: FIPS 41067 processed.

## 2209/3107: FIPS 41069 processed.

## 2210/3107: FIPS 41071 processed.

## 2211/3107: FIPS 42001 processed.

## 2212/3107: FIPS 42003 processed.

## 2213/3107: FIPS 42005 processed.

## 2214/3107: FIPS 42007 processed.

## 2215/3107: FIPS 42009 processed.

## 2216/3107: FIPS 42011 processed.

## 2217/3107: FIPS 42013 processed.

## 2218/3107: FIPS 42015 processed.

## 2219/3107: FIPS 42017 processed.

## 2220/3107: FIPS 42019 processed.

## 2221/3107: FIPS 42021 processed.

## 2222/3107: FIPS 42023 processed.

## 2223/3107: FIPS 42025 processed.

## 2224/3107: FIPS 42027 processed.

## 2225/3107: FIPS 42029 processed.

## 2226/3107: FIPS 42031 processed.

## 2227/3107: FIPS 42033 processed.

## 2228/3107: FIPS 42035 processed.

## 2229/3107: FIPS 42037 processed.

## 2230/3107: FIPS 42039 processed.

## 2231/3107: FIPS 42041 processed.

## 2232/3107: FIPS 42043 processed.

## 2233/3107: FIPS 42045 processed.

## 2234/3107: FIPS 42047 processed.

## 2235/3107: FIPS 42049 processed.

## 2236/3107: FIPS 42051 processed.

## 2237/3107: FIPS 42053 processed.

## 2238/3107: FIPS 42055 processed.

## 2239/3107: FIPS 42057 processed.

## 2240/3107: FIPS 42059 processed.

## 2241/3107: FIPS 42061 processed.

## 2242/3107: FIPS 42063 processed.

## 2243/3107: FIPS 42065 processed.

## 2244/3107: FIPS 42067 processed.

## 2245/3107: FIPS 42069 processed.

## 2246/3107: FIPS 42071 processed.

## 2247/3107: FIPS 42073 processed.

## 2248/3107: FIPS 42075 processed.

## 2249/3107: FIPS 42077 processed.

## 2250/3107: FIPS 42079 processed.

## Checkpoint saved at iteration 2250

## 2251/3107: FIPS 42081 processed.

## 2252/3107: FIPS 42083 processed.

## 2253/3107: FIPS 42085 processed.

## 2254/3107: FIPS 42087 processed.

## 2255/3107: FIPS 42089 processed.

## 2256/3107: FIPS 42091 processed.

## 2257/3107: FIPS 42093 processed.

## 2258/3107: FIPS 42095 processed.

## 2259/3107: FIPS 42097 processed.

## 2260/3107: FIPS 42099 processed.

## 2261/3107: FIPS 42101 processed.

## 2262/3107: FIPS 42103 processed.

## 2263/3107: FIPS 42105 processed.

## 2264/3107: FIPS 42107 processed.

## 2265/3107: FIPS 42109 processed.

## 2266/3107: FIPS 42111 processed.

## 2267/3107: FIPS 42113 processed.

## 2268/3107: FIPS 42115 processed.

## 2269/3107: FIPS 42117 processed.

## 2270/3107: FIPS 42119 processed.

## 2271/3107: FIPS 42121 processed.

## 2272/3107: FIPS 42123 processed.

## 2273/3107: FIPS 42125 processed.

## 2274/3107: FIPS 42127 processed.

## 2275/3107: FIPS 42129 processed.

## 2276/3107: FIPS 42131 processed.

## 2277/3107: FIPS 42133 processed.

## 2278/3107: FIPS 44001 processed.

## 2279/3107: FIPS 44003 processed.

## 2280/3107: FIPS 44005 processed.

## 2281/3107: FIPS 44007 processed.

## 2282/3107: FIPS 44009 processed.

## 2283/3107: FIPS 45001 processed.

## 2284/3107: FIPS 45003 processed.

## 2285/3107: FIPS 45005 processed.

## 2286/3107: FIPS 45007 processed.

## 2287/3107: FIPS 45009 processed.

## 2288/3107: FIPS 45011 processed.

## 2289/3107: FIPS 45013 processed.

## 2290/3107: FIPS 45015 processed.

## 2291/3107: FIPS 45017 processed.

## 2292/3107: FIPS 45019 processed.

## 2293/3107: FIPS 45021 processed.

## 2294/3107: FIPS 45023 processed.

## 2295/3107: FIPS 45025 processed.

## 2296/3107: FIPS 45027 processed.

## 2297/3107: FIPS 45029 processed.

## 2298/3107: FIPS 45031 processed.

## 2299/3107: FIPS 45033 processed.

## 2300/3107: FIPS 45035 processed.

## Checkpoint saved at iteration 2300

## 2301/3107: FIPS 45037 processed.

## 2302/3107: FIPS 45039 processed.

## 2303/3107: FIPS 45041 processed.

## 2304/3107: FIPS 45043 processed.

## 2305/3107: FIPS 45045 processed.

## 2306/3107: FIPS 45047 processed.

## 2307/3107: FIPS 45049 processed.

## 2308/3107: FIPS 45051 processed.

## 2309/3107: FIPS 45053 processed.

## 2310/3107: FIPS 45055 processed.

## 2311/3107: FIPS 45057 processed.

## 2312/3107: FIPS 45059 processed.

## 2313/3107: FIPS 45061 processed.

## 2314/3107: FIPS 45063 processed.

## 2315/3107: FIPS 45065 processed.

## 2316/3107: FIPS 45067 processed.

## 2317/3107: FIPS 45069 processed.

## 2318/3107: FIPS 45071 processed.

## 2319/3107: FIPS 45073 processed.

## 2320/3107: FIPS 45075 processed.

## 2321/3107: FIPS 45077 processed.

## 2322/3107: FIPS 45079 processed.

## 2323/3107: FIPS 45081 processed.

## 2324/3107: FIPS 45083 processed.

## 2325/3107: FIPS 45085 processed.

## 2326/3107: FIPS 45087 processed.

## 2327/3107: FIPS 45089 processed.

## 2328/3107: FIPS 45091 processed.

## 2329/3107: FIPS 46003 processed.

## 2330/3107: FIPS 46005 processed.

## 2331/3107: FIPS 46007 processed.

## 2332/3107: FIPS 46009 processed.

## 2333/3107: FIPS 46011 processed.

## 2334/3107: FIPS 46013 processed.

## 2335/3107: FIPS 46015 processed.

## 2336/3107: FIPS 46017 processed.

## 2337/3107: FIPS 46019 processed.

## 2338/3107: FIPS 46021 processed.

## 2339/3107: FIPS 46023 processed.

## 2340/3107: FIPS 46025 processed.

## 2341/3107: FIPS 46027 processed.

## 2342/3107: FIPS 46029 processed.

## 2343/3107: FIPS 46031 processed.

## 2344/3107: FIPS 46033 processed.

## 2345/3107: FIPS 46035 processed.

## 2346/3107: FIPS 46037 processed.

## 2347/3107: FIPS 46039 processed.

## 2348/3107: FIPS 46041 processed.

## 2349/3107: FIPS 46043 processed.

## 2350/3107: FIPS 46045 processed.

## Checkpoint saved at iteration 2350

## 2351/3107: FIPS 46047 processed.

## 2352/3107: FIPS 46049 processed.

## 2353/3107: FIPS 46051 processed.

## 2354/3107: FIPS 46053 processed.

## 2355/3107: FIPS 46055 processed.

## 2356/3107: FIPS 46057 processed.

## 2357/3107: FIPS 46059 processed.

## 2358/3107: FIPS 46061 processed.

## 2359/3107: FIPS 46063 processed.

## 2360/3107: FIPS 46065 processed.

## 2361/3107: FIPS 46067 processed.

## 2362/3107: FIPS 46069 processed.



## 2363/3107: FIPS 46071 processed.

## 2364/3107: FIPS 46073 processed.

## 2365/3107: FIPS 46075 processed.

## 2366/3107: FIPS 46077 processed.

## 2367/3107: FIPS 46079 processed.

## 2368/3107: FIPS 46081 processed.

## 2369/3107: FIPS 46083 processed.

## 2370/3107: FIPS 46085 processed.

## 2371/3107: FIPS 46087 processed.

## 2372/3107: FIPS 46089 processed.

## 2373/3107: FIPS 46091 processed.

## 2374/3107: FIPS 46093 processed.

## 2375/3107: FIPS 46095 processed.

## 2376/3107: FIPS 46097 processed.

## 2377/3107: FIPS 46099 processed.

## 2378/3107: FIPS 46101 processed.

## 2379/3107: FIPS 46103 processed.

## 2380/3107: FIPS 46105 processed.

## 2381/3107: FIPS 46107 processed.

## 2382/3107: FIPS 46109 processed.

## 2383/3107: FIPS 46111 processed.

## 2384/3107: FIPS 46115 processed.

## 2385/3107: FIPS 46117 processed.

## 2386/3107: FIPS 46119 processed.

## 2387/3107: FIPS 46121 processed.

## 2388/3107: FIPS 46123 processed.

## 2389/3107: FIPS 46125 processed.

## 2390/3107: FIPS 46127 processed.

## 2391/3107: FIPS 46129 processed.

## 2392/3107: FIPS 46135 processed.

## 2393/3107: FIPS 46137 processed.

## 2394/3107: FIPS 47001 processed.

## 2395/3107: FIPS 47003 processed.

## 2396/3107: FIPS 47005 processed.

## 2397/3107: FIPS 47007 processed.

## 2398/3107: FIPS 47009 processed.

## 2399/3107: FIPS 47011 processed.

## 2400/3107: FIPS 47013 processed.

## Checkpoint saved at iteration 2400

## 2401/3107: FIPS 47015 processed.

## 2402/3107: FIPS 47017 processed.

## 2403/3107: FIPS 47019 processed.

## 2404/3107: FIPS 47021 processed.

## 2405/3107: FIPS 47023 processed.

## 2406/3107: FIPS 47025 processed.

## 2407/3107: FIPS 47027 processed.

## 2408/3107: FIPS 47029 processed.

## 2409/3107: FIPS 47031 processed.

## 2410/3107: FIPS 47033 processed.

## 2411/3107: FIPS 47035 processed.

## 2412/3107: FIPS 47037 processed.

## 2413/3107: FIPS 47039 processed.

## 2414/3107: FIPS 47041 processed.

## 2415/3107: FIPS 47043 processed.

## 2416/3107: FIPS 47045 processed.

## 2417/3107: FIPS 47047 processed.

## 2418/3107: FIPS 47049 processed.

## 2419/3107: FIPS 47051 processed.

## 2420/3107: FIPS 47053 processed.

## 2421/3107: FIPS 47055 processed.

## 2422/3107: FIPS 47057 processed.

## 2423/3107: FIPS 47059 processed.

## 2424/3107: FIPS 47061 processed.

## 2425/3107: FIPS 47063 processed.

## 2426/3107: FIPS 47065 processed.

## 2427/3107: FIPS 47067 processed.

## 2428/3107: FIPS 47069 processed.

## 2429/3107: FIPS 47071 processed.

## 2430/3107: FIPS 47073 processed.

## 2431/3107: FIPS 47075 processed.

## 2432/3107: FIPS 47077 processed.

## 2433/3107: FIPS 47079 processed.

## 2434/3107: FIPS 47081 processed.

## 2435/3107: FIPS 47083 processed.

## 2436/3107: FIPS 47085 processed.

## 2437/3107: FIPS 47087 processed.

## 2438/3107: FIPS 47089 processed.

## 2439/3107: FIPS 47091 processed.

## 2440/3107: FIPS 47093 processed.

## 2441/3107: FIPS 47095 processed.

## 2442/3107: FIPS 47097 processed.

## 2443/3107: FIPS 47099 processed.

## 2444/3107: FIPS 47101 processed.

## 2445/3107: FIPS 47103 processed.

## 2446/3107: FIPS 47105 processed.

## 2447/3107: FIPS 47107 processed.

## 2448/3107: FIPS 47109 processed.

## 2449/3107: FIPS 47111 processed.

## 2450/3107: FIPS 47113 processed.

## Checkpoint saved at iteration 2450

## 2451/3107: FIPS 47115 processed.

## 2452/3107: FIPS 47117 processed.

## 2453/3107: FIPS 47119 processed.

## 2454/3107: FIPS 47121 processed.

## 2455/3107: FIPS 47123 processed.

## 2456/3107: FIPS 47125 processed.

## 2457/3107: FIPS 47127 processed.

## 2458/3107: FIPS 47129 processed.

## 2459/3107: FIPS 47131 processed.

## 2460/3107: FIPS 47133 processed.

## 2461/3107: FIPS 47135 processed.

## 2462/3107: FIPS 47137 processed.

## 2463/3107: FIPS 47139 processed.

## 2464/3107: FIPS 47141 processed.

## 2465/3107: FIPS 47143 processed.

## 2466/3107: FIPS 47145 processed.

## 2467/3107: FIPS 47147 processed.

## 2468/3107: FIPS 47149 processed.

## 2469/3107: FIPS 47151 processed.

## 2470/3107: FIPS 47153 processed.

## 2471/3107: FIPS 47155 processed.

## 2472/3107: FIPS 47157 processed.

## 2473/3107: FIPS 47159 processed.

## 2474/3107: FIPS 47161 processed.

## 2475/3107: FIPS 47163 processed.

## 2476/3107: FIPS 47165 processed.

## 2477/3107: FIPS 47167 processed.

## 2478/3107: FIPS 47169 processed.

## 2479/3107: FIPS 47171 processed.

## 2480/3107: FIPS 47173 processed.

## 2481/3107: FIPS 47175 processed.

## 2482/3107: FIPS 47177 processed.

## 2483/3107: FIPS 47179 processed.

## 2484/3107: FIPS 47181 processed.

## 2485/3107: FIPS 47183 processed.

## 2486/3107: FIPS 47185 processed.

## 2487/3107: FIPS 47187 processed.

## 2488/3107: FIPS 47189 processed.

## 2489/3107: FIPS 48001 processed.

## 2490/3107: FIPS 48003 processed.

## 2491/3107: FIPS 48005 processed.

## 2492/3107: FIPS 48007 processed.

## 2493/3107: FIPS 48009 processed.

## 2494/3107: FIPS 48011 processed.

## 2495/3107: FIPS 48013 processed.

## 2496/3107: FIPS 48015 processed.

## 2497/3107: FIPS 48017 processed.

## 2498/3107: FIPS 48019 processed.

## 2499/3107: FIPS 48021 processed.

## 2500/3107: FIPS 48023 processed.

## Checkpoint saved at iteration 2500

## 2501/3107: FIPS 48025 processed.

## 2502/3107: FIPS 48027 processed.

## 2503/3107: FIPS 48029 processed.

## 2504/3107: FIPS 48031 processed.

## 2505/3107: FIPS 48033 processed.

## 2506/3107: FIPS 48035 processed.

## 2507/3107: FIPS 48037 processed.

## 2508/3107: FIPS 48039 processed.

## 2509/3107: FIPS 48041 processed.

## 2510/3107: FIPS 48043 processed.

## 2511/3107: FIPS 48045 processed.

## 2512/3107: FIPS 48047 processed.

## 2513/3107: FIPS 48049 processed.

## 2514/3107: FIPS 48051 processed.

## 2515/3107: FIPS 48053 processed.

## 2516/3107: FIPS 48055 processed.

## 2517/3107: FIPS 48057 processed.

## 2518/3107: FIPS 48059 processed.

## 2519/3107: FIPS 48061 processed.

## 2520/3107: FIPS 48063 processed.

## 2521/3107: FIPS 48065 processed.

## 2522/3107: FIPS 48067 processed.

## 2523/3107: FIPS 48069 processed.

## 2524/3107: FIPS 48071 processed.

## 2525/3107: FIPS 48073 processed.

## 2526/3107: FIPS 48075 processed.

## 2527/3107: FIPS 48077 processed.

## 2528/3107: FIPS 48079 processed.

## 2529/3107: FIPS 48081 processed.

## 2530/3107: FIPS 48083 processed.

## 2531/3107: FIPS 48085 processed.

## 2532/3107: FIPS 48087 processed.

## 2533/3107: FIPS 48089 processed.

## 2534/3107: FIPS 48091 processed.

## 2535/3107: FIPS 48093 processed.



## 2536/3107: FIPS 48095 processed.

## 2537/3107: FIPS 48097 processed.

## 2538/3107: FIPS 48099 processed.

## 2539/3107: FIPS 48101 processed.

## 2540/3107: FIPS 48103 processed.

## 2541/3107: FIPS 48105 processed.

## 2542/3107: FIPS 48107 processed.

## 2543/3107: FIPS 48109 processed.

## 2544/3107: FIPS 48111 processed.

## 2545/3107: FIPS 48113 processed.

## 2546/3107: FIPS 48115 processed.

## 2547/3107: FIPS 48117 processed.

## 2548/3107: FIPS 48119 processed.

## 2549/3107: FIPS 48121 processed.

## 2550/3107: FIPS 48123 processed.

## Checkpoint saved at iteration 2550

## 2551/3107: FIPS 48125 processed.

## 2552/3107: FIPS 48127 processed.

## 2553/3107: FIPS 48129 processed.

## 2554/3107: FIPS 48131 processed.

## 2555/3107: FIPS 48133 processed.

## 2556/3107: FIPS 48135 processed.

## 2557/3107: FIPS 48137 processed.

## 2558/3107: FIPS 48139 processed.

## 2559/3107: FIPS 48141 processed.

## 2560/3107: FIPS 48143 processed.

## 2561/3107: FIPS 48145 processed.

## 2562/3107: FIPS 48147 processed.

## 2563/3107: FIPS 48149 processed.

## 2564/3107: FIPS 48151 processed.

## 2565/3107: FIPS 48153 processed.

## 2566/3107: FIPS 48155 processed.

## 2567/3107: FIPS 48157 processed.

## 2568/3107: FIPS 48159 processed.

## 2569/3107: FIPS 48161 processed.

## 2570/3107: FIPS 48163 processed.

## 2571/3107: FIPS 48165 processed.

## 2572/3107: FIPS 48167 processed.

## 2573/3107: FIPS 48169 processed.

## 2574/3107: FIPS 48171 processed.

## 2575/3107: FIPS 48173 processed.

## 2576/3107: FIPS 48175 processed.

## 2577/3107: FIPS 48177 processed.

## 2578/3107: FIPS 48179 processed.

## 2579/3107: FIPS 48181 processed.

## 2580/3107: FIPS 48183 processed.

## 2581/3107: FIPS 48185 processed.

## 2582/3107: FIPS 48187 processed.

## 2583/3107: FIPS 48189 processed.

## 2584/3107: FIPS 48191 processed.

## 2585/3107: FIPS 48193 processed.

## 2586/3107: FIPS 48195 processed.

## 2587/3107: FIPS 48197 processed.

## 2588/3107: FIPS 48199 processed.

## 2589/3107: FIPS 48201 processed.

## 2590/3107: FIPS 48203 processed.

## 2591/3107: FIPS 48205 processed.

## 2592/3107: FIPS 48207 processed.

## 2593/3107: FIPS 48209 processed.

## 2594/3107: FIPS 48211 processed.

## 2595/3107: FIPS 48213 processed.

## 2596/3107: FIPS 48215 processed.

## 2597/3107: FIPS 48217 processed.

## 2598/3107: FIPS 48219 processed.

## 2599/3107: FIPS 48221 processed.

## 2600/3107: FIPS 48223 processed.

## Checkpoint saved at iteration 2600

## 2601/3107: FIPS 48225 processed.

## 2602/3107: FIPS 48227 processed.

## 2603/3107: FIPS 48229 processed.

## 2604/3107: FIPS 48231 processed.

## 2605/3107: FIPS 48233 processed.

## 2606/3107: FIPS 48235 processed.

## 2607/3107: FIPS 48237 processed.

## 2608/3107: FIPS 48239 processed.

## 2609/3107: FIPS 48241 processed.

## 2610/3107: FIPS 48243 processed.

## 2611/3107: FIPS 48245 processed.

## 2612/3107: FIPS 48247 processed.

## 2613/3107: FIPS 48249 processed.

## 2614/3107: FIPS 48251 processed.

## 2615/3107: FIPS 48253 processed.

## 2616/3107: FIPS 48255 processed.

## 2617/3107: FIPS 48257 processed.

## 2618/3107: FIPS 48259 processed.

## 2619/3107: FIPS 48261 processed.

## 2620/3107: FIPS 48263 processed.

## 2621/3107: FIPS 48265 processed.

## 2622/3107: FIPS 48267 processed.

## 2623/3107: FIPS 48269 processed.

## 2624/3107: FIPS 48271 processed.

## 2625/3107: FIPS 48273 processed.

## 2626/3107: FIPS 48275 processed.

## 2627/3107: FIPS 48277 processed.

## 2628/3107: FIPS 48279 processed.

## 2629/3107: FIPS 48281 processed.

## 2630/3107: FIPS 48283 processed.

## 2631/3107: FIPS 48285 processed.

## 2632/3107: FIPS 48287 processed.

## 2633/3107: FIPS 48289 processed.

## 2634/3107: FIPS 48291 processed.

## 2635/3107: FIPS 48293 processed.

## 2636/3107: FIPS 48295 processed.

## 2637/3107: FIPS 48297 processed.

## 2638/3107: FIPS 48299 processed.

## 2639/3107: FIPS 48301 processed.

## 2640/3107: FIPS 48303 processed.

## 2641/3107: FIPS 48305 processed.

## 2642/3107: FIPS 48307 processed.

## 2643/3107: FIPS 48309 processed.

## 2644/3107: FIPS 48311 processed.

## 2645/3107: FIPS 48313 processed.

## 2646/3107: FIPS 48315 processed.

## 2647/3107: FIPS 48317 processed.

## 2648/3107: FIPS 48319 processed.

## 2649/3107: FIPS 48321 processed.

## 2650/3107: FIPS 48323 processed.

## Checkpoint saved at iteration 2650

## 2651/3107: FIPS 48325 processed.

## 2652/3107: FIPS 48327 processed.

## 2653/3107: FIPS 48329 processed.

## 2654/3107: FIPS 48331 processed.

## 2655/3107: FIPS 48333 processed.

## 2656/3107: FIPS 48335 processed.

## 2657/3107: FIPS 48337 processed.

## 2658/3107: FIPS 48339 processed.

## 2659/3107: FIPS 48341 processed.

## 2660/3107: FIPS 48343 processed.

## 2661/3107: FIPS 48345 processed.

## 2662/3107: FIPS 48347 processed.

## 2663/3107: FIPS 48349 processed.

## 2664/3107: FIPS 48351 processed.

## 2665/3107: FIPS 48353 processed.

## 2666/3107: FIPS 48355 processed.

## 2667/3107: FIPS 48357 processed.

## 2668/3107: FIPS 48359 processed.

## 2669/3107: FIPS 48361 processed.

## 2670/3107: FIPS 48363 processed.

## 2671/3107: FIPS 48365 processed.

## 2672/3107: FIPS 48367 processed.

## 2673/3107: FIPS 48369 processed.

## 2674/3107: FIPS 48371 processed.

## 2675/3107: FIPS 48373 processed.

## 2676/3107: FIPS 48375 processed.

## 2677/3107: FIPS 48377 processed.

## 2678/3107: FIPS 48379 processed.

## 2679/3107: FIPS 48381 processed.

## 2680/3107: FIPS 48383 processed.

## 2681/3107: FIPS 48385 processed.

## 2682/3107: FIPS 48387 processed.

## 2683/3107: FIPS 48389 processed.

## 2684/3107: FIPS 48391 processed.

## 2685/3107: FIPS 48393 processed.

## 2686/3107: FIPS 48395 processed.

## 2687/3107: FIPS 48397 processed.

## 2688/3107: FIPS 48399 processed.

## 2689/3107: FIPS 48401 processed.

## 2690/3107: FIPS 48403 processed.

## 2691/3107: FIPS 48405 processed.

## 2692/3107: FIPS 48407 processed.

## 2693/3107: FIPS 48409 processed.

## 2694/3107: FIPS 48411 processed.

## 2695/3107: FIPS 48413 processed.

## 2696/3107: FIPS 48415 processed.

## 2697/3107: FIPS 48417 processed.

## 2698/3107: FIPS 48419 processed.

## 2699/3107: FIPS 48421 processed.

## 2700/3107: FIPS 48423 processed.

## Checkpoint saved at iteration 2700

## 2701/3107: FIPS 48425 processed.

## 2702/3107: FIPS 48427 processed.

## 2703/3107: FIPS 48429 processed.

## 2704/3107: FIPS 48431 processed.

## 2705/3107: FIPS 48433 processed.

## 2706/3107: FIPS 48435 processed.

## 2707/3107: FIPS 48437 processed.



## 2708/3107: FIPS 48439 processed.  
## 2709/3107: FIPS 48441 processed.  
## 2710/3107: FIPS 48443 processed.  
## 2711/3107: FIPS 48445 processed.  
## 2712/3107: FIPS 48447 processed.  
## 2713/3107: FIPS 48449 processed.  
## 2714/3107: FIPS 48451 processed.  
## 2715/3107: FIPS 48453 processed.  
## 2716/3107: FIPS 48455 processed.  
## 2717/3107: FIPS 48457 processed.  
## 2718/3107: FIPS 48459 processed.  
## 2719/3107: FIPS 48461 processed.  
## 2720/3107: FIPS 48463 processed.  
## 2721/3107: FIPS 48465 processed.  
## 2722/3107: FIPS 48467 processed.  
## 2723/3107: FIPS 48469 processed.  
## 2724/3107: FIPS 48471 processed.  
## 2725/3107: FIPS 48473 processed.  
## 2726/3107: FIPS 48475 processed.  
## 2727/3107: FIPS 48477 processed.  
## 2728/3107: FIPS 48479 processed.  
## 2729/3107: FIPS 48481 processed.

## 2730/3107: FIPS 48483 processed.

## 2731/3107: FIPS 48485 processed.

## 2732/3107: FIPS 48487 processed.

## 2733/3107: FIPS 48489 processed.

## 2734/3107: FIPS 48491 processed.

## 2735/3107: FIPS 48493 processed.

## 2736/3107: FIPS 48495 processed.

## 2737/3107: FIPS 48497 processed.

## 2738/3107: FIPS 48499 processed.

## 2739/3107: FIPS 48501 processed.

## 2740/3107: FIPS 48503 processed.

## 2741/3107: FIPS 48505 processed.

## 2742/3107: FIPS 48507 processed.

## 2743/3107: FIPS 49001 processed.

## 2744/3107: FIPS 49003 processed.

## 2745/3107: FIPS 49005 processed.

## 2746/3107: FIPS 49007 processed.

## 2747/3107: FIPS 49009 processed.

## 2748/3107: FIPS 49011 processed.

## 2749/3107: FIPS 49013 processed.

## 2750/3107: FIPS 49015 processed.

## Checkpoint saved at iteration 2750

## 2751/3107: FIPS 49017 processed.  
## 2752/3107: FIPS 49019 processed.  
## 2753/3107: FIPS 49021 processed.  
## 2754/3107: FIPS 49023 processed.  
## 2755/3107: FIPS 49025 processed.  
## 2756/3107: FIPS 49027 processed.  
## 2757/3107: FIPS 49029 processed.  
## 2758/3107: FIPS 49031 processed.  
## 2759/3107: FIPS 49033 processed.  
## 2760/3107: FIPS 49035 processed.  
## 2761/3107: FIPS 49037 processed.  
## 2762/3107: FIPS 49039 processed.  
## 2763/3107: FIPS 49041 processed.  
## 2764/3107: FIPS 49043 processed.  
## 2765/3107: FIPS 49045 processed.  
## 2766/3107: FIPS 49047 processed.  
## 2767/3107: FIPS 49049 processed.  
## 2768/3107: FIPS 49051 processed.  
## 2769/3107: FIPS 49053 processed.  
## 2770/3107: FIPS 49055 processed.  
## 2771/3107: FIPS 49057 processed.  
## 2772/3107: FIPS 50001 processed.

## 2773/3107: FIPS 50003 processed.

## 2774/3107: FIPS 50005 processed.

## 2775/3107: FIPS 50007 processed.

## 2776/3107: FIPS 50009 processed.

## 2777/3107: FIPS 50011 processed.

## 2778/3107: FIPS 50013 processed.

## 2779/3107: FIPS 50015 processed.

## 2780/3107: FIPS 50017 processed.

## 2781/3107: FIPS 50019 processed.

## 2782/3107: FIPS 50021 processed.

## 2783/3107: FIPS 50023 processed.

## 2784/3107: FIPS 50025 processed.

## 2785/3107: FIPS 50027 processed.

## 2786/3107: FIPS 51001 processed.

## 2787/3107: FIPS 51003 processed.

## 2788/3107: FIPS 51005 processed.

## 2789/3107: FIPS 51007 processed.

## 2790/3107: FIPS 51009 processed.

## 2791/3107: FIPS 51011 processed.

## 2792/3107: FIPS 51013 processed.

## 2793/3107: FIPS 51015 processed.

## 2794/3107: FIPS 51017 processed.

## 2795/3107: FIPS 51019 processed.

## 2796/3107: FIPS 51021 processed.

## 2797/3107: FIPS 51023 processed.

## 2798/3107: FIPS 51025 processed.

## 2799/3107: FIPS 51027 processed.

## 2800/3107: FIPS 51029 processed.

## Checkpoint saved at iteration 2800

## 2801/3107: FIPS 51031 processed.

## 2802/3107: FIPS 51033 processed.

## 2803/3107: FIPS 51035 processed.

## 2804/3107: FIPS 51036 processed.

## 2805/3107: FIPS 51037 processed.

## 2806/3107: FIPS 51041 processed.

## 2807/3107: FIPS 51043 processed.

## 2808/3107: FIPS 51045 processed.

## 2809/3107: FIPS 51047 processed.

## 2810/3107: FIPS 51049 processed.

## 2811/3107: FIPS 51051 processed.

## 2812/3107: FIPS 51053 processed.

## 2813/3107: FIPS 51057 processed.

## 2814/3107: FIPS 51059 processed.

## 2815/3107: FIPS 51061 processed.

## 2816/3107: FIPS 51063 processed.

## 2817/3107: FIPS 51065 processed.

## 2818/3107: FIPS 51067 processed.

## 2819/3107: FIPS 51069 processed.

## 2820/3107: FIPS 51071 processed.

## 2821/3107: FIPS 51073 processed.

## 2822/3107: FIPS 51075 processed.

## 2823/3107: FIPS 51077 processed.

## 2824/3107: FIPS 51079 processed.

## 2825/3107: FIPS 51081 processed.

## 2826/3107: FIPS 51083 processed.

## 2827/3107: FIPS 51085 processed.

## 2828/3107: FIPS 51087 processed.

## 2829/3107: FIPS 51089 processed.

## 2830/3107: FIPS 51091 processed.

## 2831/3107: FIPS 51093 processed.

## 2832/3107: FIPS 51095 processed.

## 2833/3107: FIPS 51097 processed.

## 2834/3107: FIPS 51099 processed.

## 2835/3107: FIPS 51101 processed.

## 2836/3107: FIPS 51103 processed.

## 2837/3107: FIPS 51105 processed.

## 2838/3107: FIPS 51107 processed.

## 2839/3107: FIPS 51109 processed.

## 2840/3107: FIPS 51111 processed.

## 2841/3107: FIPS 51113 processed.

## 2842/3107: FIPS 51115 processed.

## 2843/3107: FIPS 51117 processed.

## 2844/3107: FIPS 51119 processed.

## 2845/3107: FIPS 51121 processed.

## 2846/3107: FIPS 51125 processed.

## 2847/3107: FIPS 51127 processed.

## 2848/3107: FIPS 51131 processed.

## 2849/3107: FIPS 51133 processed.

## 2850/3107: FIPS 51135 processed.

## Checkpoint saved at iteration 2850

## 2851/3107: FIPS 51137 processed.

## 2852/3107: FIPS 51139 processed.

## 2853/3107: FIPS 51141 processed.

## 2854/3107: FIPS 51143 processed.

## 2855/3107: FIPS 51145 processed.

## 2856/3107: FIPS 51147 processed.

## 2857/3107: FIPS 51149 processed.

## 2858/3107: FIPS 51153 processed.

## 2859/3107: FIPS 51155 processed.

## 2860/3107: FIPS 51157 processed.

## 2861/3107: FIPS 51159 processed.

## 2862/3107: FIPS 51161 processed.

## 2863/3107: FIPS 51163 processed.

## 2864/3107: FIPS 51165 processed.

## 2865/3107: FIPS 51167 processed.

## 2866/3107: FIPS 51169 processed.

## 2867/3107: FIPS 51171 processed.

## 2868/3107: FIPS 51173 processed.

## 2869/3107: FIPS 51175 processed.

## 2870/3107: FIPS 51177 processed.

## 2871/3107: FIPS 51179 processed.

## 2872/3107: FIPS 51181 processed.

## 2873/3107: FIPS 51183 processed.

## 2874/3107: FIPS 51185 processed.

## 2875/3107: FIPS 51187 processed.

## 2876/3107: FIPS 51191 processed.

## 2877/3107: FIPS 51193 processed.

## 2878/3107: FIPS 51195 processed.

## 2879/3107: FIPS 51197 processed.

## 2880/3107: FIPS 51199 processed.



## 2881/3107: FIPS 51510 processed.

## 2882/3107: FIPS 51520 processed.

## 2883/3107: FIPS 51530 processed.

## 2884/3107: FIPS 51540 processed.

## 2885/3107: FIPS 51550 processed.

## 2886/3107: FIPS 51570 processed.

## 2887/3107: FIPS 51580 processed.

## 2888/3107: FIPS 51590 processed.

## 2889/3107: FIPS 51595 processed.

## 2890/3107: FIPS 51600 processed.

## 2891/3107: FIPS 51610 processed.

## 2892/3107: FIPS 51620 processed.

## 2893/3107: FIPS 51630 processed.

## 2894/3107: FIPS 51640 processed.

## 2895/3107: FIPS 51650 processed.

## 2896/3107: FIPS 51660 processed.

## 2897/3107: FIPS 51670 processed.

## 2898/3107: FIPS 51678 processed.

## 2899/3107: FIPS 51680 processed.

## 2900/3107: FIPS 51683 processed.

## Checkpoint saved at iteration 2900

## 2901/3107: FIPS 51685 processed.

## 2902/3107: FIPS 51690 processed.

## 2903/3107: FIPS 51700 processed.

## 2904/3107: FIPS 51710 processed.

## 2905/3107: FIPS 51720 processed.

## 2906/3107: FIPS 51730 processed.

## 2907/3107: FIPS 51735 processed.

## 2908/3107: FIPS 51740 processed.

## 2909/3107: FIPS 51750 processed.

## 2910/3107: FIPS 51760 processed.

## 2911/3107: FIPS 51770 processed.

## 2912/3107: FIPS 51775 processed.

## 2913/3107: FIPS 51790 processed.

## 2914/3107: FIPS 51800 processed.

## 2915/3107: FIPS 51810 processed.

## 2916/3107: FIPS 51820 processed.

## 2917/3107: FIPS 51830 processed.

## 2918/3107: FIPS 51840 processed.

## 2919/3107: FIPS 53001 processed.

## 2920/3107: FIPS 53003 processed.

## 2921/3107: FIPS 53005 processed.

## 2922/3107: FIPS 53007 processed.

## 2923/3107: FIPS 53009 processed.

## 2924/3107: FIPS 53011 processed.

## 2925/3107: FIPS 53013 processed.

## 2926/3107: FIPS 53015 processed.

## 2927/3107: FIPS 53017 processed.

## 2928/3107: FIPS 53019 processed.

## 2929/3107: FIPS 53021 processed.

## 2930/3107: FIPS 53023 processed.

## 2931/3107: FIPS 53025 processed.

## 2932/3107: FIPS 53027 processed.

## 2933/3107: FIPS 53029 processed.

## 2934/3107: FIPS 53031 processed.

## 2935/3107: FIPS 53033 processed.

## 2936/3107: FIPS 53035 processed.

## 2937/3107: FIPS 53037 processed.

## 2938/3107: FIPS 53039 processed.

## 2939/3107: FIPS 53041 processed.

## 2940/3107: FIPS 53043 processed.

## 2941/3107: FIPS 53045 processed.

## 2942/3107: FIPS 53047 processed.

## 2943/3107: FIPS 53049 processed.

## 2944/3107: FIPS 53051 processed.

## 2945/3107: FIPS 53053 processed.

## 2946/3107: FIPS 53055 processed.

## 2947/3107: FIPS 53057 processed.

## 2948/3107: FIPS 53059 processed.

## 2949/3107: FIPS 53061 processed.

## 2950/3107: FIPS 53063 processed.

## Checkpoint saved at iteration 2950

## 2951/3107: FIPS 53065 processed.

## 2952/3107: FIPS 53067 processed.

## 2953/3107: FIPS 53069 processed.

## 2954/3107: FIPS 53071 processed.

## 2955/3107: FIPS 53073 processed.

## 2956/3107: FIPS 53075 processed.

## 2957/3107: FIPS 53077 processed.

## 2958/3107: FIPS 54001 processed.

## 2959/3107: FIPS 54003 processed.

## 2960/3107: FIPS 54005 processed.

## 2961/3107: FIPS 54007 processed.

## 2962/3107: FIPS 54009 processed.

## 2963/3107: FIPS 54011 processed.

## 2964/3107: FIPS 54013 processed.

## 2965/3107: FIPS 54015 processed.

## 2966/3107: FIPS 54017 processed.

## 2967/3107: FIPS 54019 processed.

## 2968/3107: FIPS 54021 processed.

## 2969/3107: FIPS 54023 processed.

## 2970/3107: FIPS 54025 processed.

## 2971/3107: FIPS 54027 processed.

## 2972/3107: FIPS 54029 processed.

## 2973/3107: FIPS 54031 processed.

## 2974/3107: FIPS 54033 processed.

## 2975/3107: FIPS 54035 processed.

## 2976/3107: FIPS 54037 processed.

## 2977/3107: FIPS 54039 processed.

## 2978/3107: FIPS 54041 processed.

## 2979/3107: FIPS 54043 processed.

## 2980/3107: FIPS 54045 processed.

## 2981/3107: FIPS 54047 processed.

## 2982/3107: FIPS 54049 processed.

## 2983/3107: FIPS 54051 processed.

## 2984/3107: FIPS 54053 processed.

## 2985/3107: FIPS 54055 processed.

## 2986/3107: FIPS 54057 processed.

## 2987/3107: FIPS 54059 processed.

## 2988/3107: FIPS 54061 processed.

## 2989/3107: FIPS 54063 processed.

## 2990/3107: FIPS 54065 processed.

## 2991/3107: FIPS 54067 processed.

## 2992/3107: FIPS 54069 processed.

## 2993/3107: FIPS 54071 processed.

## 2994/3107: FIPS 54073 processed.

## 2995/3107: FIPS 54075 processed.

## 2996/3107: FIPS 54077 processed.

## 2997/3107: FIPS 54079 processed.

## 2998/3107: FIPS 54081 processed.

## 2999/3107: FIPS 54083 processed.

## 3000/3107: FIPS 54085 processed.

## Checkpoint saved at iteration 3000

## 3001/3107: FIPS 54087 processed.

## 3002/3107: FIPS 54089 processed.

## 3003/3107: FIPS 54091 processed.

## 3004/3107: FIPS 54093 processed.

## 3005/3107: FIPS 54095 processed.

## 3006/3107: FIPS 54097 processed.

## 3007/3107: FIPS 54099 processed.

## 3008/3107: FIPS 54101 processed.

## 3009/3107: FIPS 54103 processed.

## 3010/3107: FIPS 54105 processed.

## 3011/3107: FIPS 54107 processed.

## 3012/3107: FIPS 54109 processed.

## 3013/3107: FIPS 55001 processed.

## 3014/3107: FIPS 55003 processed.

## 3015/3107: FIPS 55005 processed.

## 3016/3107: FIPS 55007 processed.

## 3017/3107: FIPS 55009 processed.

## 3018/3107: FIPS 55011 processed.

## 3019/3107: FIPS 55013 processed.

## 3020/3107: FIPS 55015 processed.

## 3021/3107: FIPS 55017 processed.

## 3022/3107: FIPS 55019 processed.

## 3023/3107: FIPS 55021 processed.

## 3024/3107: FIPS 55023 processed.

## 3025/3107: FIPS 55025 processed.

## 3026/3107: FIPS 55027 processed.

## 3027/3107: FIPS 55029 processed.

## 3028/3107: FIPS 55031 processed.

## 3029/3107: FIPS 55033 processed.

## 3030/3107: FIPS 55035 processed.

## 3031/3107: FIPS 55037 processed.

## 3032/3107: FIPS 55039 processed.

## 3033/3107: FIPS 55041 processed.

## 3034/3107: FIPS 55043 processed.

## 3035/3107: FIPS 55045 processed.

## 3036/3107: FIPS 55047 processed.

## 3037/3107: FIPS 55049 processed.

## 3038/3107: FIPS 55051 processed.

## 3039/3107: FIPS 55053 processed.

## 3040/3107: FIPS 55055 processed.

## 3041/3107: FIPS 55057 processed.

## 3042/3107: FIPS 55059 processed.

## 3043/3107: FIPS 55061 processed.

## 3044/3107: FIPS 55063 processed.

## 3045/3107: FIPS 55065 processed.

## 3046/3107: FIPS 55067 processed.

## 3047/3107: FIPS 55069 processed.

## 3048/3107: FIPS 55071 processed.

## 3049/3107: FIPS 55073 processed.

## 3050/3107: FIPS 55075 processed.

## Checkpoint saved at iteration 3050

## 3051/3107: FIPS 55077 processed.

## 3052/3107: FIPS 55078 processed.



## 3053/3107: FIPS 55079 processed.

## 3054/3107: FIPS 55081 processed.

## 3055/3107: FIPS 55083 processed.

## 3056/3107: FIPS 55085 processed.

## 3057/3107: FIPS 55087 processed.

## 3058/3107: FIPS 55089 processed.

## 3059/3107: FIPS 55091 processed.

## 3060/3107: FIPS 55093 processed.

## 3061/3107: FIPS 55095 processed.

## 3062/3107: FIPS 55097 processed.

## 3063/3107: FIPS 55099 processed.

## 3064/3107: FIPS 55101 processed.

## 3065/3107: FIPS 55103 processed.

## 3066/3107: FIPS 55105 processed.

## 3067/3107: FIPS 55107 processed.

## 3068/3107: FIPS 55109 processed.

## 3069/3107: FIPS 55111 processed.

## 3070/3107: FIPS 55113 processed.

## 3071/3107: FIPS 55115 processed.

## 3072/3107: FIPS 55117 processed.

## 3073/3107: FIPS 55119 processed.

## 3074/3107: FIPS 55121 processed.

## 3075/3107: FIPS 55123 processed.

## 3076/3107: FIPS 55125 processed.

## 3077/3107: FIPS 55127 processed.

## 3078/3107: FIPS 55129 processed.

## 3079/3107: FIPS 55131 processed.

## 3080/3107: FIPS 55133 processed.

## 3081/3107: FIPS 55135 processed.

## 3082/3107: FIPS 55137 processed.

## 3083/3107: FIPS 55139 processed.

## 3084/3107: FIPS 55141 processed.

## 3085/3107: FIPS 56001 processed.

## 3086/3107: FIPS 56003 processed.

## 3087/3107: FIPS 56005 processed.

## 3088/3107: FIPS 56007 processed.

## 3089/3107: FIPS 56009 processed.

## 3090/3107: FIPS 56011 processed.

## 3091/3107: FIPS 56013 processed.

## 3092/3107: FIPS 56015 processed.

## 3093/3107: FIPS 56017 processed.

## 3094/3107: FIPS 56019 processed.

## 3095/3107: FIPS 56021 processed.

## 3096/3107: FIPS 56023 processed.

```
## 3097/3107: FIPS 56025 processed.

## 3098/3107: FIPS 56027 processed.

## 3099/3107: FIPS 56029 processed.

## 3100/3107: FIPS 56031 processed.

## Checkpoint saved at iteration 3100

## 3101/3107: FIPS 56033 processed.

## 3102/3107: FIPS 56035 processed.

## 3103/3107: FIPS 56037 processed.

## 3104/3107: FIPS 56039 processed.

## 3105/3107: FIPS 56041 processed.

## 3106/3107: FIPS 56043 processed.

## 3107/3107: FIPS 56045 processed.

## Checkpoint saved at iteration 3107
```

```
# Save and export results
df_causal_arima <- bind_rows(results_list)
message("All counties processed. Final results saved.")
```

```
## All counties processed. Final results saved.
```

Export df\_causal\_arima

```
write_csv(df_causal_arima, "~/df_causal_arima.csv")
```

Read df\_causal\_arima

```
df_causal_arima <- read_csv("df_causal_arima.csv")

## Rows: 3107 Columns: 12
## -- Column specification -----
## Delimiter: ","
## chr (1): fips
## dbl (11): avg_effect, cum_effect, rel_cum_effect, sd_norm, ci_lower_norm, ci...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Create df\_map by adding polygons to df\_causal\_arima

```
# Join shapefile with df_causal_arima
df_map <- counties_sf %>%
  inner_join(df_causal_arima, by = "fips")
```

Filter for significant counties ( $p\_value\_boot < 0.05$ )

```
df_map_sig <- df_map %>%
  filter(p_value_boot < 0.05)
```

Plot choropleth map of U.S. with cumulative effect at each county, sequential

```
# Define sequential color palette
seq_colors <- c("#021B40", "#1F78B4", "#D6ECF5")

# Compute global limits for cum_effect
global_min <- min(df_map_sig$cum_effect, na.rm = TRUE)
global_max <- max(df_map_sig$cum_effect, na.rm = TRUE)

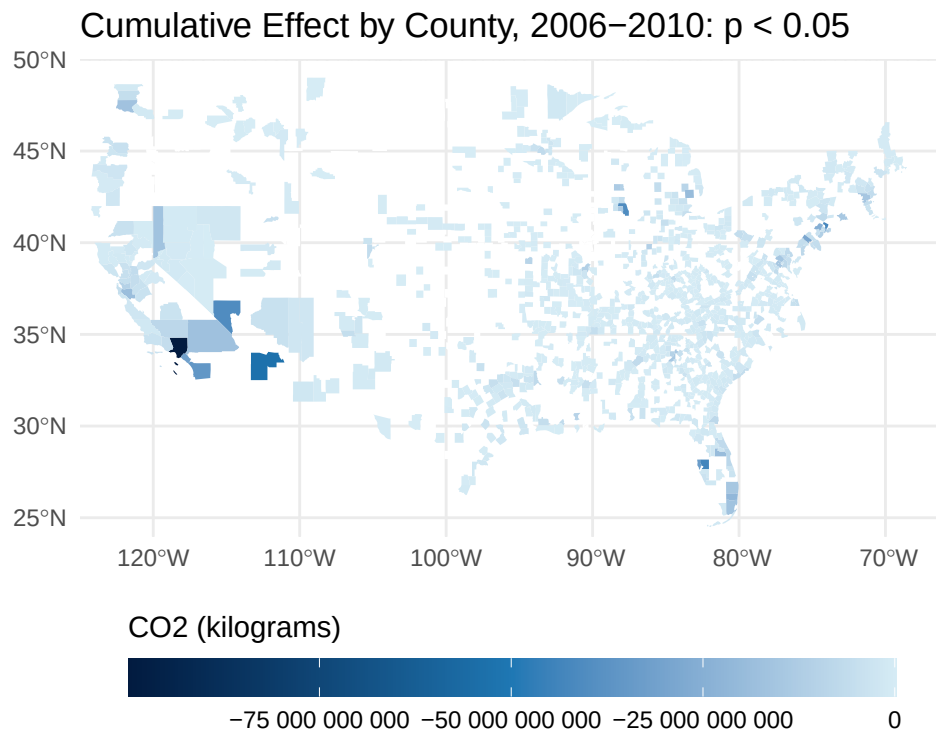
# Create graph of cumulative effects
sig_cum_effect_map <- ggplot(data = df_map_sig) +
  geom_sf(aes(fill = cum_effect), color = NA) +
  scale_fill_gradientn(colors = seq_colors,
    na.value = "white",
    limits = c(global_min, 0),
    labels = label_number(accuracy = 1),
    guide = guide_colorbar(barwidth = 20, # 50
```

```

        barheight = 1,
        title.position = "top",
        title.vjust = 1)) +
labs(title = glue("Cumulative Effect by County, 2006–2010: p < 0.05"),
     fill = "CO2 (kilograms)") +
theme_minimal() +
# theme(legend.position = "bottom",
#       plot.title = element_text(size = 50),
#       axis.text = element_text(size = 40),
#       legend.title = element_text(size = 50),
#       legend.text = element_text(size = 45)) +
theme(legend.position = "bottom") +
coord_sf(xlim = c(-125, -66), ylim = c(24, 50), expand = FALSE)

# Plot and save graph
sig_cum_effect_map

```



```

ggsave("sig_cum_effect_map.png",
       plot = sig_cum_effect_map,
       width = 20, height = 18, dpi = 300,
       bg = "white")

```

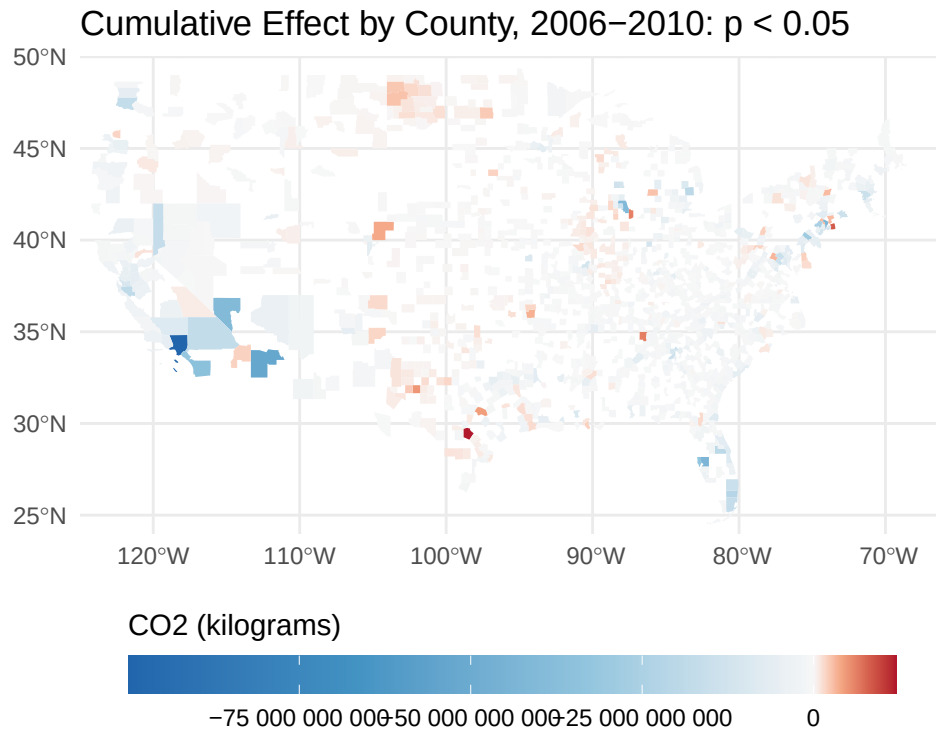
Plot choropleth map of U.S. with cumulative effect at each county, diverging

```
# Define diverging color palette
div_colors <- c("#2166AC", "#4393C3", "#92C5DE",
               "#F7F7F7",
               "#F4A582", "#D6604D", "#B2182B")

# Compute global limits for cum_effect
global_min <- min(df_map_sig$cum_effect, na.rm = TRUE)
global_max <- max(df_map_sig$cum_effect, na.rm = TRUE)

# Create graph of cumulative effects
sig_cum_effect_map <- ggplot(data = df_map_sig) +
  geom_sf(aes(fill = cum_effect), color = NA) +
  scale_fill_gradientn(colors = div_colors,
                      na.value = "white",
                      limits = c(global_min, global_max),
                      values = scales::rescale(c(global_min, 0, global_max)),
                      labels = label_number(accuracy = 1),
                      guide = guide_colorbar(barwidth = 20, # 50
                                             barheight = 1,
                                             title.position = "top",
                                             title.vjust = 1)) +
  labs(title = glue("Cumulative Effect by County, 2006-2010: p < 0.05"),
       fill = "CO2 (kilograms)") +
  theme_minimal() +
  # theme(legend.position = "bottom",
  #       plot.title = element_text(size = 50),
  #       axis.text = element_text(size = 40),
  #       legend.title = element_text(size = 50),
  #       legend.text = element_text(size = 45)) +
  theme(legend.position = "bottom") +
  coord_sf(xlim = c(-125, -66), ylim = c(24, 50), expand = FALSE)

# Plot and save graph
sig_cum_effect_map
```



```
ggsave("sig_cum_effect_map.png",
  plot = sig_cum_effect_map,
  width = 20, height = 18, dpi = 300,
  bg = "white")
```

## Data wrangling

Transform df from county-year level to county level

```
df <- df %>%
  select(county, year, pm25, co2) %>%
  pivot_wider(id_cols = county,
    names_from = year,
    values_from = c(pm25, co2),
    names_glue = "{.value}_{year}",
    values_fn = list(pm25 = ~first(na.omit(.)),
      co2 = ~first(na.omit(.))),
    values_fill = NA_real_) %>%
  select(county,
    all_of(paste0("pm25_", 2000:2010)),
    all_of(paste0("co2_", 2000:2010)))
```

Combine df with df\_causal\_arima (already at county level)

```
df <- df %>%
  left_join(df_causal_arima %>% select(fips,
                                     rel_cum_effect,
                                     avg_effect,
                                     sd_boot,
                                     p_value_boot),
            by = c("county" = "fips"))
```

Calculate 2005-2010 change in PM2.5 concentration for each county

```
df <- df %>%
  mutate(pm25_change_2005_2010 = pm25_2010 - pm25_2005,
         pm25_rel_change_2005_2010 = pm25_change_2005_2010 / pm25_2005)
```

Add variable for significance (p\_value\_boot < 0.05)

```
df <- df %>%
  mutate(significance = case_when(p_value_boot < 0.05 ~ 1,
                                p_value_boot >= 0.05 ~ 0))
```

Regression analysis (for counties with significance == 1)

Filter for significant counties (significance == 1)

```
df_sig <- df %>%
  filter(significance == 1)
```

Plot scatterplots

```
# Visualize rel_cum_effect and pm25_2005
plot1 <- ggplot(df_sig, aes(y = rel_cum_effect,
                           x = pm25_2005)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
```

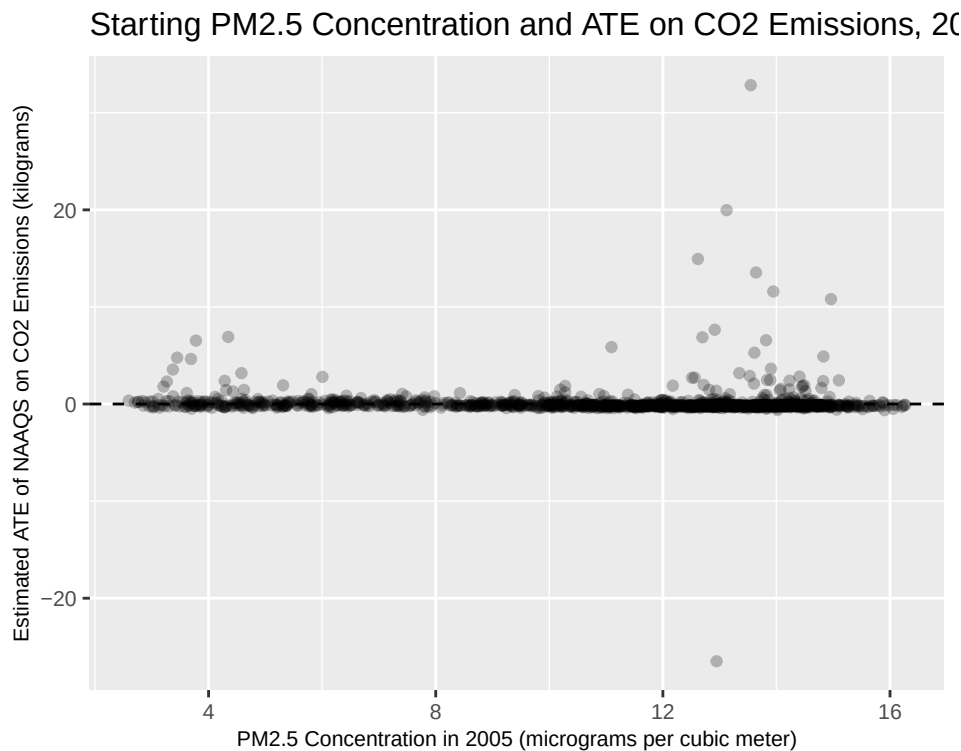


```

size = "Weight (estimated inverse variance)",
title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
theme(plot.title = element_text(size = 12),
axis.title.x = element_text(size = 8),
axis.title.y = element_text(size = 8),
axis.text = element_text(size = 8))

```

plot1



```
ggsave("plot1.png", plot = plot1)
```

## Saving 5 x 4 in image

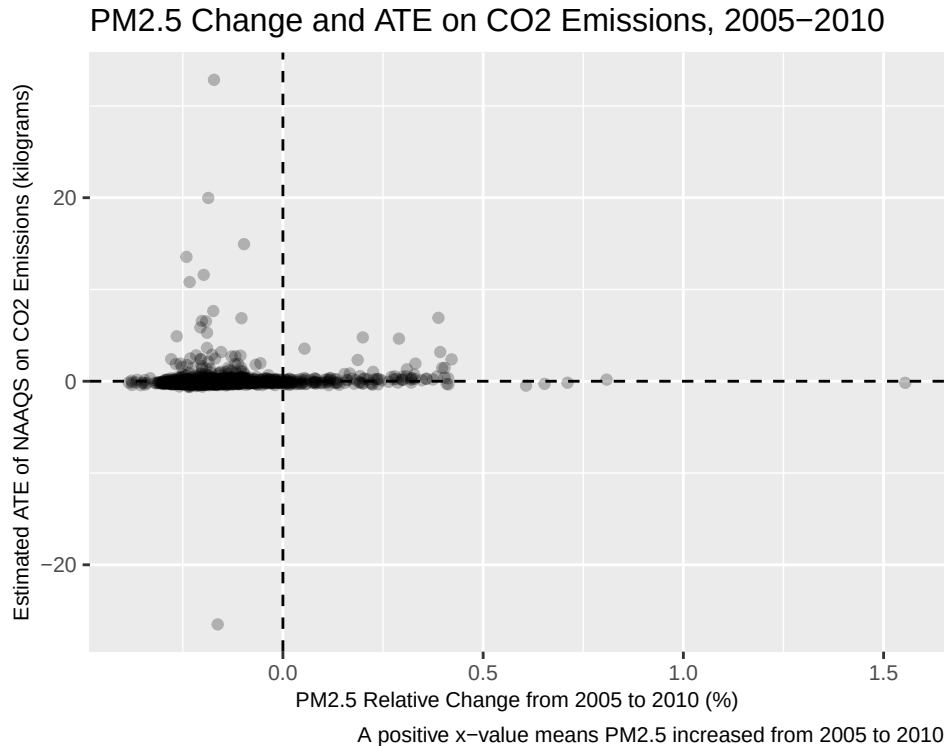
```

# Visualize rel_cum_effect and pm25_rel_change_2005_2010
plot2 <- ggplot(df_sig, aes(y = rel_cum_effect,
                             x = pm25_rel_change_2005_2010)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       size = "Weight (estimated inverse variance)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),

```

```
axis.title.y = element_text(size = 8),
axis.text = element_text(size = 8),
plot.caption = element_text(size = 8))
```

plot2



```
ggsave("plot2.png", plot = plot2)
```

## Saving 5 x 4 in image

Run OLS linear regression with average effect as dependent variable

```
# Fit model
model_ols <- lm(avg_effect ~ pm25_2005 + pm25_rel_change_2005_2010,
  data = df_sig)

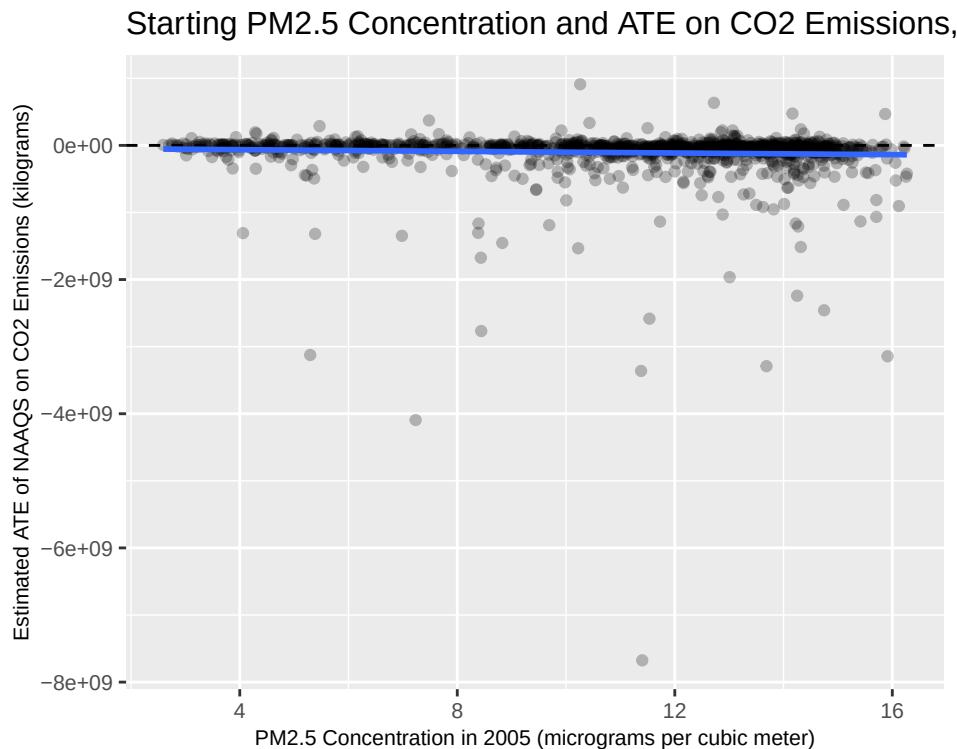
get_regression_table(model_ols)
```

```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>                <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept          -7.68e7 38705282.    -1.98  0.047  -1.53e8 -8.64e5
## 2 pm25_2005           1.64e5  4135809.     0.04  0.968  -7.95e6  8.28e6
## 3 pm25_rel_change_2005_2~ 2.19e8 97754099.     2.24  0.025   2.74e7  4.11e8
```

## Plot scatterplots with OLS model

```
# Visualize avg_effect and pm25_2005
plot1 <- ggplot(df_sig, aes(y = avg_effect,
                           x = pm25_2005)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       size = "Weight (estimated inverse variance)",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

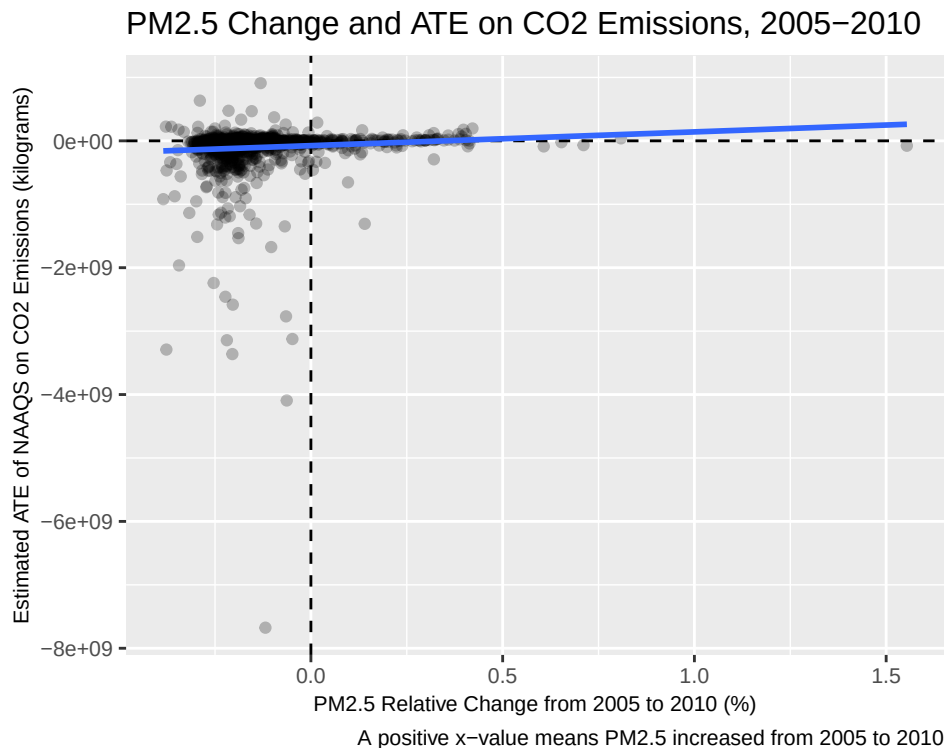


```
ggsave("plot1.png", plot = plot1)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

```
# Visualize avg_effect and pm25_rel_change_2005_2010
plot2 <- ggplot(df_sig, aes(y = avg_effect,
                           x = pm25_rel_change_2005_2010)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       size = "Weight (estimated inverse variance)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8),
        plot.caption = element_text(size = 8))
plot2
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggsave("plot2.png", plot = plot2)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

## Run WLS linear regression with average effect as dependent variable

```
# Create weight from inverse variance
df_sig <- df_sig %>%
  mutate(weight = 1 / (sd_boot)^2)

# Fit full model
model_wls <- lm(avg_effect ~ pm25_2005 + pm25_rel_change_2005_2010,
               weights = weight,
               data = df_sig)
get_regression_table(model_wls)
```

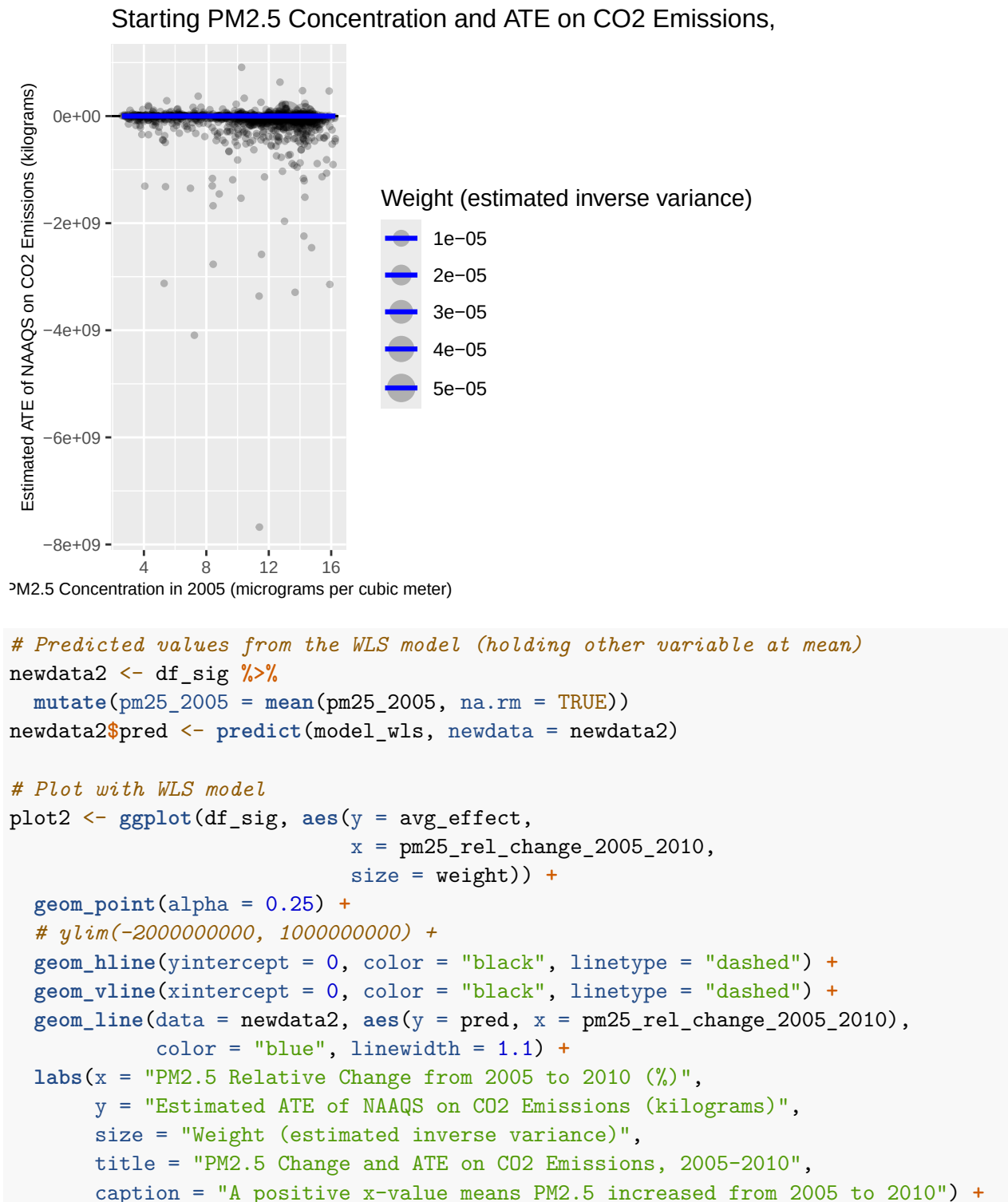
```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>              <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept          402032.   444549.     0.904   0.366 -470050. 1274115.
## 2 pm25_2005         -33289.    36824.    -0.904   0.366 -105527.  38949.
## 3 pm25_rel_change_2005_2~ -258519.   285174.    -0.907   0.365 -817953.  300915.
```

## Plot scatterplots with WLS model

```
# Predicted values from the WLS model (holding other variable at mean)
newdata1 <- df_sig %>%
  mutate(pm25_rel_change_2005_2010 = mean(pm25_rel_change_2005_2010, na.rm = TRUE))
newdata1$pred <- predict(model_wls, newdata = newdata1)

# Plot with WLS model
plot1 <- ggplot(df_sig, aes(y = avg_effect,
                           x = pm25_2005,
                           size = weight)) +
  # ylim(-2000000000, 1000000000) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_line(data = newdata1, aes(y = pred, x = pm25_2005),
           color = "blue", linewidth = 1.1) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       size = "Weight (estimated inverse variance)",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

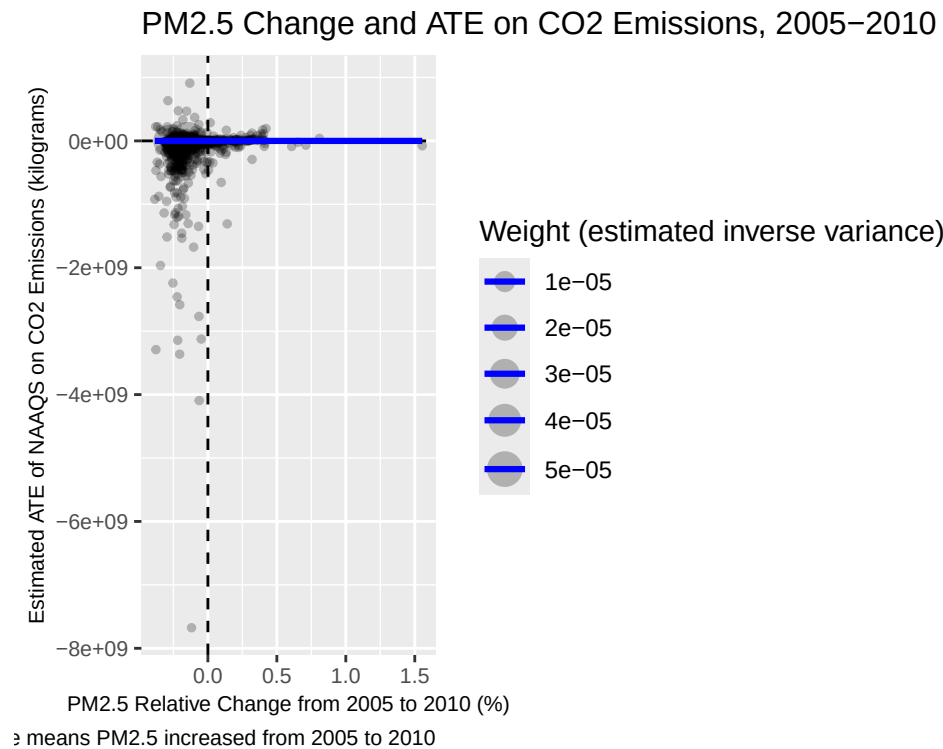


```

theme(plot.title = element_text(size = 12),
      axis.title.x = element_text(size = 8),
      axis.title.y = element_text(size = 8),
      axis.text = element_text(size = 8),
      plot.caption = element_text(size = 8))

```

plot2



## Fit smoothing splines

```

# Fit model
gam_model <- gam(avg_effect ~ s(pm25_2005 + pm25_rel_change_2005_2010), weights = weight,
                 data = df_sig)

```

```

## Warning in term[i] <- attr(terms(reformulate(term[i])), "term.labels"): number
## of items to replace is not a multiple of replacement length

```

```

# Summarize model
summary(gam_model)

```

```

##
## Family: gaussian
## Link function: identity
##

```

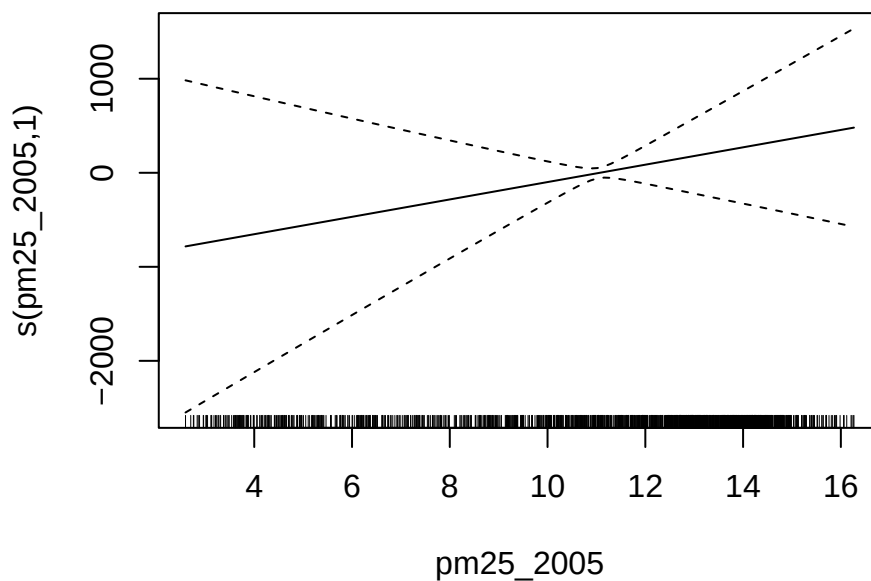
```
## Formula:
## avg_effect ~ s(pm25_2005 + pm25_rel_change_2005_2010)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   60.26    263.28   0.229   0.819
##
## Approximate significance of smooth terms:
##             edf Ref.df      F p-value
## s(pm25_2005)   1      1 0.814   0.367
##
## R-sq.(adj) = -6.83e-05   Deviance explained = 0.0621%
## GCV = 0.26078   Scale est. = 0.26039    n = 1351
```

```
tidy(gam_model)
```

```
## # A tibble: 1 x 5
##   term          edf ref.df statistic p.value
##   <chr>        <dbl> <dbl>      <dbl>   <dbl>
## 1 s(pm25_2005)  1.00   1.00      0.814   0.367
```

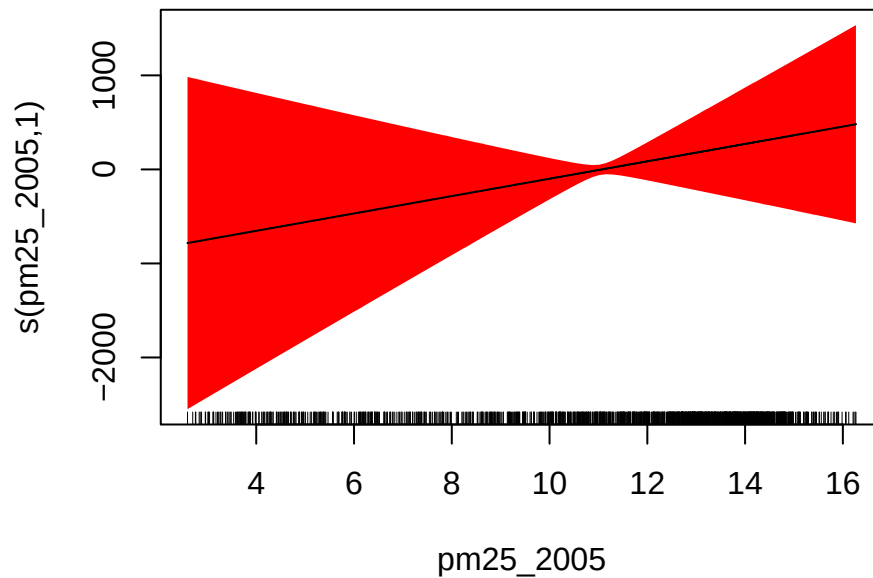
```
# Visualize model
```

```
plot(gam_model)
```



```
plot(gam_model, pages = 1, shade = TRUE, shade.col = "red",
     residuals = TRUE, pch = 16, cex = 0.1, scheme = 1, rug = TRUE)
```

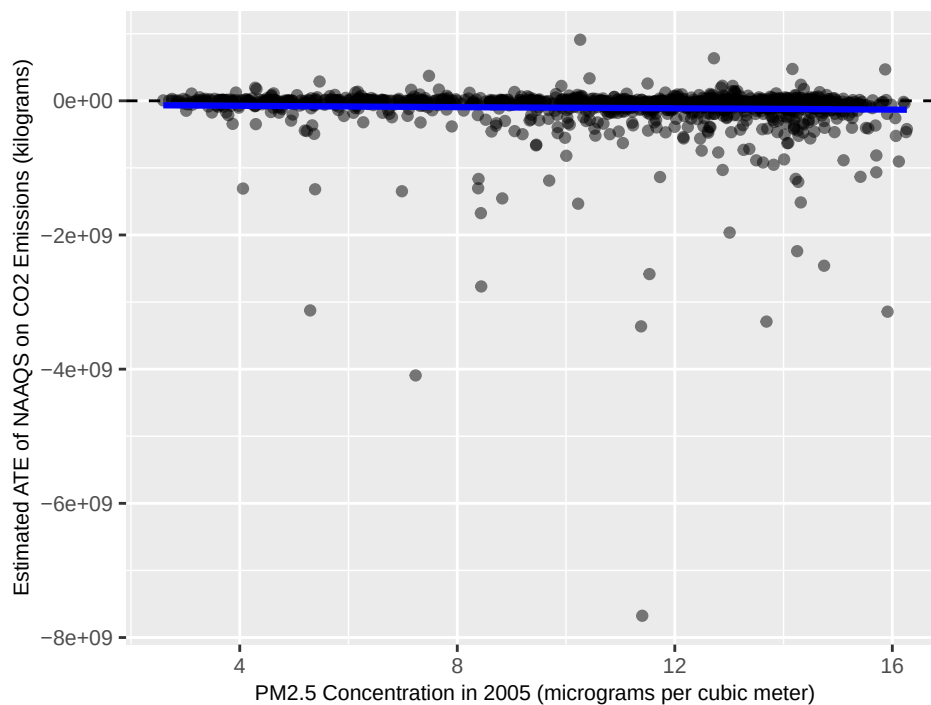




Plot scatterplots with smoothing splines

```
# Visualize avg_effect and pm25_2005 with smoothing spline
plot1 <- ggplot(df_sig, aes(y = avg_effect,
                           x = pm25_2005)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(aes(x = pm25_2005, y = avg_effect, group = 1),
             method = "gam",
             formula = y ~ s(x, bs = "cs"),
             color = "blue",
             fill = "grey80",
             linewidth = 1.1,
             alpha = 0.2,
             inherit.aes = FALSE) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1
```

## Starting PM2.5 Concentration and ATE on CO2 Emissions,



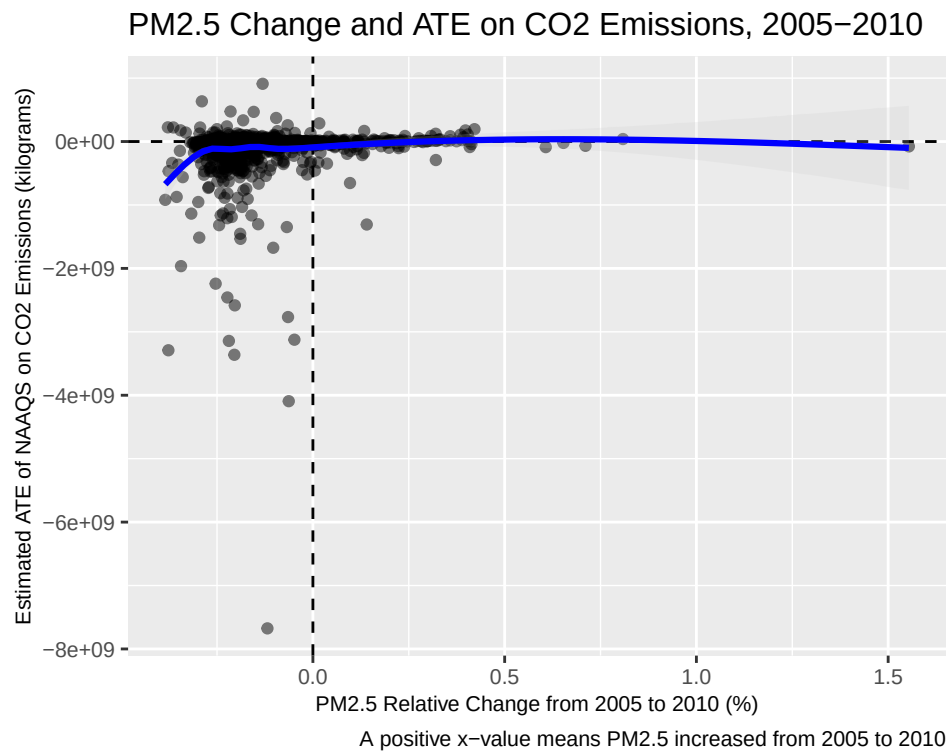
```
ggsave("plot1.png", plot = plot1)
```

```
## Saving 5 x 4 in image
```

```
# Visualize avg_effect and pm25_rel_change_2005_2010 with smoothing spline
plot2 <- ggplot(df_sig, aes(y = avg_effect,
                             x = pm25_rel_change_2005_2010)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(aes(x = pm25_rel_change_2005_2010, y = avg_effect, group = 1),
              method = "gam",
              formula = y ~ s(x, bs = "cs"),
              color = "blue",
              fill = "grey80",
              linewidth = 1.1,
              alpha = 0.2,
              inherit.aes = FALSE) +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
```

```
axis.text = element_text(size = 8),
plot.caption = element_text(size = 8))
```

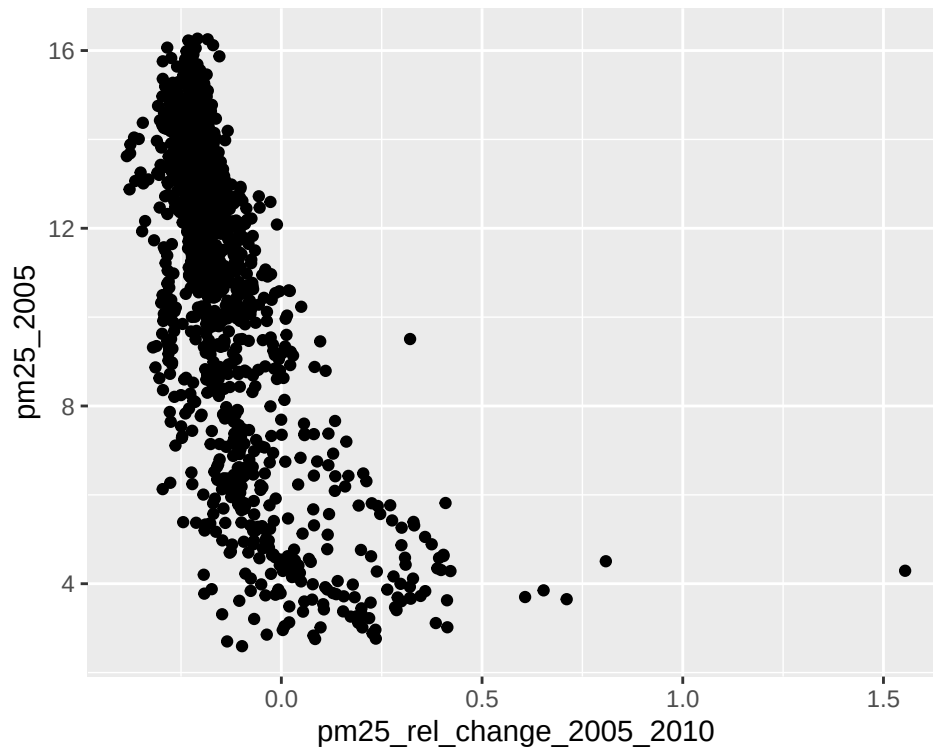
plot2



```
ggsave("plot2.png", plot = plot2)
```

## Saving 5 x 4 in image

```
# Visualize pm25_2005 and pm25_change_2005_2010
ggplot(df_sig, aes(y = pm25_2005, x = pm25_rel_change_2005_2010)) +
  geom_point()
```

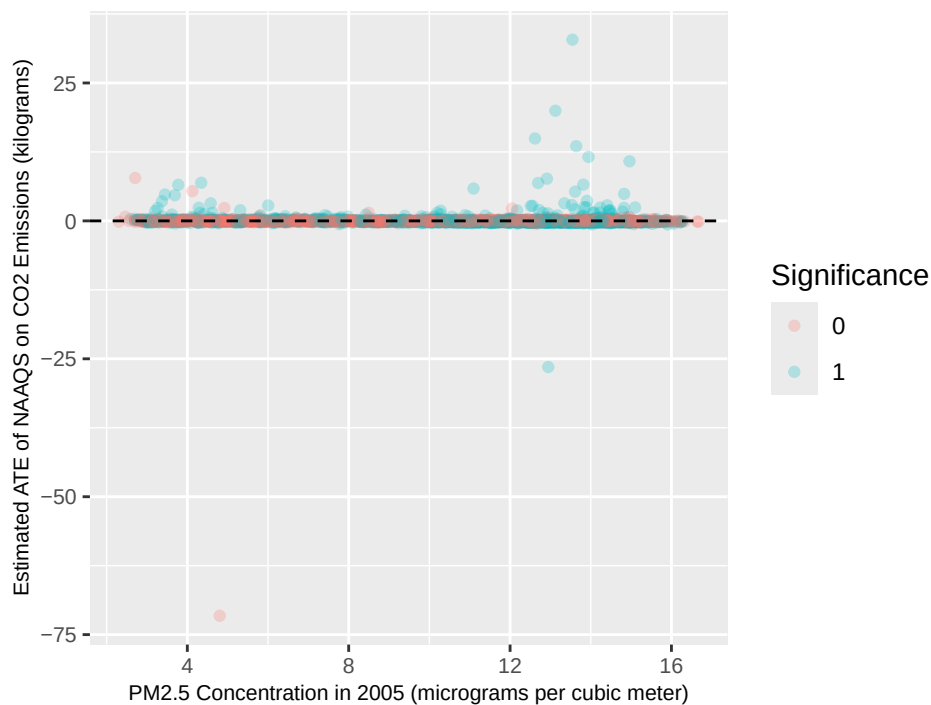


## Regression analysis (for all counties)

### Plot scatterplots

```
# Visualize rel_cum_effect and pm25_2005
plot1 <- ggplot(df, aes(y = rel_cum_effect,
                        x = pm25_2005,
                        color = factor(significance))) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       size = "Weight (estimated inverse variance)",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1
```

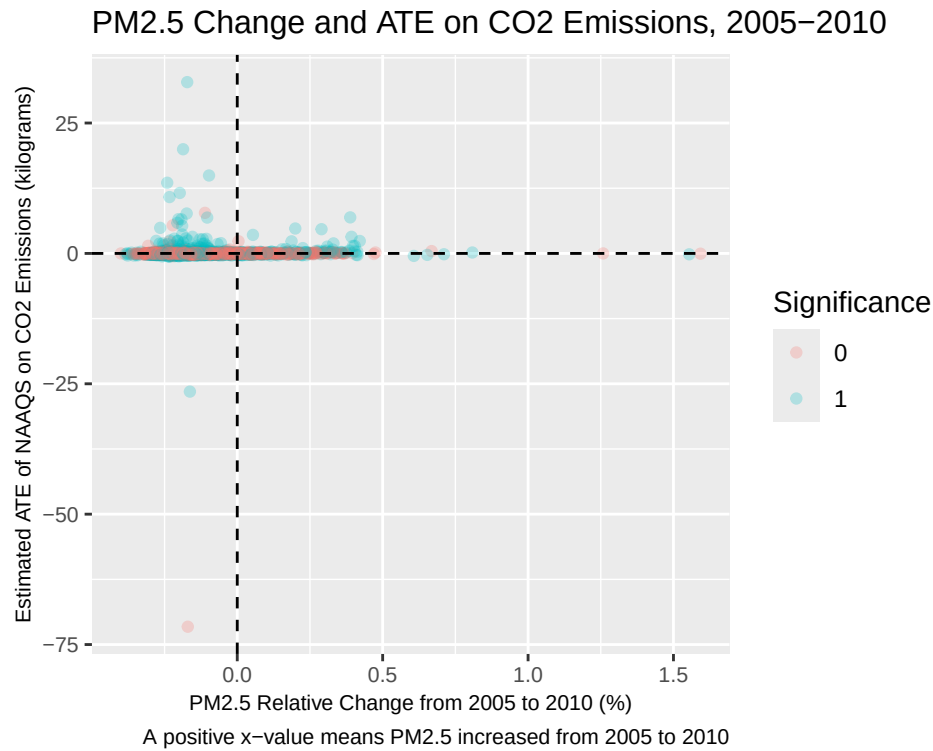
## Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010



```
ggsave("plot1.png", plot = plot1)
```

```
## Saving 5 x 4 in image
```

```
# Visualize rel_cum_effect and pm25_rel_change_2005_2010
plot2 <- ggplot(df, aes(y = rel_cum_effect,
                        x = pm25_rel_change_2005_2010,
                        color = factor(significance))) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       size = "Weight (estimated inverse variance)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8),
        plot.caption = element_text(size = 8))
plot2
```



```
ggsave("plot2.png", plot = plot2)
```

```
## Saving 5 x 4 in image
```

Run OLS linear regression with average effect as dependent variable

```
# Fit model
model_ols <- lm(avg_effect ~ pm25_2005 + pm25_rel_change_2005_2010,
  data = df)

get_regression_table(model_ols)
```

```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>                <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept          -1.43e7 16630717.   -0.859   0.39  -4.69e7  1.83e7
## 2 pm25_2005           -3.13e6  1822216.   -1.72    0.086 -6.71e6  4.40e5
## 3 pm25_rel_change_2005_2~  9.60e7 45117388.    2.13    0.034  7.49e6  1.84e8
```

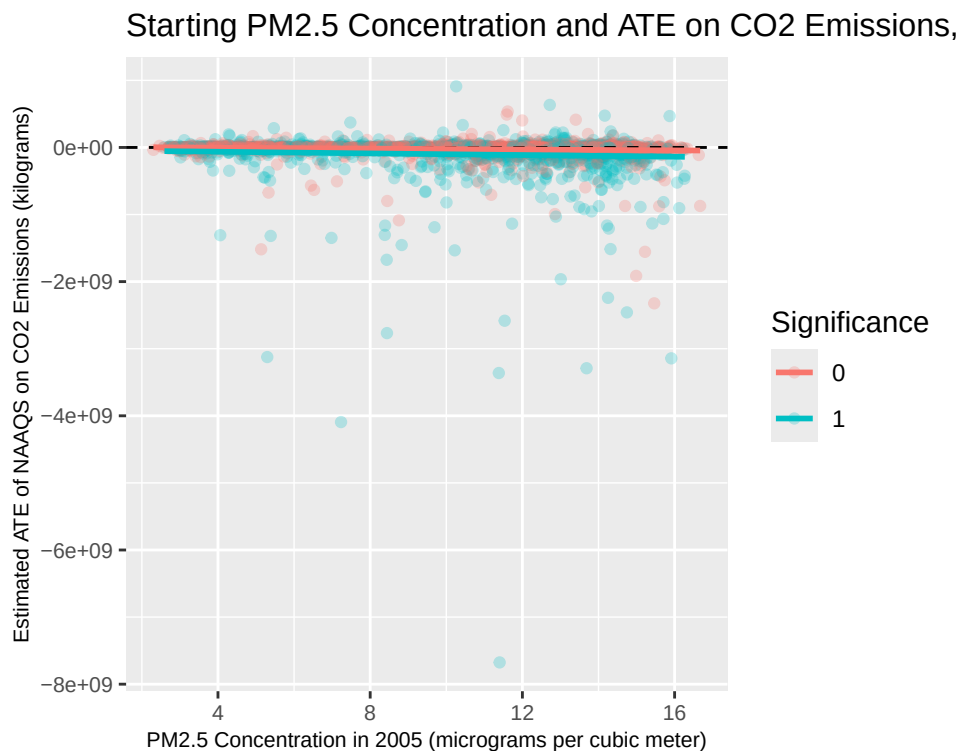
Plot scatterplots with OLS model

```

# Visualize avg_effect and pm25_2005
plot1 <- ggplot(df, aes(y = avg_effect,
                        x = pm25_2005,
                        color = factor(significance))) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       size = "Weight (estimated inverse variance)",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

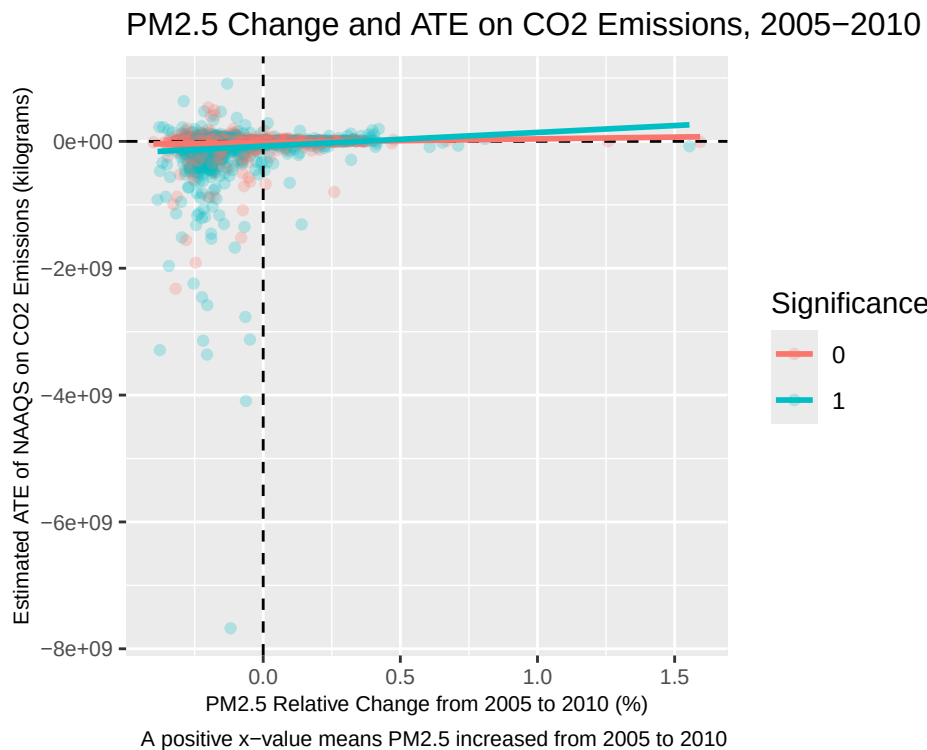


```
ggsave("plot1.png", plot = plot1)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

```
# Visualize avg_effect and pm25_rel_change_2005_2010
plot2 <- ggplot(df, aes(y = avg_effect,
                        x = pm25_rel_change_2005_2010,
                        color = factor(significance))) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       size = "Weight (estimated inverse variance)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8),
        plot.caption = element_text(size = 8))
plot2
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```





```
ggsave("plot2.png", plot = plot2)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

Run WLS linear regression with average effect as dependent variable

```
# Create weight from inverse variance
df <- df %>%
  mutate(weight = 1 / (sd_boot)^2)

# Fit full model
model_wls <- lm(avg_effect ~ pm25_2005 + pm25_rel_change_2005_2010,
               weights = weight,
               data = df)
get_regression_table(model_wls)
```

```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>              <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept         -997.      757.    -1.32    0.188  -2480.    487.
## 2 pm25_2005           95.2      62.1     1.53    0.126   -26.6    217.
## 3 pm25_rel_change_2005_2~ 22.0      690.     0.032   0.975  -1330.   1374.
```

Plot scatterplots with WLS model

```
# Predicted values from the WLS model (holding other variable at mean)
newdata1 <- df %>%
  mutate(pm25_rel_change_2005_2010 = mean(pm25_rel_change_2005_2010, na.rm = TRUE))
newdata1$pred <- predict(model_wls, newdata = newdata1)

# Plot with WLS model
plot1 <- ggplot(df, aes(y = avg_effect,
                       x = pm25_2005,
                       color = factor(significance),
                       size = weight)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_line(data = newdata1, aes(y = pred, x = pm25_2005),
           color = "blue", linewidth = 1.1) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
```

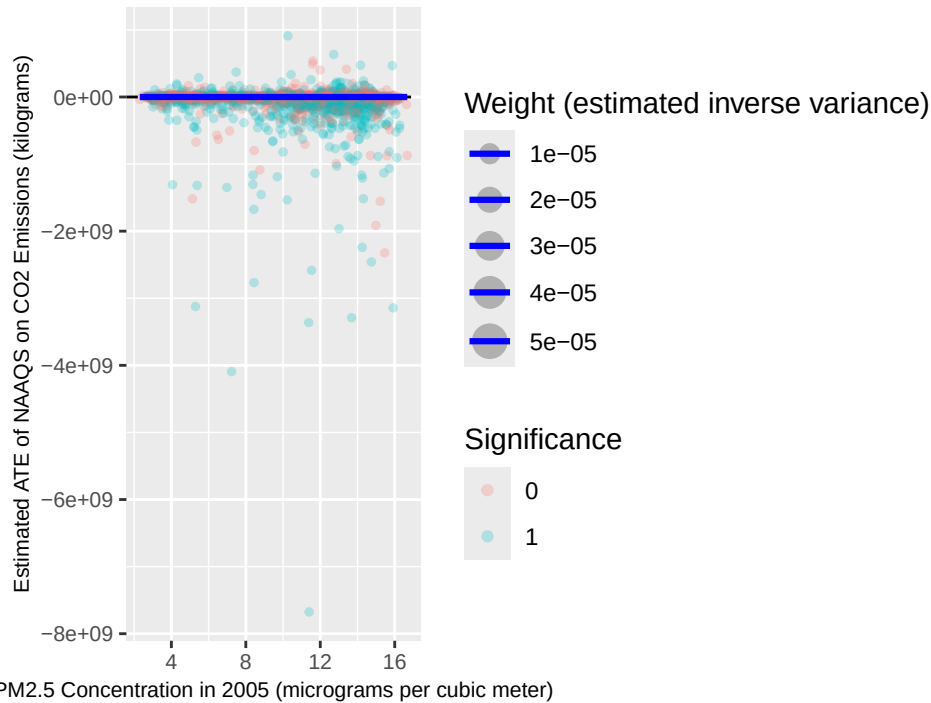
```

size = "Weight (estimated inverse variance)",
title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
theme(plot.title = element_text(size = 12),
axis.title.x = element_text(size = 8),
axis.title.y = element_text(size = 8),
axis.text = element_text(size = 8))

```

plot1

Starting PM2.5 Concentration and ATE on CO2 Emissions,



```

# Predicted values from the WLS model (holding other variable at mean)
newdata2 <- df %>%
  mutate(pm25_2005 = mean(pm25_2005, na.rm = TRUE))
newdata2$pred <- predict(model_wls, newdata = newdata2)

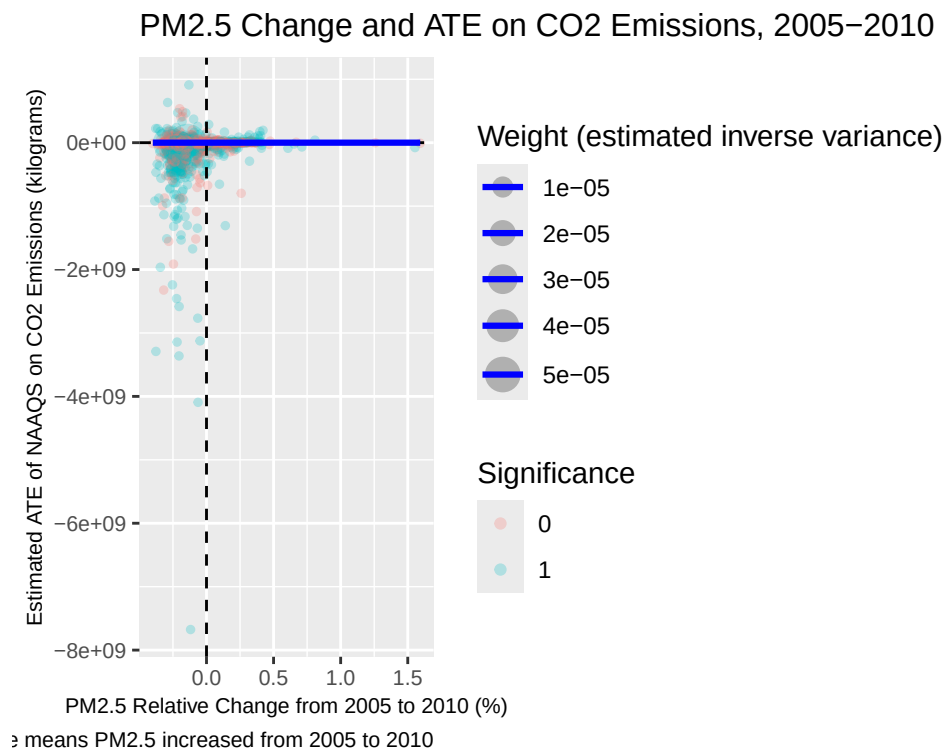
# Plot with WLS model
plot2 <- ggplot(df, aes(y = avg_effect,
                        x = pm25_rel_change_2005_2010,
                        color = factor(significance),
                        size = weight)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_line(data = newdata2, aes(y = pred, x = pm25_rel_change_2005_2010),
            color = "blue", linewidth = 1.1) +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",

```

```

color = "Significance",
size = "Weight (estimated inverse variance)",
title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
theme(plot.title = element_text(size = 12),
axis.title.x = element_text(size = 8),
axis.title.y = element_text(size = 8),
axis.text = element_text(size = 8),
plot.caption = element_text(size = 8))
plot2

```



## Fit smoothing splines

```

# Fit model
gam_model <- gam(avg_effect ~ s(pm25_2005 + pm25_rel_change_2005_2010), weights = weight,
data = df)

```

```

## Warning in term[i] <- attr(terms(reformulate(term[i])), "term.labels"): number
## of items to replace is not a multiple of replacement length

```

```

# Summarize model
summary(gam_model)

```

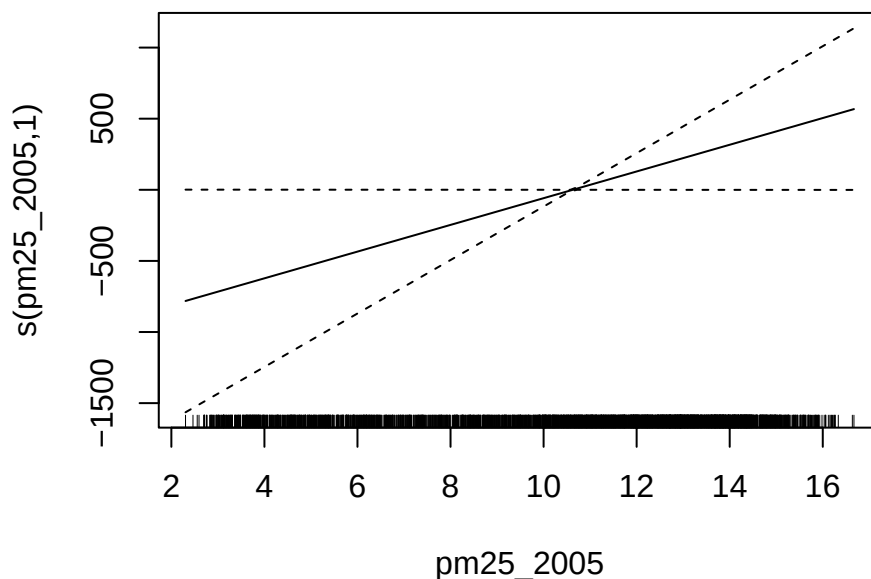
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## avg_effect ~ s(pm25_2005 + pm25_rel_change_2005_2010)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   14.48    138.33   0.105   0.917
##
## Approximate significance of smooth terms:
##             edf Ref.df    F p-value
## s(pm25_2005)   1      1 3.989  0.0459 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.000918   Deviance explained = 0.128%
## GCV = 0.11659   Scale est. = 0.11652   n = 3107
```

```
tidy(gam_model)
```

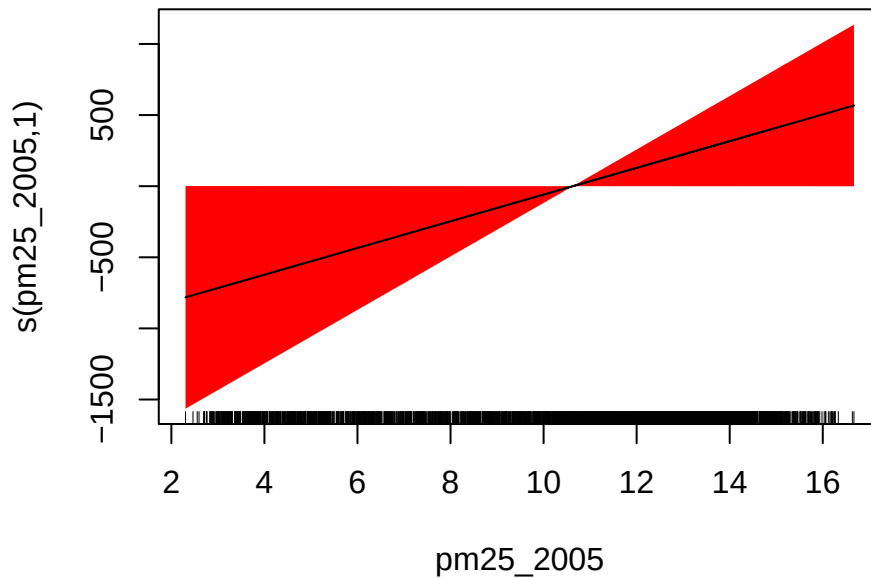
```
## # A tibble: 1 x 5
##   term          edf ref.df statistic p.value
##   <chr>         <dbl> <dbl>      <dbl>   <dbl>
## 1 s(pm25_2005)  1.00   1.00      3.99  0.0459
```

```
# Visualize model
```

```
plot(gam_model)
```



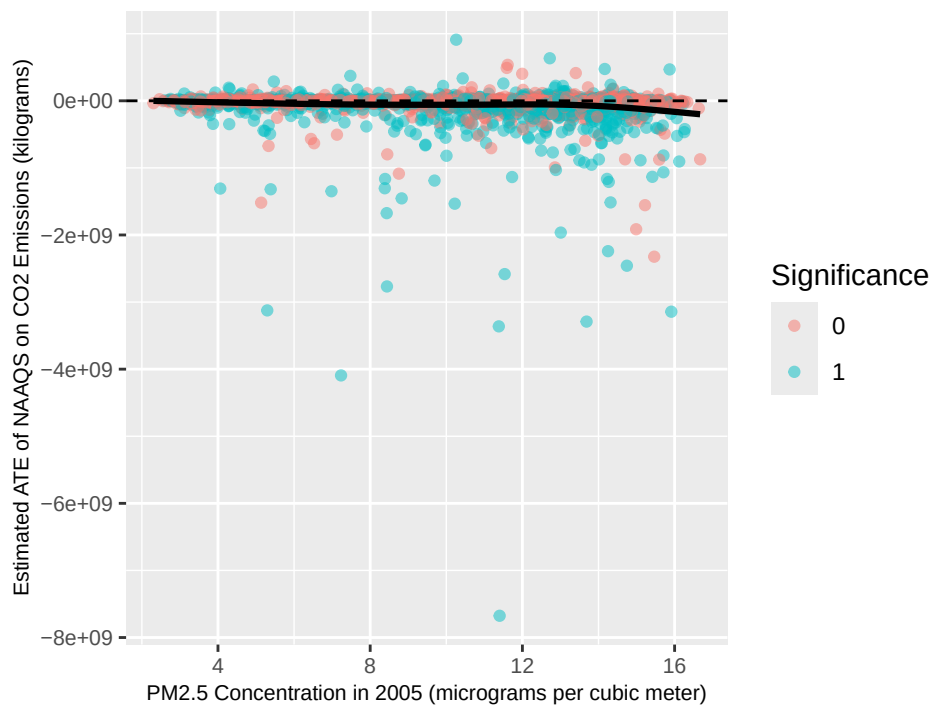
```
plot(gam_model, pages = 1, shade = TRUE, shade.col = "red",
     residuals = TRUE, pch = 16, cex = 0.1, scheme = 1, rug = TRUE)
```



Plot scatterplots with smoothing splines

```
# Visualize avg_effect and pm25_2005 with smoothing spline
plot1 <- ggplot(df, aes(y = avg_effect,
                        x = pm25_2005,
                        color = factor(significance))) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(aes(x = pm25_2005, y = avg_effect, group = 1),
             method = "gam",
             formula = y ~ s(x, bs = "cs"),
             color = "black",
             fill = "grey80",
             linewidth = 1.1,
             alpha = 0.2,
             inherit.aes = FALSE) +
  labs(x = "PM2.5 Concentration in 2005 (micrograms per cubic meter)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       title = "Starting PM2.5 Concentration and ATE on CO2 Emissions, 2005-2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8))
plot1
```

## Starting PM2.5 Concentration and ATE on CO2 Emissions,



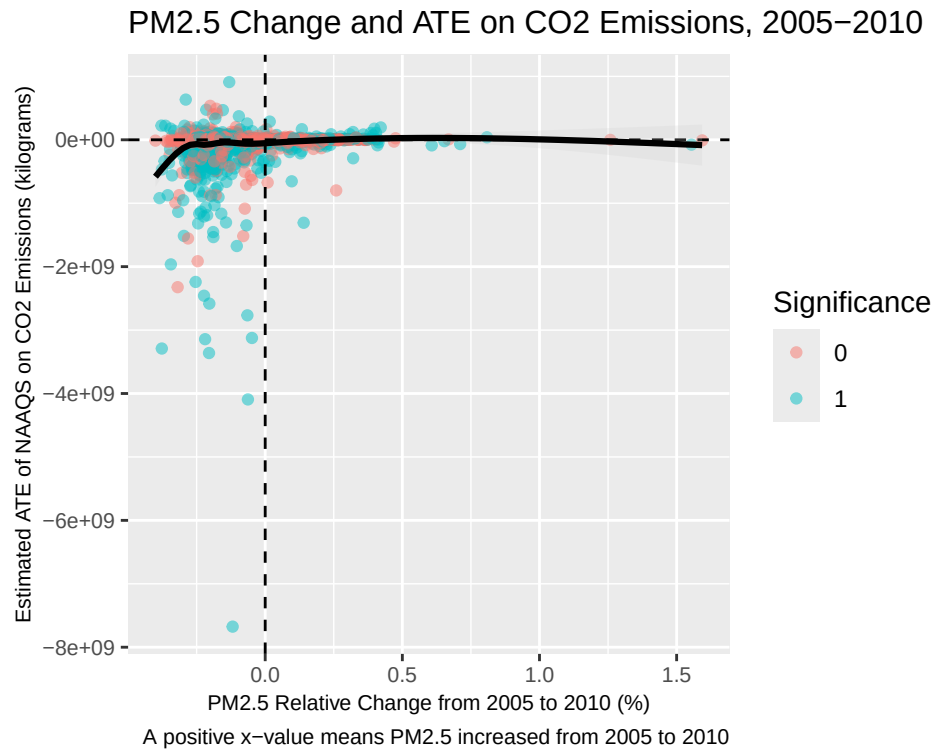
```
ggsave("plot1.png", plot = plot1)
```

```
## Saving 5 x 4 in image
```

```
# Visualize avg_effect and pm25_rel_change_2005_2010 with smoothing spline
plot2 <- ggplot(df, aes(y = avg_effect,
                        x = pm25_rel_change_2005_2010,
                        color = factor(significance))) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(aes(x = pm25_rel_change_2005_2010, y = avg_effect, group = 1),
              method = "gam",
              formula = y ~ s(x, bs = "cs"),
              color = "black",
              fill = "grey80",
              linewidth = 1.1,
              alpha = 0.2,
              inherit.aes = FALSE) +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       color = "Significance",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
```

```
axis.title.x = element_text(size = 8),
axis.title.y = element_text(size = 8),
axis.text = element_text(size = 8),
plot.caption = element_text(size = 8))
```

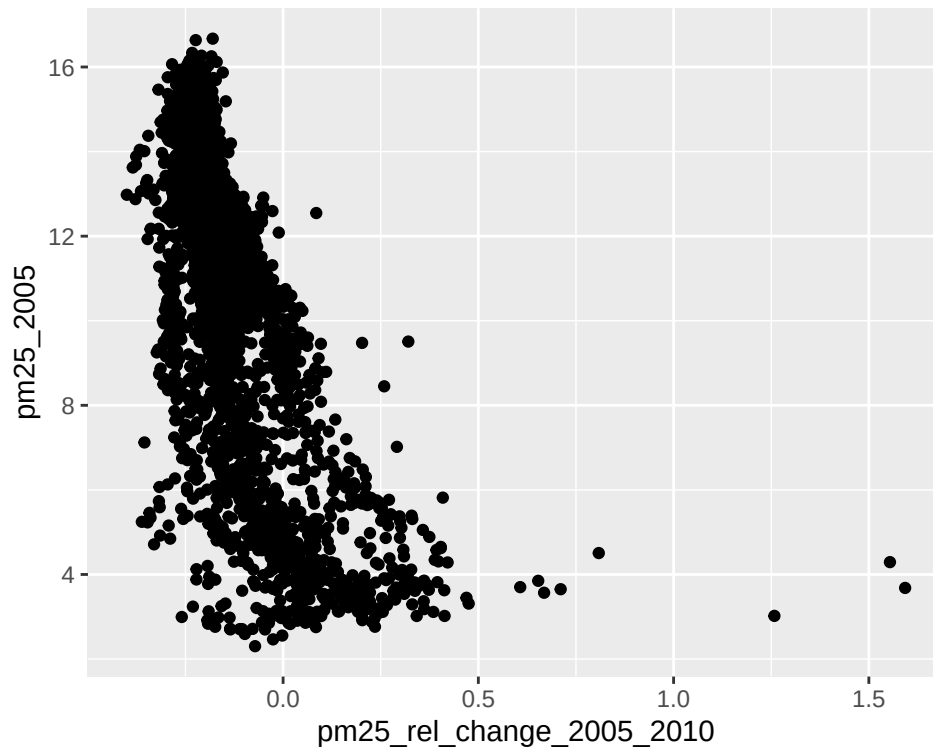
plot2



```
ggsave("plot2.png", plot = plot2)
```

## Saving 5 x 4 in image

```
# Visualize pm25_2005 and pm25_change_2005_2010
ggplot(df, aes(y = pm25_2005, x = pm25_rel_change_2005_2010)) +
  geom_point()
```



Model and visualize with only counties that reduced PM2.5 from 2005 to 2010

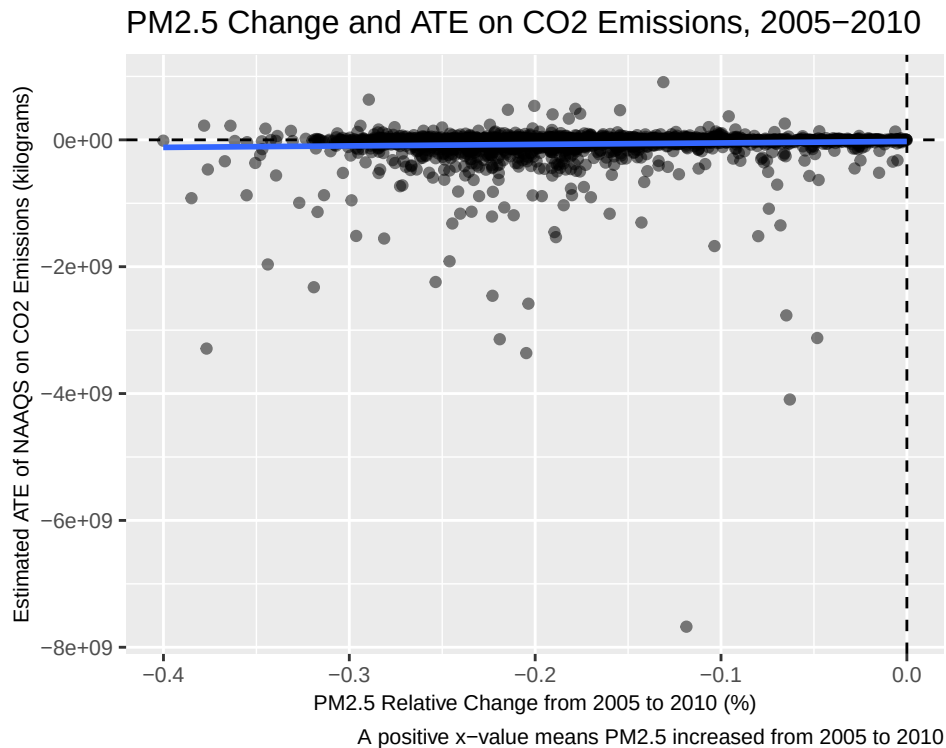
```
# Filter for counties that reduced PM2.5 from 2005 to 2010
df_red <- df %>%
  filter(pm25_change_2005_2010 < 0) %>%
  mutate(weight = 1 / (sd_boot)^2)

# Visualize avg_effect and pm25_rel_change_2005_2010 with OLS model
plot2 <- ggplot(df_red, aes(y = avg_effect,
                           x = pm25_rel_change_2005_2010)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm") +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
  theme(plot.title = element_text(size = 12),
        axis.title.x = element_text(size = 8),
        axis.title.y = element_text(size = 8),
        axis.text = element_text(size = 8),
        plot.caption = element_text(size = 8))

plot2
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggsave("plot2.png", plot = plot2)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

```
# Fit model with OLS
model_test_ols <- lm(avg_effect ~ pm25_rel_change_2005_2010, data = df_red)
get_regression_table(model_test_ols)
```

```
## # A tibble: 2 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>              <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept         -2.53e7 15096770.    -1.68  0.093  -5.49e7  4.27e6
## 2 pm25_rel_change_2005_2~  2.33e8 77962366.     2.99  0.003   8.05e7  3.86e8
```

```
# Fit model with WLS
model_test_wls <- lm(avg_effect ~ pm25_rel_change_2005_2010, weights = weight, data = df_red)
get_regression_table(model_test_wls)
```

```
## # A tibble: 2 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>              <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept           151.    107.     1.41  0.159   -59.2   360.
## 2 pm25_rel_change_2005_2~  -670.    517.    -1.30  0.195  -1684.   343.
```

```
# Fit two-predictor model with OLS
model_test_two_ols <- lm(avg_effect ~ pm25_rel_change_2005_2010 + pm25_2005,
                        data = df_red)
get_regression_table(model_test_two_ols)
```

```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>                <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept            -9.62e6 21918557.   -0.439   0.661  -5.26e7  3.34e7
## 2 pm25_rel_change_2005_2~  1.86e8 91379826.    2.04    0.042   7.10e6  3.65e8
## 3 pm25_2005            -2.12e6 2144451.   -0.989   0.323  -6.33e6  2.08e6
```

```
# Fit two-predictor model with WLS
model_test_two_wls <- lm(avg_effect ~ pm25_rel_change_2005_2010 + pm25_2005,
                        weights = weight,
                        data = df_red)
get_regression_table(model_test_two_wls)
```

```
## # A tibble: 3 x 7
##   term                estimate std_error statistic p_value lower_ci upper_ci
##   <chr>                <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 intercept           -1000.    749.    -1.34    0.182  -2469.    468.
## 2 pm25_rel_change_2005_2~   22.6   683.    0.033   0.974  -1316.   1361.
## 3 pm25_2005              95.5    61.5    1.55    0.121   -25.1    216.
```

Model and visualize with only significant counties that reduced PM2.5 from 2005 to 2010

```
# Filter for significant counties that reduced PM2.5 from 2005 to 2010
df_red_sig <- df %>%
  filter(pm25_change_2005_2010 < 0, significance == 1) %>%
  mutate(weight = 1 / (sd_boot)^2)

# Visualize avg_effect and pm25_rel_change_2005_2010 with OLS model
plot2 <- ggplot(df_red_sig, aes(y = avg_effect,
                                x = pm25_rel_change_2005_2010)) +
  geom_point(alpha = 0.5) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed") +
  geom_vline(xintercept = 0, color = "black", linetype = "dashed") +
  geom_smooth(method = "lm") +
  labs(x = "PM2.5 Relative Change from 2005 to 2010 (%)",
       y = "Estimated ATE of NAAQS on CO2 Emissions (kilograms)",
       title = "PM2.5 Change and ATE on CO2 Emissions, 2005-2010",
       caption = "A positive x-value means PM2.5 increased from 2005 to 2010") +
```

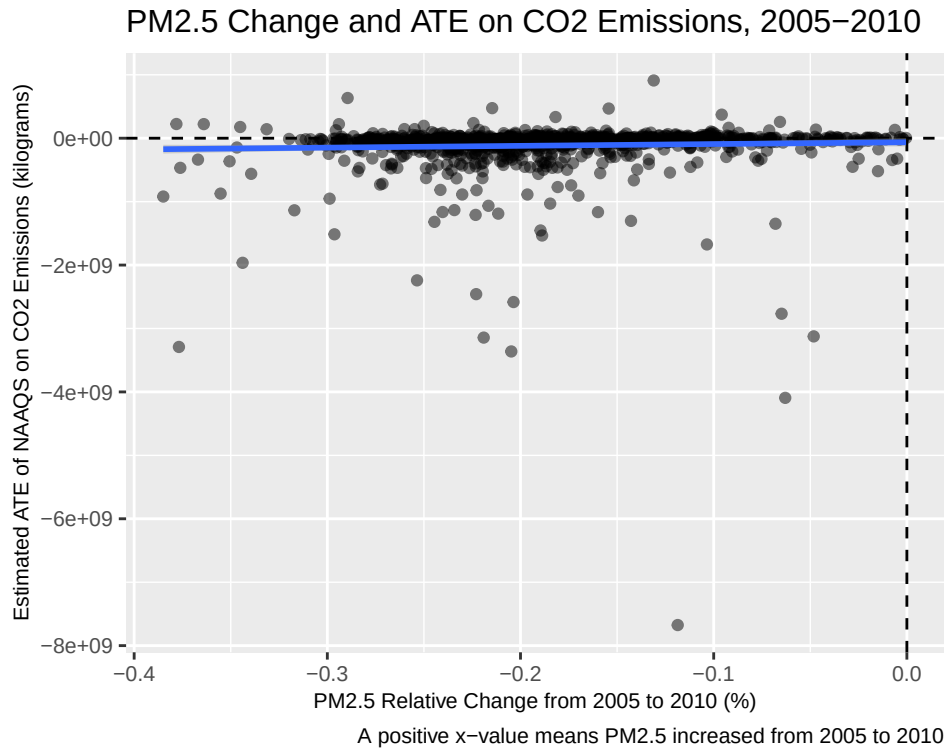
```

theme(plot.title = element_text(size = 12),
      axis.title.x = element_text(size = 8),
      axis.title.y = element_text(size = 8),
      axis.text = element_text(size = 8),
      plot.caption = element_text(size = 8))

```

plot2

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggsave("plot2.png", plot = plot2)
```

```
## Saving 5 x 4 in image
## 'geom_smooth()' using formula = 'y ~ x'
```

```

# Fit model with OLS
model_test_ols <- lm(avg_effect ~ pm25_rel_change_2005_2010, data = df_red_sig)
get_regression_table(model_test_ols)

```

```
## # A tibble: 2 x 7
```

| ##   | term                    | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|------|-------------------------|----------|-----------|-----------|---------|----------|----------|
| ##   | <chr>                   | <dbl>    | <dbl>     | <dbl>     | <dbl>   | <dbl>    | <dbl>    |
| ## 1 | intercept               | -6.34e7  | 3.36e7    | -1.89     | 0.059   | -1.29e8  | 2.47e6   |
| ## 2 | pm25_rel_change_2005_2~ | 2.84e8   | 1.68e8    | 1.68      | 0.092   | -4.68e7  | 6.14e8   |

```
# Fit model with WLS
```

```
model_test_wls <- lm(avg_effect ~ pm25_rel_change_2005_2010, weights = weight, data = df_red_sig)
get_regression_table(model_test_wls)
```

```
## # A tibble: 2 x 7
```

| ##   | term                    | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|------|-------------------------|----------|-----------|-----------|---------|----------|----------|
| ##   | <chr>                   | <dbl>    | <dbl>     | <dbl>     | <dbl>   | <dbl>    | <dbl>    |
| ## 1 | intercept               | 152.     | 157.      | 0.968     | 0.333   | -156.    | 461.     |
| ## 2 | pm25_rel_change_2005_2~ | -719.    | 761.      | -0.945    | 0.345   | -2212.   | 774.     |

```
# Fit two-predictor model with OLS
```

```
model_test_two_ols <- lm(avg_effect ~ pm25_rel_change_2005_2010 + pm25_2005,
  data = df_red_sig)
get_regression_table(model_test_two_ols)
```

```
## # A tibble: 3 x 7
```

| ##   | term                    | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|------|-------------------------|----------|-----------|-----------|---------|----------|----------|
| ##   | <chr>                   | <dbl>    | <dbl>     | <dbl>     | <dbl>   | <dbl>    | <dbl>    |
| ## 1 | intercept               | -8.18e7  | 4.87e7    | -1.68     | 0.094   | -1.77e8  | 1.38e7   |
| ## 2 | pm25_rel_change_2005_2~ | 3.44e8   | 2.05e8    | 1.68      | 0.093   | -5.77e7  | 7.47e8   |
| ## 3 | pm25_2005               | 2.54e6   | 4.87e6    | 0.521     | 0.603   | -7.02e6  | 1.21e7   |

```
# Fit two-predictor model with WLS
```

```
model_test_two_wls <- lm(avg_effect ~ pm25_rel_change_2005_2010 + pm25_2005,
  weights = weight,
  data = df_red_sig)
get_regression_table(model_test_two_wls)
```

```
## # A tibble: 3 x 7
```

| ##   | term                    | estimate | std_error | statistic | p_value | lower_ci | upper_ci |
|------|-------------------------|----------|-----------|-----------|---------|----------|----------|
| ##   | <chr>                   | <dbl>    | <dbl>     | <dbl>     | <dbl>   | <dbl>    | <dbl>    |
| ## 1 | intercept               | -114895. | 475885.   | -0.241    | 0.809   | -1.05e6  | 818758.  |
| ## 2 | pm25_rel_change_2005_2~ | 73082.   | 305276.   | 0.239     | 0.811   | -5.26e5  | 672012.  |
| ## 3 | pm25_2005               | 9530.    | 39419.    | 0.242     | 0.809   | -6.78e4  | 86868.   |